程序鉴别材料

软件名称：**VersePC**
版本号：**V1.0**

前1500行代码（第1-30页）

```
============================================================文件: .source-backup\agent-
engine.js============================================================
/**
 * VersePC Agent Engine
 *
 * 核心设计:
 * 1. 状态机驱动: IDLE → RUNNING → STREAMING → ACTING → OBSERVING → REFLECTING → DONE
 * 2. 增量流式输出: 仅发送新字符, 消息去重 (OutputManager)
 * 3. 事件驱动: 统一 say/ask 消息格式
 * 4. 全功能保留: 意图检测、计划生成、卡死检测、反思、被动检测、并行工具、进度轮询
 */

const https = require('https');
const http = require('http');
const { URL } = require('url');
const { getPluginManager } = require('./plugin-manager');

// ============================================================================
// Agent State Machine
// ============================================================================

const AgentState = {
    IDLE: 'idle',
    RUNNING: 'running',
    STREAMING: 'streaming',
    ACTING: 'acting',
    OBSERVING: 'observing',
    REFLECTING: 'reflecting',
    RESPONDING: 'responding',
    WAITING_FOR_INPUT: 'waiting_for_input',
    DONE: 'done',
    STUCK: 'stuck',
    ERROR: 'error'
};

const SayType = {
    TEXT: 'text',
    REASONING: 'reasoning',
    TOOL_START: 'tool_start',
    TOOL_RESULT: 'tool_result',
    TOOL_END: 'tool_end',
    ERROR: 'error',
    COMPLETION: 'completion',
    API_STARTED: 'api_req_started',
    API_FINISHED: 'api_req_finished',
    FOLLOWUP: 'followup',
    PLAN_CREATED: 'plan_created',
```

```
    PLAN_STEP_UPDATE: 'plan_step_update',
    PLAN_DONE: 'plan_done',
```

```
    THINKING_STEP: 'thinking_step',
    REFLECTION: 'reflection',
    PROGRESS: 'progress'
};

const AskType = {
    TOOL_APPROVAL: 'tool_approval',
    FOLLOWUP: 'followup'
};

// ================================================================================
// Output Manager (增量输出 + 去重)
// ================================================================================

class OutputManager {
    constructor() {
        this.displayedMessages = new Map();
        this.streamedContent = new Map();
        this.currentlyStreamingTs = null;
        this.tsCounter = 0;
    }

    _nextTs() { return ++this.tsCounter; }

    _streamDelta(ts, text) {
        const previous = this.streamedContent.get(ts);
        if (!previous) {
            this.streamedContent.set(ts, { text, headerShown: true });
            this.currentlyStreamingTs = ts;
            return { action: 'full', text };
        }
        if (text.length > previous.text.length && text.startsWith(previous.text)) {
            const delta = text.slice(previous.text.length);
            this.streamedContent.set(ts, { text, headerShown: true });
            return { action: 'delta', text: delta };
        }
        return { action: 'skip' };
    }

    _finishStream(ts) {
        if (this.currentlyStreamingTs === ts) {
            this.currentlyStreamingTs = null;
        }
    }

    processMessage(msg) {
        const text = msg.text || '';
        const isPartial = msg.partial === true;
        let ts;
```

```
        if (isPartial && msg.type === 'say' && (msg.say === SayType.TEXT || msg.say ===
SayType.REASONING || msg.say === SayType.COMPLETION)) {
            if (this.currentlyStreamingTs) {
                const activeMsg = this.displayedMessages.get(this.currentlyStreamingTs);
                if (activeMsg && activeMsg.say === msg.say && activeMsg.partial) {
                    ts = this.currentlyStreamingTs;
                } else {
                    ts = this._nextTs();
                }
            } else {
                ts = this._nextTs();
            }
        } else {
            if (!isPartial && msg.type === 'say' && this.currentlyStreamingTs) {
                const activeMsg = this.displayedMessages.get(this.currentlyStreamingTs);
                if (activeMsg && activeMsg.say === msg.say) {
                    ts = this.currentlyStreamingTs;
                } else {
                    ts = msg.ts || this._nextTs();
                }
            } else {
                ts = msg.ts || this._nextTs();
            }
        }

        const previous = this.displayedMessages.get(ts);
        const alreadyComplete = previous && !previous.partial;

        if (msg.type === 'say') {
            return this._processSay(ts, msg.say, text, isPartial, alreadyComplete, msg);
        }
        if (msg.type === 'ask') {
            return this._processAsk(ts, msg.ask, text, isPartial, alreadyComplete, msg);
        }
        return null;
    }

    _processSay(ts, say, text, isPartial, alreadyComplete, msg) {
        switch (say) {
            case SayType.TEXT:
            case SayType.REASONING:
            case SayType.COMPLETION:
                if (isPartial && text) {
                    const delta = this._streamDelta(ts, text);
                    this.displayedMessages.set(ts, { ts, say, text, partial: true });
                    return { type: 'say', say, ts, text: delta.text, partial: true, delta:
delta.action === 'delta' };
                }
                if (!isPartial && !alreadyComplete) {
                    const streamed = this.streamedContent.get(ts);
                    if (streamed && text && text.length > streamed.text.length &&
text.startsWith(streamed.text)) {
                        const remaining = text.slice(streamed.text.length);
```

```
                        this._finishStream(ts);
                        this.displayedMessages.set(ts, { ts, say, text, partial: false });
                        return { type: 'say', say, ts, text: remaining, partial: false,
trailing: true };
                    }
                    this._finishStream(ts);
                    this.displayedMessages.set(ts, { ts, say, text, partial: false });
                    return { type: 'say', say, ts, text: text || '', partial: false };
                }
                break;

            case SayType.TOOL_START:
            case SayType.TOOL_RESULT:
            case SayType.TOOL_END:
            case SayType.ERROR:
            case SayType.PLAN_CREATED:
            case SayType.PLAN_STEP_UPDATE:
            case SayType.PLAN_DONE:
            case SayType.THINKING_STEP:
            case SayType.REFLECTION:
            case SayType.PROGRESS:
                this.displayedMessages.set(ts, { ts, text, partial: false });
                return { type: 'say', say, ts, text, partial: false };

            case SayType.API_STARTED:
                this.displayedMessages.set(ts, { ts, text, partial: true });
                return { type: 'say', say, ts, text, partial: false };

            case SayType.API_FINISHED:
            case SayType.FOLLOWUP:
                this.displayedMessages.set(ts, { ts, text, partial: false });
                return { type: 'say', say, ts, text, partial: false };
        }
        return null;
    }

    _processAsk(ts, ask, text, isPartial, alreadyComplete, fullMsg) {
        this.displayedMessages.set(ts, { ts, text, partial: false });
        return { type: 'ask', ask, ts, text, partial: false, args: fullMsg.args, toolName:
fullMsg.toolName, risk: fullMsg.risk };
    }

    clear() {
        this.displayedMessages.clear();
        this.streamedContent.clear();
        this.currentlyStreamingTs = null;
        this.tsCounter = 0;
    }
}

// ============================================================================
// AI Platform & Provider 配置（所有平台）
```

```
// ============================================================================
// 结构：
//   PLATFORMS: providerKey → { name, baseUrl, authType, apiFormat, thinkingParams }
//   MODELS: modelId → { provider, name, free }
//
// apiFormat 可选值：'openai'(默认) | 'anthropic' | 'google'
// authType 可选值：'bearer'(默认) | 'x-api-key' | 'url_key'

const PLATFORMS = {
    zhipu: {
        name: '智谱 GLM',
        baseUrl: 'https://open.bigmodel.cn/api/paas/v4',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    deepseek: {
        name: 'DeepSeek',
        baseUrl: 'https://api.deepseek.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    qwen: {
        name: '通义千问',
        baseUrl: 'https://dashscope.aliyuncs.com/compatible-mode/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    moonshot: {
        name: 'Moonshot Kimi',
        baseUrl: 'https://api.moonshot.cn/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    yi: {
        name: '零一万物',
        baseUrl: 'https://api.lingyiwanwu.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    baichuan: {
        name: '百川智能',
        baseUrl: 'https://api.baichuan-ai.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
```

```
    },
    minimax: {
        name: 'MiniMax',
        baseUrl: 'https://api.minimax.chat/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    stepfun: {
        name: '阶跃星辰',
        baseUrl: 'https://api.stepfun.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    siliconflow: {
        name: 'SiliconFlow',
        baseUrl: 'https://api.siliconflow.cn/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    openrouter: {
        name: 'OpenRouter',
        baseUrl: 'https://openrouter.ai/api/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    groq: {
        name: 'Groq',
        baseUrl: 'https://api.groq.com/openai/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    openai: {
        name: 'OpenAI',
        baseUrl: 'https://api.openai.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    anthropic: {
        name: 'Anthropic',
        baseUrl: 'https://api.anthropic.com',
        authType: 'x-api-key',
        apiFormat: 'anthropic',
        thinkingParams: {}
    },
```

```
    google: {
        name: 'Google Gemini',
        baseUrl: 'https://generativelanguage.googleapis.com/v1beta',
        authType: 'url_key',
        apiFormat: 'google',
        thinkingParams: {}
    }
};

const MODELS = {
    // 智谱 GLM
    'glm-5.1': { provider: 'zhipu', name: 'GLM-5.1' },
    'glm-5': { provider: 'zhipu', name: 'GLM-5' },
    'glm-5-plus': { provider: 'zhipu', name: 'GLM-5-Plus' },
    'glm-5-air': { provider: 'zhipu', name: 'GLM-5-Air' },
    'glm-5-flash': { provider: 'zhipu', name: 'GLM-5-Flash', free: true },
    'glm-4.7': { provider: 'zhipu', name: 'GLM-4.7' },
    // DeepSeek
    'deepseek-v4-pro': { provider: 'deepseek', name: 'DeepSeek-V4-Pro' },
    'deepseek-v4-flash': { provider: 'deepseek', name: 'DeepSeek-V4-Flash' },
    'deepseek-chat': { provider: 'deepseek', name: 'DeepSeek-V3.2' },
    'deepseek-reasoner': { provider: 'deepseek', name: 'DeepSeek-R1' },
    // 通义千问
    'qwen3.6-max-preview': { provider: 'qwen', name: 'Qwen3.6-Max' },
    'qwen3.6-plus': { provider: 'qwen', name: 'Qwen3.6-Plus' },
    'qwen3.6-flash': { provider: 'qwen', name: 'Qwen3.6-Flash', free: true },
    'qwen3-235b-a22b': { provider: 'qwen', name: 'Qwen3-235B' },
    'qwen3-30b-a3b': { provider: 'qwen', name: 'Qwen3-30B', free: true },
    'qwq-plus': { provider: 'qwen', name: 'QwQ-Plus' },
    // Moonshot Kimi
    'kimi-k2.6': { provider: 'moonshot', name: 'Kimi-K2.6' },
    'kimi-k2.5': { provider: 'moonshot', name: 'Kimi-K2.5' },
    'moonshot-v1-128k': { provider: 'moonshot', name: 'Moonshot-v1-128k' },
    'moonshot-v1-32k': { provider: 'moonshot', name: 'Moonshot-v1-32k' },
    // 零一万物 Yi
    'yi-lightning': { provider: 'yi', name: 'Yi-Lightning', free: true },
    'yi-large': { provider: 'yi', name: 'Yi-Large' },
    'yi-large-turbo': { provider: 'yi', name: 'Yi-Large-Turbo' },
    'yi-medium': { provider: 'yi', name: 'Yi-Medium' },
    // 百川
    'Baichuan4-Turbo': { provider: 'baichuan', name: 'Baichuan4-Turbo' },
    'Baichuan4-Air': { provider: 'baichuan', name: 'Baichuan4-Air' },
    'Baichuan4': { provider: 'baichuan', name: 'Baichuan4' },
    'Baichuan3-Turbo': { provider: 'baichuan', name: 'Baichuan3-Turbo' },
    // MiniMax
    'MiniMax-M2.7': { provider: 'minimax', name: 'MiniMax-M2.7' },
    'MiniMax-M2.5': { provider: 'minimax', name: 'MiniMax-M2.5' },
    'MiniMax-M2.1': { provider: 'minimax', name: 'MiniMax-M2.1' },
    // 阶跃星辰
    'step-2-16k': { provider: 'stepfun', name: 'Step-2-16K' },
```

```
    'step-1-8k': { provider: 'stepfun', name: 'Step-1-8K', free: true },
    // SiliconFlow
    'Pro/deepseek-ai/DeepSeek-V4-Pro': { provider: 'siliconflow', name: 'DeepSeek-V4-Pro'
},
    'deepseek-ai/DeepSeek-V4-Flash': { provider: 'siliconflow', name: 'DeepSeek-V4-Flash',
free: true },
    'Qwen/Qwen3-235B-A22B': { provider: 'siliconflow', name: 'Qwen3-235B', free: true },
    'Qwen/Qwen3-30B-A3B': { provider: 'siliconflow', name: 'Qwen3-30B', free: true },
    'Qwen/Qwen3.6-Flash': { provider: 'siliconflow', name: 'Qwen3.6-Flash', free: true },
    'Pro/zai-org/GLM-5.1': { provider: 'siliconflow', name: 'GLM-5.1', free: true },
    // OpenRouter
    'deepseek/deepseek-v4-pro': { provider: 'openrouter', name: 'DeepSeek V4 Pro' },
    'deepseek/deepseek-v4-flash:free': { provider: 'openrouter', name: 'DeepSeek V4 Flash',
free: true },
    'qwen/qwen3-235b-a22b': { provider: 'openrouter', name: 'Qwen3 235B' },
    'google/gemini-2.5-flash': { provider: 'openrouter', name: 'Gemini 2.5 Flash', free:
true },
    'anthropic/claude-sonnet-4-6': { provider: 'openrouter', name: 'Claude Sonnet 4.6' },
    'anthropic/claude-3.5-sonnet': { provider: 'openrouter', name: 'Claude 3.5 Sonnet',
free: true },
    // Groq
    'llama-3.3-70b-versatile': { provider: 'groq', name: 'Llama 3.3 70B', free: true },
    'qwen/qwen3-32b': { provider: 'groq', name: 'Qwen3 32B', free: true },
    'deepseek-r1-distill-llama-70b': { provider: 'groq', name: 'DeepSeek R1 70B', free:
true },
    // OpenAI
    'gpt-4.1': { provider: 'openai', name: 'GPT-4.1' },
    'gpt-4.1-mini': { provider: 'openai', name: 'GPT-4.1-mini' },
    'gpt-4.1-nano': { provider: 'openai', name: 'GPT-4.1-nano' },
    'gpt-4o': { provider: 'openai', name: 'GPT-4o' },
    'gpt-4o-mini': { provider: 'openai', name: 'GPT-4o-mini' },
    'o3-mini': { provider: 'openai', name: 'o3-mini' },
    'o3': { provider: 'openai', name: 'o3' },
    // Anthropic Claude
    'claude-sonnet-4-6-20250217': { provider: 'anthropic', name: 'Claude Sonnet 4.6' },
    'claude-opus-4-7-20260416': { provider: 'anthropic', name: 'Claude Opus 4.7' },
    'claude-sonnet-4-5-20250929': { provider: 'anthropic', name: 'Claude Sonnet 4.5' },
    'claude-opus-4-5-20251124': { provider: 'anthropic', name: 'Claude Opus 4.5' },
    'claude-haiku-4-5-20251001': { provider: 'anthropic', name: 'Claude Haiku 4.5' },
    // Google Gemini
    'gemini-3.5-flash': { provider: 'google', name: 'Gemini 3.5 Flash' },
    'gemini-2.5-pro': { provider: 'google', name: 'Gemini 2.5 Pro' },
    'gemini-2.5-flash': { provider: 'google', name: 'Gemini 2.5 Flash', free: true },
    'gemini-2.5-flash-lite': { provider: 'google', name: 'Gemini 2.5 Flash-Lite', free:
true },
    'gemini-2.0-flash': { provider: 'google', name: 'Gemini 2.0 Flash', free: true }
};

function getProviderInfo(modelId) {
    const modelInfo = MODELS[modelId];
    if (!modelInfo) return { platform: PLATFORMS.zhipu, modelInfo: { name: modelId } };
    return { platform: PLATFORMS[modelInfo.provider] || PLATFORMS.zhipu, modelInfo };
}

function getProviderForModel(modelId) {
```

```
    return getProviderInfo(modelId).platform;
}
```

```
function buildApiHeaders(platform, apiKey) {
    const headers = { 'Content-Type': 'application/json' };
    switch (platform.authType) {
        case 'x-api-key':
            headers['x-api-key'] = apiKey;
            headers['anthropic-version'] = '2023-06-01';
            break;
        case 'url_key':
            headers['x-goog-api-key'] = apiKey;
            break;
        case 'bearer':
        default:
            headers['Authorization'] = `Bearer ${apiKey}`;
            break;
    }
    if (platform.apiFormat === 'openrouter') {
        headers['HTTP-Referer'] = 'https://versepc.app';
        headers['X-Title'] = 'VersePC';
    }
    return headers;
}

function buildChatEndpoint(platform, modelId, apiKey) {
    const base = platform.baseUrl;
    switch (platform.apiFormat) {
        case 'anthropic':
            return { url: base + '/v1/messages', method: 'POST' };
        case 'google':
            return { url: base + '/models/' + modelId + ':streamGenerateContent?
alt=sse&key=' + encodeURIComponent(apiKey), method: 'POST' };
        case 'openai':
        default:
            return { url: base + '/chat/completions', method: 'POST' };
    }
}

function buildNonStreamingEndpoint(platform, modelId, apiKey) {
    const base = platform.baseUrl;
    switch (platform.apiFormat) {
        case 'google':
            return { url: base + '/models/' + modelId + ':generateContent?key=' +
encodeURIComponent(apiKey), method: 'POST' };
        default:
            return buildChatEndpoint(platform, modelId, apiKey);
    }
}

// Anthropic 消息格式转换
function _toAnthropicMessages(messages) {
    const system = messages.find(m => m.role === 'system');
    const others = messages.filter(m => m.role !== 'system');
```

```
    const result = { messages: [] };
    if (system) result.system = system.content;

    let i = 0;
    while (i < others.length) {
        const m = others[i];

        if (m.role === 'tool') {
            const toolResults = [];
            while (i < others.length && others[i].role === 'tool') {
                const toolContent = typeof others[i].content === 'string' ?
others[i].content : '';
                toolResults.push({ type: 'tool_result', tool_use_id: others[i].tool_call_id
|| 'unknown', content: toolContent });
                i++;
            }
            if (toolResults.length > 0) {
                result.messages.push({ role: 'user', content: toolResults });
            }
            continue;
        }

        const content = [];
        if (m.content) content.push({ type: 'text', text: m.content });
        if (m.tool_calls) {
            for (const tc of m.tool_calls) {
                let input = {};
                try { input = JSON.parse(tc.function.arguments); } catch (e) {}
                content.push({ type: 'tool_use', id: tc.id, name: tc.function.name, input
});
            }
        }

        if (content.length > 0) {
            result.messages.push({ role: m.role === 'assistant' ? 'assistant' : 'user',
content });
        }
        i++;
    }

    if (result.messages.length === 0 && others.length > 0) {
        result.messages = others.map(m => ({ role: m.role === 'assistant' ? 'assistant' :
'user', content: m.content || '...' }));
    }
    return result;
}

function _toAnthropicTools(tools) {
    if (!tools || !tools.length) return undefined;
    return tools.map(t => ({ name: t.function.name, description: t.function.description,
input_schema: t.function.parameters }));
}

// Google Gemini 消息格式转换
```

```
function _toGoogleMessages(messages) {
    const systemMsg = messages.find(m => m.role === 'system');
```

```
    const nonSystem = messages.filter(m => m.role !== 'system');
    const result = { contents: [] };
    if (systemMsg) result.system_instruction = { parts: [{ text: systemMsg.content }] };

    for (const m of nonSystem) {
        const parts = [];
        if (m.content) parts.push({ text: m.content });
        if (m.tool_calls) {
            for (const tc of m.tool_calls) {
                let args = {};
                try { args = JSON.parse(tc.function.arguments); } catch (e) {}
                parts.push({ functionCall: { name: tc.function.name, args } });
            }
        }
        if (m.role === 'tool') {
            try {
                const parsed = JSON.parse(m.content);
                parts.push({ functionResponse: { name: m.name || m.tool_name || '',
response: parsed } });
            } catch (e) {
                parts.push({ functionResponse: { name: m.name || m.tool_name || '',
response: { result: m.content || '' } } });
            }
        }
        if (parts.length > 0) {
            result.contents.push({
                role: m.role === 'assistant' ? 'model' : (m.role === 'tool' ? 'function' :
'user'),
                parts
            });
        }
    }
    return result;
}

function _toGoogleTools(tools) {
    if (!tools || !tools.length) return undefined;
    return [{ functionDeclarations: tools.map(t => ({ name: t.function.name, description:
t.function.description, parameters: t.function.parameters })) }];
}

// ============================================================================
// Tool 定义
// ============================================================================

const AI_TOOLS = [
    { type: 'function', function: { name: 'bash', description: 'Execute a shell command.
Supports any CLI tool (git, npm, node, python, pip, etc.). For long-running processes (dev
servers, watchers), use background=true. Use manage_processes(action="list") to see running
processes, manage_processes(action="output", pid=N) to read output,
manage_processes(action="stop", pid=N) to stop.', parameters: { type: 'object', properties:
{ command: { type: 'string', description: 'The command to execute' }, cwd: { type:
'string', description: 'Working directory (absolute path). Defaults to previous cwd or
project root.' }, timeout: { type: 'number', description: 'Timeout in seconds (default 120,
max 600). For long-running commands use background=true instead.' }, background: { type:
'boolean', description: 'Run in background (for dev servers, watchers). Returns immediately
```

```
with process info. Use bash(command="taskkill /PID <pid> /F") to stop.' }, restart: { type:
'boolean', description: 'Reset shell session state (cwd, env)' } }, required: ['command'] }
} },
    { type: 'function', function: { name: 'str_replace_based_edit_tool', description: 'File
editing tool: view, create, and edit files. Commands: view (read file), create (new file),
str_replace (exact string replacement), insert (insert at line). create cannot be used on
existing files.', parameters: { type: 'object', properties: { command: { type: 'string',
enum: ['view', 'create', 'str_replace', 'insert'], description: 'Command to execute' },
path: { type: 'string', description: 'Absolute path to file or directory' }, file_text: {
type: 'string', description: 'Required for create: file content' }, old_str: { type:
'string', description: 'Required for str_replace: exact string to replace (must be unique)'
}, new_str: { type: 'string', description: 'New string for str_replace/insert' },
insert_line: { type: 'integer', description: 'Required for insert: line number to insert
after' }, view_range: { type: 'array', items: { type: 'integer' }, description: 'Optional
for view: line range [start, end]' } }, required: ['command', 'path'] } } },
    { type: 'function', function: { name: 'json_edit_tool', description: 'JSON file editing
tool using JSONPath expressions. Operations: view, set, add, remove. Examples:
$.users[0].name, $.config.database', parameters: { type: 'object', properties: { operation:
{ type: 'string', enum: ['view', 'set', 'add', 'remove'], description: 'Operation to
perform' }, file_path: { type: 'string', description: 'Absolute path to JSON file' },
json_path: { type: 'string', description: 'JSONPath expression' }, value: { type: 'object',
description: 'Value to set or add (required for set/add)' }, pretty_print: { type:
'boolean', description: 'Format output (default true)' } }, required: ['operation',
'file_path'] } } },
    { type: 'function', function: { name: 'sequential_thinking', description: 'Break down
complex problems into sequential thinking steps. Each step produces a conclusion. Supports
revising previous steps. Use when deep analysis is needed.', parameters: { type: 'object',
properties: { thought: { type: 'string', description: 'Thinking content for current step'
}, thought_number: { type: 'number', description: 'Current step number' }, total_thoughts:
{ type: 'number', description: 'Estimated total steps' }, next_thought_needed: { type:
'boolean', description: 'Whether another step is needed' }, is_revision: { type: 'boolean',
description: 'Whether revising a previous step' }, revises_thought: { type: 'number',
description: 'Step number being revised (when is_revision=true)' }, branch_from_thought: {
type: 'number', description: 'Branch from this step (optional)' }, branch_id: { type:
'string', description: 'Branch identifier (optional)' } }, required: ['thought',
'thought_number', 'total_thoughts', 'next_thought_needed'] } } },
    { type: 'function', function: { name: 'attempt_completion', description: 'Report task
completion. Only call after verifying the task is done. The result will be presented to the
user for confirmation.', parameters: { type: 'object', properties: { result: { type:
'string', description: 'Final result message with summary of completed work' } }, required:
['result'] } } },
    { type: 'function', function: { name: 'ckg', description: 'Code Knowledge Graph: search
for functions, classes, and class methods in the codebase.', parameters: { type: 'object',
properties: { command: { type: 'string', enum: ['search_function', 'search_class',
'search_class_method'], description: 'Search command' }, path: { type: 'string',
description: 'Codebase path' }, identifier: { type: 'string', description:
'Function/class/method name to search' }, print_body: { type: 'boolean', description:
'Print function/class body (default true)' } }, required: ['command', 'path', 'identifier']
} } },
    { type: 'function', function: { name: 'update_todo_list', description: 'Create or
update a task plan. MUST be called as the FIRST action for any task requiring tools.
Decomposes the user request into concrete, actionable tasks and tracks progress. Format: [
] pending, [-] in progress, [x] completed.', parameters: { type: 'object', properties: {
todos: { type: 'string', description: 'Complete task list in Markdown format. Example:\n- [
] Task 1: Analyze code structure\n- [-] Task 2: Modify CSS styles (currently executing)\n-
[x] Task 3: Bug fix completed' } }, required: ['todos'] } } },
    {
```

```
        type: 'function',
        function: {
            name: 'sub_agent_dispatch',
            description: '派遣子代理执行特定任务。file_search: 在代码库中搜索文件和目录,
code_analysis: 分析代码结构和调用链, resource_download: 搜索Minecraft资源, crash_analysis: 分
析崩溃日志, code_completion: 代码补全和优化, auto: 自动选择最合适的子代理类型',
            parameters: {
                type: 'object',
                properties: {
                    agent_type: {
                        type: 'string',
                        enum: ['file_search', 'code_analysis', 'resource_download',
'crash_analysis', 'code_completion', 'auto'],
                        description: '子代理类型。使用 auto 可自动根据任务描述选择最合适的子代
理'
                    },
                    task: {
                        type: 'string',
                        description: '子代理要执行的具体任务描述'
                    }
                },
                required: ['agent_type', 'task']
            }
        }
    },
    { type: 'function', function: { name: 'start_preview', description: 'Start a local HTTP
server to preview web files (HTML/CSS/JS). Opens a browser preview panel in the UI. Use
this when the user wants to see how their web code looks. Use command="stop" to stop the
server.', parameters: { type: 'object', properties: { command: { type: 'string', enum:
['start', 'stop'], description: 'start to begin preview, stop to end it' }, root: { type:
'string', description: 'Absolute path to the directory to serve (required for start)' },
port: { type: 'number', description: 'Port number (default 8080)' } }, required:
['command'] } } },
    { type: 'function', function: { name: 'manage_processes', description: 'List and manage
background processes started by bash. Use to check dev server output, list running
processes, or stop processes.', parameters: { type: 'object', properties: { action: { type:
'string', enum: ['list', 'output', 'stop', 'stop_all'], description: 'list: show running
processes. output: get stdout/stderr of a process. stop: kill a process by PID. stop_all:
kill all background processes.' }, pid: { type: 'number', description: 'Process ID
(required for output and stop actions)' }, tail: { type: 'number', description: 'Number of
last lines to return for output (default 50)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'ask_user', description: 'Ask the user a question
to get clarification or make a decision. Use when you need user input to proceed. Supports
free text or multiple choice.', parameters: { type: 'object', properties: { question: {
type: 'string', description: 'The question to ask the user' }, options: { type: 'array',
items: { type: 'string' }, description: 'Optional multiple choice options. If provided,
user can select one. If omitted, user can type free text.' }, context: { type: 'string',
description: 'Optional context or explanation for why you are asking' } }, required:
['question'] } } },
    { type: 'function', function: { name: 'undo_edit', description: 'Manage file edit
backups. List recent AI file changes, view diffs, and restore previous versions. Use
action="list" to see history, action="restore" to rollback, action="diff" to compare
versions.', parameters: { type: 'object', properties: { action: { type: 'string', enum:
['list', 'restore', 'diff', 'session'], description: 'list: show backup history. restore:
rollback to a backup. diff: compare backup with current. session: show session summary.' },
backup_id: { type: 'string', description: 'Backup ID (required for restore and diff
```

```
actions)' }, file_path: { type: 'string', description: 'Filter backups by file path
(optional for list)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'view_history', description: 'View change history
and audit log. Track all AI file modifications, tool executions, and session activity.',
parameters: { type: 'object', properties: { action: { type: 'string', enum: ['changes',
'audit', 'summary'], description: 'changes: file modification history. audit: tool
execution log. summary: session statistics.' }, file_path: { type: 'string', description:
'Filter changes by file path (optional)' }, tool_name: { type: 'string', description:
'Filter by tool name (optional)' }, limit: { type: 'number', description: 'Max results to
return (default 20)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'validate_code', description: 'Validate code
syntax before writing. Checks JS/TS/JSON/CSS/HTML for syntax errors. Use before large file
writes to catch errors early.', parameters: { type: 'object', properties: { content: {
type: 'string', description: 'The code content to validate' }, file_path: { type: 'string',
description: 'File path (used to determine language for validation)' } }, required:
['content', 'file_path'] } } },
    { type: 'function', function: { name: 'build_index', description: 'Build a searchable
index of the codebase. Must be called before semantic_search. Indexes all code files in the
directory for fast semantic search.', parameters: { type: 'object', properties: { root_dir:
{ type: 'string', description: 'Absolute path to the root directory to index' } },
required: ['root_dir'] } } },
    { type: 'function', function: { name: 'semantic_search', description: 'Search the
codebase using natural language queries. Finds files by meaning, not just keywords.
Requires build_index to be called first. Use for queries like "find authentication code",
"where is the database connection", "error handling logic".', parameters: { type: 'object',
properties: { query: { type: 'string', description: 'Natural language search query
describing what you are looking for' }, root_dir: { type: 'string', description: 'Limit
search to this directory (optional)' }, max_results: { type: 'number', description:
'Maximum results to return (default 10)' } }, required: ['query'] } } },
    { type: 'function', function: { name: 'index_stats', description: 'Get statistics about
the current code index (number of files, tokens, memory usage).', parameters: { type:
'object', properties: {} } } },
];

let _pluginManager = null;
function _getPluginTools() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getTools();
    } catch (e) { return []; }
}

function _getPluginDisplayNames() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getToolDisplayNames();
    } catch (e) { return {}; }
}

function _getPluginRisks() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
```

```
        return _pluginManager.getToolRisks();
    } catch (e) { return {}; }
}

function _getPluginPromptExtensions() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getPromptExtensions();
    } catch (e) { return []; }
}

function _isPluginTool(name) {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.isPluginTool(name);
    } catch (e) { return false; }
}

function _getAllTools() {
    return [...AI_TOOLS, ..._getPluginTools()];
}

const toolDescriptions = AI_TOOLS.map(t => `- ${t.function.name}:
${t.function.description}`).join('\n');

const TOOL_RISK = {
    str_replace_based_edit_tool: 'safe', json_edit_tool: 'safe', ckg: 'safe',
    sequential_thinking: 'safe', attempt_completion: 'safe',
    bash: 'safe', update_todo_list: 'safe',
    sub_agent_dispatch: 'safe', start_preview: 'safe', manage_processes: 'safe',
    ask_user: 'safe',
    undo_edit: 'safe',
    view_history: 'safe',
    validate_code: 'safe',
    build_index: 'safe',
    semantic_search: 'safe',
    index_stats: 'safe'
};

const TOOL_DISPLAY_NAMES = {
    bash: '执行命令', str_replace_based_edit_tool: '编辑文件',
    json_edit_tool: '编辑JSON', sequential_thinking: '分步思考',
    attempt_completion: '完成任务', ckg: '代码图谱',
    update_todo_list: '更新计划',
    sub_agent_dispatch: '派遣子代理',
    start_preview: '启动预览',
    manage_processes: '进程管理',
    ask_user: '询问用户',
    undo_edit: '撤销编辑',
    view_history: '查看历史',
    validate_code: '验证代码',
```

```
    build_index: '构建索引',
    semantic_search: '语义搜索',
    index_stats: '索引统计'
};

const AGENT_META = {
    file_search: {
        name: '文件搜索代理', role: 'File Search', avatar: 'robot', color: '#4caf50',
        systemPrompt: `你是 VersePC 项目的文件搜索代理。你的任务是在代码库中快速定位文件和目录。

## 工作流程
1. 首先理解搜索目标（文件名、关键词、文件类型等）
2. 使用 bash 工具执行搜索命令：
   - 按文件名搜索：find . -name "pattern" -type f
   - 按内容搜索：grep -r "pattern" --include="*.js" --include="*.css" --include="*.html" -l
   - 查看目录结构：ls -la，tree（限制深度）
   - 查看文件信息：wc -l，file，stat
3. 对搜索结果进行筛选和排序
4. 输出结构化结果

## 输出格式
搜索完成后，用以下格式总结：
- 搜索目标：xxx
- 找到 N 个文件
- 关键文件列表（路径 + 用途说明）
- 相关代码片段（如有）

## 注意事项
- 优先搜索项目根目录下的 js/、css/、agent-engine.js 等核心文件
- 搜索时排除 node_modules、dist、.git 目录
- 如果搜索结果过多，按相关性排序，只展示最相关的
- 用中文输出结果`
    },
    code_analysis: {
        name: '代码分析代理', role: 'Code Analysis', avatar: 'robot', color: '#9c27b0',
        systemPrompt: `你是 VersePC 项目的代码分析代理。你的任务是分析代码结构、理解项目架构、追踪函数调用链。

## 工作流程
1. 首先确定分析目标（文件、函数、模块）
2. 使用 bash 工具阅读代码：
   - cat 查看文件内容
   - grep 搜索函数调用
   - find 查找相关文件
3. 分析代码结构和依赖关系
4. 输出分析报告

## 分析维度
- 文件职责：该文件的主要功能
- 依赖关系：依赖了哪些模块，被哪些模块依赖
- 函数调用链：关键函数的调用路径
```

```
- 数据流：数据如何在模块间传递
- 潜在问题：发现的 bug、性能问题、代码异味

## 输出格式
分析完成后，用以下格式总结：
- 分析目标：xxx
- 文件结构：列出相关文件及其职责
- 核心逻辑：关键函数和调用链
- 发现的问题（如有）：问题描述 + 建议修复方案

## 注意事项
- 用中文输出分析结果
- 代码片段保留原始格式
- 行号引用格式：文件名:行号`
    },
    resource_download: {
        name: '资源搜索代理', role: 'Resource Search', avatar: 'robot', color: '#8d6e63',
        systemPrompt: `你是 Minecraft 资源搜索代理。你的任务是搜索和推荐 Minecraft 相关资源。

## 支持的资源类型
- Mod（模组）
- 整合包（Modpack）
- 材质包（Texture Pack）
- 光影包（Shader Pack）

## 工作流程
1. 理解用户需求（版本、类型、功能偏好）
2. 搜索 CurseForge 和 Modrinth 上的资源
3. 筛选和推荐

## 推荐标准
- 版本兼容性：优先推荐与用户 Minecraft 版本兼容的资源
- 稳定性：优先推荐更新频繁、bug 少的资源
- 下载量和评分：作为参考指标
- 兼容性：推荐的资源之间不要有冲突

## 输出格式
- 资源名称 + 简介
- 版本兼容信息
- 下载量/评分
- 安装建议
- 注意事项（如有）

## 注意事项
- 用中文输出推荐结果
- 如果资源需要前置依赖，一并说明`
    },
    crash_analysis: {
        name: '崩溃分析代理', role: 'Crash Analysis', avatar: 'robot', color: '#9e9e9e',
        systemPrompt: `你是 Minecraft 崩溃分析代理。你的任务是分析崩溃日志，定位错误原因并提供修复建议。
```

```
## 工作流程
1. 定位日志文件（crash-reports/、logs/ 目录）
2. 使用 bash 工具读取和分析日志
3. 识别崩溃类型和原因
4. 提供修复建议

## 崩溃类型识别
- Mod 冲突：检查多个 Mod 的兼容性
- 内存不足：检查 JVM 参数和内存分配
- 配置错误：检查配置文件格式和值
- 版本不兼容：检查 Mod 与 Minecraft 版本的兼容性
- 驱动问题：检查显卡驱动版本
- Java 版本：检查 Java 版本是否匹配

## 输出格式
- 崩溃类型：xxx
- 错误原因：详细描述
- 涉及文件：相关日志文件和配置文件
- 修复步骤：按优先级排列的修复方案
- 预防建议：避免再次发生的方法

## 注意事项
- 用中文输出分析结果
- 引用关键日志行（文件:行号）
- 修复步骤要具体可操作`
    },
    code_completion: {
        name: '代码补全代理', role: 'Code Completion', avatar: 'robot', color: '#00bcd4',
        systemPrompt: `你是 VersePC 项目的代码补全代理。你的任务是代码重写、优化、补全和重构。

## 工作流程
1. 阅读并理解目标代码的上下文和意图
2. 分析代码结构、变量作用域、依赖关系
3. 生成高质量的补全/优化代码
4. 验证生成的代码语法正确性

## 支持的任务类型
- 代码补全：根据上下文补全未完成的代码
- 代码重写：重构现有代码以提升可读性或性能
- 代码优化：优化算法、减少冗余、提升效率
- 模式匹配：按照项目现有代码风格生成新代码

## 输出格式
- 原始代码（简要引用）
- 修改后的完整代码
- 修改说明：改了什么、为什么改
- 影响范围：可能受影响的文件和函数

## 注意事项
```

```
- 用中文输出说明，代码本身保持原语言
- 保持与项目现有代码风格一致
- 不要引入项目中未使用的新依赖
- 代码片段保留原始缩进格式`
    }
};



// ============================================================================
// HTTP API 工具
// ============================================================================

function makeApiStreamRequest(apiUrl, bodyStr, headers) {
    const options = {
        hostname: apiUrl.hostname,
        path: apiUrl.pathname + (apiUrl.search || ''),
        method: 'POST',
        headers: { ...headers, 'Content-Length': Buffer.byteLength(bodyStr), 'Connection':
'close' },
        agent: false
    };
    const proto = apiUrl.protocol === 'https:' ? https : http;
    return new Promise((resolve, reject) => {
        const req = proto.request(options, (res) => {
            if (res.statusCode >= 400) {
                let errData = '';
                res.on('data', chunk => errData += chunk);
                res.on('end', () => {
                    let errMsg = `HTTP ${res.statusCode}`;
                    try {
                        const parsed = JSON.parse(errData);
                        errMsg = parsed.error?.message || parsed.error?.code ||
parsed.message || errMsg;
                    } catch (e) {}
                    reject(new Error(errMsg));
                });
                return;
            }
            resolve(res);
        });
        req.on('error', reject);
        req.setTimeout(60000, () => { req.destroy(); reject(new Error('API连接超时(60s)'));
});
        req.write(bodyStr);
        req.end();
    });
}

function makeApiRequest(apiUrl, bodyStr, headers) {
    const options = {
        hostname: apiUrl.hostname,
        path: apiUrl.pathname + (apiUrl.search || ''),
```

```
        method: 'POST',
        headers: { ...headers, 'Content-Length': Buffer.byteLength(bodyStr), 'Connection':
'close' },
        agent: false
    };
    const proto = apiUrl.protocol === 'https:' ? https : http;
    return new Promise((resolve, reject) => {
        const req = proto.request(options, (res) => {
            let data = '';
            res.on('data', chunk => data += chunk);
            res.on('end', () => {
                if (res.statusCode >= 400) {
                    let errMsg = `HTTP ${res.statusCode}`;
                    try {
                        const parsed = JSON.parse(data);
                        errMsg = parsed.error?.message || parsed.error?.code ||
parsed.message || errMsg;
                    } catch (e) {}
                    reject(new Error(errMsg));
                    return;
                }
                try {
                    const parsed = JSON.parse(data);
                    if (parsed.error) {
                        reject(new Error(parsed.error.message || parsed.error.code ||
JSON.stringify(parsed.error)));
                        return;
                    }
                    resolve(parsed.choices?.[0]?.message?.content || '');
                } catch (e) {
                    reject(e);
                }
            });
        });
        req.on('error', reject);
        req.setTimeout(30000, () => { req.destroy(); reject(new Error('API请求超时(30s)'));
});
        req.write(bodyStr);
        req.end();
    });
}

// ============================================================================
// Agent Engine (全功能)
// ============================================================================

class AgentEngine {
    constructor(options = {}) {
        this.onChunk = options.onChunk || (() => {});
        this.onRequestApproval = options.onRequestApproval || null;
        this.onAskUser = options.onAskUser || null;
        this.executeTool = options.executeTool || null;
        this.output = new OutputManager();
```

```
        this.enablePlanning = options.enablePlanning !== false;
        this.enableReflection = options.enableReflection !== false;
        this.enableStuckDetection = options.enableStuckDetection !== false;
        this.enablePassiveDetection = options.enablePassiveDetection !== false;
        this.maxRounds = options.maxRounds || 24;
        this.maxConsecutiveFailures = options.maxConsecutiveFailures || 3;
        this.maxPassiveDetections = options.maxPassiveDetections || 2;
        this.maxRepeatText = options.maxRepeatText || 2;

        this._aborted = false;
        this._actionHistory = [];
        this._lastTextContent = '';
        this._repeatTextCount = 0;
        this._consecutiveFailures = 0;
        this._consecutiveMistakes = 0;
        this._passiveDetectionCount = 0;
        this._activePlan = null;
        this._planStepResults = {};
        this._provider = options.provider || null;
        this._apiUrl = options.apiUrl || null;
        this._apiHeaders = options.apiHeaders || null;
        this._model = options.model || null;
    }

    abort() {
        this._aborted = true;
    }

    _send(msg) {
        const passthroughTypes = ['subagent_start', 'subagent_chunk', 'subagent_end'];
        if (passthroughTypes.includes(msg.type)) {
            this.onChunk(msg);
            return;
        }
        const processed = this.output.processMessage(msg);
        if (processed) {
            this.onChunk(processed);
        }
    }

    _initReasoningThrottle() {
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        this._reasoningBuffer = null;
        this._reasoningStarted = false;
    }

    _sendReasoningDelta(delta, fullReasoning) {
```

```
        if (!this._reasoningStarted) {
            this._reasoningStarted = true;
            this.onChunk({ type: 'reasoning_start', content: '' });
        }
        this._reasoningBuffer = (this._reasoningBuffer || '') + delta;
        if (!this._reasoningFlushTimer) {
            this._reasoningFlushTimer = setTimeout(() => {
                this._reasoningFlushTimer = null;
                if (this._reasoningBuffer) {
                    const buf = this._reasoningBuffer;
                    this._reasoningBuffer = null;
                    this.onChunk({ type: 'reasoning_content', content: buf, partial: true
});
                }
            }, 80);
        }
    }

    _flushReasoningThrottle() {
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        if (this._reasoningStarted && this._reasoningBuffer) {
            const buf = this._reasoningBuffer;
            this._reasoningBuffer = null;
            this.onChunk({ type: 'reasoning_content', content: buf, partial: true });
        }
    }

    _finishReasoningThrottle() {
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        this._reasoningBuffer = null;
        this._reasoningStarted = false;
    }

    /**
     * 主入口：处理聊天
     */
    async processChat({ apiKey, model, messages, temperature, enableTools, apiFormat:
customApiFormat, baseUrl: customBaseUrl, language, maxRounds }) {
        this._aborted = false;
        this.output.clear();
        this._actionHistory = [];
        this._lastTextContent = '';
        this._repeatTextCount = 0;
        this._consecutiveFailures = 0;
        this._consecutiveMistakes = 0;
        this._passiveDetectionCount = 0;
```

```
        this._activePlan = null;
        this._planStepResults = {};
        this._thinkingSteps = [];
        this._apiKey = apiKey;
        this._model = model || 'glm-5-flash';

        if (!apiKey) {
            this._send({ type: 'say', say: SayType.ERROR, text: '未配置 API Key，请在设置中填
写' });
            return;
        }
        if (maxRounds) this.maxRounds = maxRounds;

        let apiFormat;
        let requestModel = this._model;
        if (this._apiUrl && this._apiHeaders && !customBaseUrl) {
            apiFormat = this._provider?.apiFormat || 'openai';
        } else if (customBaseUrl && customApiFormat) {
            const authType = customApiFormat === 'anthropic' ? 'x-api-key' : 'bearer';
            this._provider = { name: 'Custom', baseUrl: customBaseUrl, authType, apiFormat:
customApiFormat, thinkingParams: {} };
            const endpoint = buildChatEndpoint(this._provider, this._model, apiKey);
            this._apiUrl = new URL(endpoint.url);
            this._apiHeaders = buildApiHeaders(this._provider, apiKey);
            apiFormat = customApiFormat;
            const customMatch = this._model.match(/^custom:(https?:\/\/.+):(.+)$/);
            if (customMatch) requestModel = customMatch[2];
        } else {
            this._provider = getProviderForModel(this._model);
            const endpoint = buildChatEndpoint(this._provider, this._model, apiKey);
            this._apiUrl = new URL(endpoint.url);
            this._apiHeaders = buildApiHeaders(this._provider, apiKey);
            apiFormat = this._provider.apiFormat || 'openai';
        }
        this._requestModel = requestModel;
        const tools = enableTools !== false ? _getAllTools() : undefined;

        let conversation = [...messages];        this.conversation = conversation;
        const lastUserMsg = [...messages].reverse().find(m => m.role === 'user');

        const contextSummary = this._buildContextSummary();
        if (contextSummary) conversation.push({ role: 'system', content: contextSummary });

        conversation.push({ role: 'system', content: `OS: Windows. You can use both CMD and
Unix-style commands (git, npm, node, python, pip, npx, etc. all work in the bash tool).
Common commands: dir/ls, type/cat, findstr/grep, copy/cp, move/mv, del/rm, mkdir. Use bash
tool for all CLI operations including git, npm, python, testing, and formatting.` });

        const pluginPrompts = _getPluginPromptExtensions();
        if (pluginPrompts.length > 0) {
            conversation.push({ role: 'system', content: '## Plugin Capabilities\n\n' +
pluginPrompts.join('\n\n') });
        }

        conversation.push({
```

```
            role: 'system',
            content: `## 全栈开发能力

你具备完整的全栈开发能力。所有 CLI 操作通过 bash 工具执行。

### 文件操作
- 查看文件：str_replace_based_edit_tool(command="view", path="绝对路径")
- 创建文件：str_replace_based_edit_tool(command="create", path="绝对路径", file_text="内容")
- 编辑文件：str_replace_based_edit_tool(command="str_replace", path="绝对路径", old_str="原
文", new_str="新文")
- 写入/覆盖文件：str_replace_based_edit_tool(command="create", ...) 或通过 bash 使用 echo/重定
向
- 搜索文件：bash(command="dir /s /b *.js") 或 bash(command="findstr /s /n \"pattern\" *.js")
- 搜索文件内容：bash(command="findstr /s /n \"keyword\" *.*")

### 版本控制 (Git)
- 查看状态：bash(command="git status")
- 查看差异：bash(command="git diff") 或 bash(command="git diff --cached")
- 提交更改：bash(command="git add . && git commit -m \"message\"")
- 查看日志：bash(command="git log --oneline -20")
- 分支操作：bash(command="git branch") / bash(command="git checkout -b new-branch")
- 合并分支：bash(command="git merge branch-name")

### 包管理
- npm: bash(command="npm install package-name") / bash(command="npm run build") /
bash(command="npm test")
- pip: bash(command="pip install package-name") / bash(command="pip list")
- npx: bash(command="npx create-react-app my-app")

### 开发服务器
- 启动开发服务器（后台运行）：bash(command="npm run dev", background=true)
- 启动 HTTP 预览：start_preview(command="start", root="项目目录路径")
- 停止预览：start_preview(command="stop")

### 代码质量
- 运行测试：bash(command="npm test") 或 bash(command="npx jest --verbose")
- 代码检查：bash(command="npx eslint src/") 或 bash(command="npx eslint --fix src/")
- 代码格式化：bash(command="npx prettier --write \"src/**/*.{js,ts,css,json}\"")
- TypeScript 检查：bash(command="npx tsc --noEmit")

### 构建与部署
- 构建项目：bash(command="npm run build")
- 查看构建输出：bash(command="dir dist") 或 bash(command="ls dist")
- 启动生产服务器：bash(command="node server.js", background=true)

### 工作流最佳实践
1. 收到开发任务后，先用 str_replace_based_edit_tool(view) 了解项目结构
2. 使用 update_todo_list 创建任务计划
3. 逐步执行：编辑文件 → 运行测试 → 修复错误 → 提交代码
4. 对于需要长时间运行的服务（dev server、watcher），使用 background=true
5. 完成后用 attempt_completion 提交总结`
        });
```

```
        if (this.enablePlanning && lastUserMsg && tools) {
            const intent = this._detectIntent(lastUserMsg.content);
            if (intent.intent === 'complex') {
                const stepHint = intent.steps_needed >= 3
                    ? `这是一个复杂的多步骤任务（检测到 ${intent.steps_needed}+ 个步骤）。`
                    : '这是一个需要多步骤执行的任务。';
                conversation.push({
                    role: 'system',
                    content: `${stepHint}

## 任务工作流

请使用 update_todo_list 工具创建任务计划，将用户的请求分解为具体的、可执行的任务。

### 执行流程
1. 调用 update_todo_list 创建任务列表
   - [ ] 任务1：具体描述
   - [ ] 任务2：具体描述
   - [ ] 任务3：具体描述
2. 逐个执行任务：
   a. 将当前任务标记为 [-] 进行中（调用 update_todo_list）
   b. 执行该任务所需的所有工具调用
   c. 任务完成后，将任务标记为 [x] 已完成（调用 update_todo_list）
3. 所有任务完成后，调用 attempt_completion 提交最终总结

### 关键规则
- 每个任务应该是自包含的，将相关的工具调用归组到一起
- attempt_completion 的 result 字段必须包含清晰的中文总结
- 如果某个任务失败，标记为 [x] 并注明错误，然后继续下一个任务
- 永远不要在回复中使用 emoji

### 子代理派遣规则
当你需要搜索文件、分析代码、搜索资源或分析崩溃日志时，你必须调用 sub_agent_dispatch 工具，而不是自己执行这些任务。
- 搜索文件/目录 → sub_agent_dispatch(agent_type="file_search", task="具体任务描述")
- 分析代码结构 → sub_agent_dispatch(agent_type="code_analysis", task="具体任务描述")
- 搜索Minecraft资源 → sub_agent_dispatch(agent_type="resource_download", task="具体任务描述")
- 分析崩溃日志 → sub_agent_dispatch(agent_type="crash_analysis", task="具体任务描述")
- 代码补全/优化 → sub_agent_dispatch(agent_type="code_completion", task="具体任务描述")

**推荐：使用 agent_type="auto" 可自动选择最合适的子代理类型**，系统会根据任务描述智能匹配。例如：
- sub_agent_dispatch(agent_type="auto", task="在项目中搜索所有配置文件")
- sub_agent_dispatch(agent_type="auto", task="分析这个崩溃日志的错误原因")

绝对不要在文本中写"[执行] 调用子代理"，而是必须实际调用 sub_agent_dispatch 工具。

### 子代理使用场景指南
- **复杂搜索任务** → file_search：查找文件、搜索代码、定位资源
- **代码理解任务** → code_analysis：分析函数调用链、理解模块依赖、代码审查
- **Minecraft资源任务** → resource_download：搜索模组、整合包、材质包、光影包
- **崩溃诊断任务** → crash_analysis：分析崩溃日志、定位错误原因、提供修复建议
```

```
- **代码补全任务** → code_completion：代码重写、优化、补全建议

**最佳实践**：对于大多数任务，推荐使用 agent_type="auto"，系统会根据任务描述自动选择最合适的子代理类型。`
                });
            } else {
                conversation.push({
                    role: 'system',
                    content: `## 执行指南

这是一个简短的问候或确认，直接回复即可。

### 执行流程
1. 直接回复用户
2. 如果需要执行操作，使用工具完成

### 关键规则
- 用中文回复
- 如果执行了操作，调用 attempt_completion 提交结果总结
- 永远不要在回复中使用 emoji`
                });
            }
        }

        if (language && language !== 'en') {
            const _langNames = { 'zh-CN': '简体中文', 'zh-TW': '繁體中文', 'ja': '日本語',
'ko': '한국어' };
            const _langName = _langNames[language] || '简体中文';
            conversation.push({
                role: 'system',
                content: `FINAL REMINDER — Language: You MUST respond entirely in
${_langName}. All explanations, todo descriptions, plan text, and completion summaries must
be in ${_langName}. Only code, file paths, and technical identifiers stay in English.`
            });
        }

        let totalUsage = { prompt_tokens: 0, completion_tokens: 0, total_tokens: 0, rounds:
0 };        let toolResults = [];        for (let round = 0; round < this.maxRounds;
round++) {
            if (this._aborted) break;
            await new Promise(resolve => setImmediate(resolve));

            if (apiFormat === 'openai') {
                const toRemove = new Set();
                for (let i = 0; i < conversation.length; i++) {
                    const msg = conversation[i];
                    if (msg.role === 'assistant' && msg.tool_calls && msg.tool_calls.length
> 0) {
                        const expectedIds = new Set(msg.tool_calls.map(tc => tc.id));
                        const foundIds = new Set();
                        for (let j = i + 1; j < conversation.length; j++) {
                            const next = conversation[j];
                            if (next.role === 'assistant') break;
                            if (next.role === 'tool' && expectedIds.has(next.tool_call_id))
{
```

```
                            foundIds.add(next.tool_call_id);
                        }
                    }
                    if (foundIds.size < expectedIds.size) {
                        toRemove.add(i);
                        for (let j = i + 1; j < conversation.length; j++) {
                            if (conversation[j].role === 'assistant') break;
                            if (conversation[j].role === 'tool') toRemove.add(j);
                        }
                    }
                }
            }
            for (let i = conversation.length - 1; i >= 0; i--) {
                if (toRemove.has(i)) conversation.splice(i, 1);
            }
            for (let i = conversation.length - 1; i >= 0; i--) {
                if (conversation[i].role === 'tool') {
                    let hasAssistantBefore = false;
                    for (let j = i - 1; j >= 0; j--) {
                        if (conversation[j].role === 'assistant') {
                            hasAssistantBefore = conversation[j].tool_calls &&
conversation[j].tool_calls.length > 0;
                            break;
                        }
                    }
                    if (!hasAssistantBefore) conversation.splice(i, 1);
                }
            }
        }

        let bodyStr;
        const hasTools = tools && tools.length > 0;

        if (apiFormat === 'anthropic') {
            const anthropicMessages = _toAnthropicMessages(conversation);
            const reqBody = {
                model: this._requestModel,
                max_tokens: 8192,
                stream: true,
                ...(anthropicMessages.system ? { system: anthropicMessages.system } :
{}),
                messages: anthropicMessages.messages
            };
            if (hasTools) reqBody.tools = _toAnthropicTools(tools);
            bodyStr = JSON.stringify(reqBody);
        } else if (apiFormat === 'google') {
            const googleMessages = _toGoogleMessages(conversation);
            const reqBody = {
                ...(googleMessages.system_instruction ? { system_instruction:
googleMessages.system_instruction } : {}),
                contents: googleMessages.contents,
                generationConfig: { temperature: temperature != null ? temperature :
0.7 }
            };
```

```
                if (hasTools) reqBody.tools = _toGoogleTools(tools);
                bodyStr = JSON.stringify(reqBody);
            } else {
                const reqBody = {
                    model: this._requestModel,
                    messages: conversation,
                    temperature: temperature != null ? temperature : 0.7,
                    stream: true,
                    stream_options: { include_usage: true }
                };
                if (hasTools) { reqBody.tools = tools; reqBody.tool_choice = 'auto'; }
                if (this._provider.thinkingParams &&
Object.keys(this._provider.thinkingParams).length > 0) {
                    Object.assign(reqBody, this._provider.thinkingParams);
                }
                if (reqBody.enable_thinking === true) delete reqBody.temperature;
                bodyStr = JSON.stringify(reqBody);
            }

            let res;
            const MAX_RETRIES = 2;
            let lastError = null;
            for (let attempt = 0; attempt <= MAX_RETRIES; attempt++) {
                try {
                    this._send({ type: 'say', say: SayType.API_STARTED, text: '' });
                    res = await makeApiStreamRequest(this._apiUrl, bodyStr,
this._apiHeaders);
                    lastError = null;
                    break;
                } catch (e) {
                    lastError = e;
                    const isRetryable = e.message && (e.message.includes('ECONNRESET') ||
e.message.includes('ECONNREFUSED') || e.message.includes('ETIMEDOUT') ||
e.message.includes('socket hang up'));
                    if (isRetryable && attempt < MAX_RETRIES) {
                        this._send({ type: 'say', say: SayType.HEARTBEAT, text: `连接中断，
正在重试 (${attempt + 1}/${MAX_RETRIES})...` });
                        await new Promise(r => setTimeout(r, 2000 * (attempt + 1)));
                        continue;
                    }
                    this._send({ type: 'say', say: SayType.ERROR, text: e.message });
                    this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
                    return;
                }
            }
            if (lastError) {
                this._send({ type: 'say', say: SayType.ERROR, text: lastError.message });
                this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
                return;
            }

            const roundData = await this._processStream(res, apiFormat);
            if (this._aborted) break;

            if (roundData.usage) {
```

```
                totalUsage.prompt_tokens += roundData.usage.prompt_tokens || 0;
                totalUsage.completion_tokens += roundData.usage.completion_tokens || 0;
                totalUsage.total_tokens += roundData.usage.total_tokens || 0;
            }
            totalUsage.rounds++;

            if (roundData.toolCalls.length > 0 && roundData.finishReason === 'tool_calls')
{
                const assistantMsg = {
                    role: 'assistant',
                    content: roundData.fullContent || '',
                    tool_calls: roundData.toolCalls.map(tc => ({
                        id: tc.id, type: 'function',
                        function: { name: tc.name, arguments: tc.argsStr }
                    }))
                };
                if (this._provider.thinkingParams &&
this._provider.thinkingParams.enable_thinking) {
                    assistantMsg.reasoning_content = roundData.fullReasoning || '';
                } else if (roundData.fullReasoning) {
                    assistantMsg.reasoning_content = roundData.fullReasoning;
                }
                conversation.push(assistantMsg);

                const stuckDetected = this.enableStuckDetection && this._detectStuck();
                if (stuckDetected) {
                    this._send({ type: 'say', say: SayType.ERROR, text: '检测到循环调用，已自
动中断' });
                    this._send({
                        type: 'say', say: SayType.TOOL_START,
                        text: JSON.stringify(roundData.toolCalls.map(tc => ({
                            id: tc.id, name: tc.name,
                            displayName: TOOL_DISPLAY_NAMES[tc.name] ||
_getPluginDisplayNames()[tc.name] || tc.name,
                            args: tc.argsStr
                        })))
                    });
                    for (const tc of roundData.toolCalls) {
                        this._send({
                            type: 'say', say: SayType.TOOL_RESULT,
                            text: JSON.stringify({ id: tc.id, name: tc.name, error: '检测到
循环调用，已自动中断' })
                        });
                    }
                    this._send({ type: 'say', say: SayType.TOOL_END, text: '' });
                    conversation.push({
                        role: 'system',
                        content: 'AGENT STATE: STUCK — Loop detected. You must try a
completely different approach, or call attempt_completion to report current progress and
difficulties.'
                    });
                    continue;
                }

                toolResults = await this._executeTools(roundData.toolCalls, lastUserMsg);
```

```
                if (this._aborted) break;
```

```
                for (const tr of toolResults) {
                    conversation.push({
                        role: 'tool',
                        tool_call_id: tr.id,
                        name: tr.name,
                        content: tr.result
                    });
                }

                await this._evaluateAndGuide(toolResults, conversation, lastUserMsg,
round);

                if (toolResults.some(r => r.isCompletion)) break;

                this._compressIfNeeded(conversation);
                continue;
            }

            if (roundData.fullContent && roundData.toolCalls.length === 0) {
                if (this.enablePassiveDetection) {
                    const isPassive = this._detectPassive(roundData.fullContent,
lastUserMsg);
                    if (isPassive && this._passiveDetectionCount <
this.maxPassiveDetections) {
                        this._passiveDetectionCount++;
                        conversation.push({
                            role: 'system',
                            content: `Your response is too passive. You have tools to
autonomously gather information and execute actions. NEVER ask the user for information you
can obtain yourself.

Review the user's request. Think about what information you need and which tool can provide
it. Call the tool immediately.

Available tools: bash, str_replace_based_edit_tool, json_edit_tool, ckg,
sequential_thinking, attempt_completion, update_todo_list.
Take action now. Do not explain your limitations.`
                        });
                        continue;
                    }
                }

                const similarity = this._computeTextSimilarity(roundData.fullContent,
this._lastTextContent);
                if (similarity > 0.4 && this._lastTextContent.length > 10) {
                    this._repeatTextCount++;
                    if (this._repeatTextCount >= this.maxRepeatText) {
                        this._send({ type: 'say', say: SayType.COMPLETION, text: '' });
                        return;
                    }
                } else if (this._detectInternalRepetition(roundData.fullContent)) {
                    this._repeatTextCount++;
                    if (this._repeatTextCount >= this.maxRepeatText) {
                        this._send({ type: 'say', say: SayType.COMPLETION, text: '' });
                        return;
                    }
```

```
                } else {
                    this._repeatTextCount = 0;
```

```
                }
                this._lastTextContent = roundData.fullContent;
            }

            if (roundData.fullContent || roundData.fullReasoning) {
                const finalAssistantMsg = { role: 'assistant', content:
roundData.fullContent || '' };
                if (roundData.fullReasoning) finalAssistantMsg.reasoning_content =
roundData.fullReasoning;
                conversation.push(finalAssistantMsg);
            }

            if (roundData.fullContent) {                conversation.push({ role:
'assistant', content: roundData.fullContent });            }            const
completionFromTool = toolResults.find(r => r.isCompletion);
            const completionText = completionFromTool ? completionFromTool.completionText :
'';
            this._send({ type: 'say', say: SayType.COMPLETION, text: roundData.fullContent
|| completionText || '' });
            if (totalUsage.total_tokens > 0) {
                this._send({ type: 'usage', usage: totalUsage });
            }
            this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
            return;
        }

        const finalCompletion = toolResults.find(r => r.isCompletion);
        this._send({ type: 'say', say: SayType.COMPLETION, text: finalCompletion ?
finalCompletion.completionText : '' });
        if (totalUsage.total_tokens > 0) {
            this._send({ type: 'usage', usage: totalUsage });
        }
        this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
    }
    // Context Builder
    // ============================================================================

    _buildContextSummary() {
        try {
            const os = require('os');
            const path = require('path');
            const fs = require('fs');
            const settingsPath = path.join(os.homedir(), '.versepc', 'settings.json');
            if (fs.existsSync(settingsPath)) {
                const settings = JSON.parse(fs.readFileSync(settingsPath, 'utf-8'));
                let s = '## 当前工作区状态（自动获取，无需询问用户）\n';
                if (settings.selectedVersion) s += `- 当前版本:
${settings.selectedVersion}\n`;
                if (settings.javaPath) s += `- Java路径: ${settings.javaPath}\n`;
                if (settings.maxMemory) s += `- 最大内存: ${settings.maxMemory}\n`;
                if (settings.gameDir) s += `- 游戏目录: ${settings.gameDir}\n`;
                s += '\n以上信息已自动获取。如需更详细的信息（模组列表、版本列表等），请使用工具获
取。';
                return s;
            }
```

```
        } catch (e) {}
        return null;
    }

    // ============================================================================
    // Intent Detection
    // ============================================================================

    _detectIntent(userMessage) {
        const lower = userMessage.toLowerCase();
        const greetingOnly = /^(你好|hi|hello|hey|谢谢|感谢|ok|好的|知道了|你是谁|你叫什么|你是|说说|聊聊|讲讲)\s*[!! ?? 。.]*$/i;
        if (greetingOnly.test(lower.trim())) return { intent: 'simple', reason: 'greeting', steps_needed: 0 };
        if (userMessage.trim().length <= 6 && !/[安装|创建|修改|删除|修复|配置|写|做|运行|执行|添加|更新]/.test(lower)) return { intent: 'simple', reason: 'short', steps_needed: 0 };

        const explicitMultiStep = /然后|接着|之后再|先.*再|先.*然后|第一步.*第二步|首先.*然后.*最后/;
        if (explicitMultiStep.test(lower)) {
            const conjunctionCount = (lower.match(/然后|接着|之后再|再(?!次)/g) || []).length;
            return { intent: 'complex', reason: 'explicit_multi_step', steps_needed: conjunctionCount + 1 };
        }

        const actionVerbs = /安装|卸载|创建|删除|修改|编辑|配置|部署|构建|编译|调试|修复|优化|迁移|升级|写|做|运行|执行|添加|更新|实现|重构|迁移|集成|生成|重写/;
        const actionMatches = lower.match(new RegExp(actionVerbs.source, 'g'));
        if (actionMatches && actionMatches.length >= 2) {
            return { intent: 'complex', reason: 'multi_action', steps_needed: actionMatches.length };
        }

        return { intent: 'complex', reason: 'general_task', steps_needed: 2 };
    }

    // ============================================================================
    // Stream Processing
    // ============================================================================

    async _processStream(res, apiFormat) {
        if (apiFormat === 'anthropic') return this._processAnthropicStream(res);
        if (apiFormat === 'google') return this._processGoogleStream(res);
        return this._processOpenAIStream(res);
    }

    // OpenAI-compatible SSE
    async _processOpenAIStream(res) {
        let buffer = '';
        let fullContent = '';
        let prevContentLen = 0;
        let fullReasoning = '';
        let prevReasoningLen = 0;
        let toolCalls = [];
        let finishReason = null;
```

```
        let reasoningStarted = false;
        let doneReceived = false;
```


后1500行代码（第31-60页）

```
            };

            res.on('close', cleanup);

            let _resClosed = false;
            res.on('close', () => { _resClosed = true; });
            const sendChunk = (data) => {
                if (_resClosed) return;
                try { res.write(`data: ${JSON.stringify(data)}\n\n`); } catch (e) {
_resClosed = true; }
            };

            const onChunk = (processed) => {
                if (!processed) return;
                if (processed.type && processed.type !== 'say') {
                    sendChunk(processed);
                    return;
                }
                sendChunk(processed);
            };

            const TOOL_RISK_MAP = {
                write_to_file: 'dangerous', replace_in_file: 'dangerous', delete_file:
'dangerous',
                execute_command: 'moderate', spawn_process: 'moderate',
                browser_action: 'moderate', mcp_tool: 'moderate',
                search_files: 'safe', list_files: 'safe', list_code_definition_names:
'safe',
                read_file: 'safe', ask_followup_question: 'safe', attempt_completion:
'safe',
                update_todo_list: 'safe', sequential_thinking: 'safe',
            };
            const onRequestApproval = (toolName, argsStr) => {
                const risk = TOOL_RISK_MAP[toolName] || 'moderate';
                const aid =
`apv_${Date.now().toString(36)}_${Math.random().toString(36).substr(2, 6)}`;
                sendChunk({ type: 'approval_requested', approvalId: aid, toolName, risk,
args: argsStr });
                return new Promise((resolve) => {
                    const t = setTimeout(() => {
                        if (approvalPendingMap[aid]) { delete approvalPendingMap[aid];
resolve({ approved: false, toolName, timeout: true }); }
                    }, 60000);
                    approvalPendingMap[aid] = { resolve, timeout: t, chatId };
                });
            };

            try {
                const { AgentEngine: EngineClass } = require('./agent-engine');
```

```
                const engine = new EngineClass({ executeTool, onRequestApproval, onChunk,
logger: console });
                await engine.processChat({ apiKey, model, messages, temperature,
enableTools, maxRounds: 24 });
            } catch (e) {
                sendChunk({ type: 'error', error: e.message });
            } finally {
                cleanup();
                try { res.end(); } catch (e) {}
            }
```

```
            return;
        }

        res.writeHead(404, { 'Content-Type': 'application/json' });
        res.end(JSON.stringify({ error: 'not found' }));
    });

    function tryListen(port) {
        server.removeAllListeners('error');
        server.on('error', (err) => {
            if (err.code === 'EADDRINUSE' && port < BASE_PORT + 100) {
                console.log(`[SSE] 端口 ${port} 被占用，尝试 ${port + 1}`);
                tryListen(port + 1);
            } else {
                console.error(`[SSE] 启动失败:`, err.message);
            }
        });
        server.listen(port, () => {
            server._actualPort = port;
            console.log(`[SSE] 启动: http://localhost:${port}`);
        });
    }
    tryListen(BASE_PORT);
    return { server, get PORT() { return server._actualPort || BASE_PORT; } };
}

module.exports = { createSSEServer };
============================================================文件: versepc-
promo\index.html============================================================
<!DOCTYPE html><html lang="zh-CN"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width, initial-scale=1.0"><title>VersePC Promo</title><link
href="https://fonts.googleapis.com/css2?
family=Schibsted+Grotesk:wght@700;900&family=JetBrains+Mono:wght@400;700&display=swap"
rel="stylesheet"><style>*, *::before, *::after { margin:0; padding:0; box-sizing:border-
box; }body { background:#0A0D14; overflow:hidden; }[data-composition-id="versepc-promo"] {
width:1920px; height:1080px; position:relative; overflow:hidden; background:#0A0D14;}.scene
{  position:absolute; inset:0;  display:flex; align-items:center; justify-content:center;
visibility:hidden; opacity:0;}.scene-inner {  width:100%; height:100%;
```

```
  display:flex; padding:100px 120px;  gap:40px; position:relative; z-index:1;}.left {
flex:1; display:flex; flex-direction:column; justify-content:center; gap:16px; }.right {
flex:1; display:flex; align-items:center; justify-content:center; }.scene-label {  font-
family:"JetBrains Mono",monospace; font-size:16px;  color:#FFB347; letter-spacing:.1em;
margin-bottom:8px;}.scene-title {  font-family:"Schibsted Grotesk",sans-serif; font-
weight:900;  font-size:80px; color:#E8ECF2; line-height:1.1; letter-spacing:-.02em;}.scene-
title .hl-g { color:#6AAA3A; }.scene-title .hl-o { color:#FFB347; }.scene-desc {  font-
size:26px; color:#7790A8; line-height:1.5;}.badge-row { display:flex; gap:12px; margin-
top:8px; }.badge {  padding:8px 18px; border:1.5px solid rgba(255,255,255,.1);
background:rgba(20,25,36,.8); font-family:"JetBrains Mono",monospace;  font-size:16px;
color:#FFB347; letter-spacing:.04em;}.stat-row { display:flex; gap:48px; margin-top:8px;
}.stat-val {  font-family:"Schibsted Grotesk",sans-serif; font-weight:900;  font-size:56px;
color:#FFB347; line-height:1;}.stat-lbl { font-size:15px; color:#7790A8; font-
family:"JetBrains Mono",monospace; margin-top:4px; }.bg-grid {  position:absolute; inset:0;
z-index:0;  background-image:    linear-gradient(rgba(255,255,255,.015) 1px,transparent
1px),    linear-gradient(90deg,rgba(255,255,255,.015) 1px,transparent 1px);  background-
size:80px 80px;}.bg-glow {  position:absolute; border-radius:50%; filter:blur(120px); z-
index:0;}.glow-gold { background:#FFB347; width:600px; height:600px; top:50%; right:-100px;
transform:translateY(-50%); opacity:.08; }.glow-green { background:#6AAA3A; width:500px;
height:500px; bottom:-100px; left:20%; opacity:.08; }.glow-mid { background:#FFB347;
width:700px; height:700px; top:50%; left:50%; transform:translate(-50%,-50%); opacity:.06;
}/* Scene 0 - Title */.scene-0 .s0-center { flex:1; display:flex; flex-direction:column;
align-items:center; justify-content:center; gap:32px; padding:100px; }.s0-logo {
width:130px; height:130px; border:3px solid #FFB347; background:rgba(20,25,36,.9);
display:flex; align-items:center; justify-content:center; box-shadow:0 0 60px
rgba(255,179,71,.2); }.s0-logo img { width:70px; height:70px; image-rendering:pixelated;
}.s0-name { font-family:"Schibsted Grotesk",sans-serif; font-weight:900; font-size:130px;
color:#E8ECF2; letter-spacing:.04em; line-height:1; text-align:center; }.s0-name .hl-o {
color:#FFB347; }
```

```
.s0-tag { font-size:34px; color:#7790A8; letter-spacing:.08em; text-align:center; }.block-dot {  position:absolute; z-index:2;  width:18px; height:18px; background:rgba(106,170,58,.3);  border:1px solid rgba(106,170,58,.4);}.block-dot:nth-child(1) { right:120px; top:140px; }.block-dot:nth-child(2) { left:140px; bottom:180px; width:24px; height:24px; background:rgba(255,179,71,.3); border-color:rgba(255,179,71,.4); }/* Mod cards */.mod-card {  background:#141924; border:1px solid rgba(255,255,255,.06); padding:18px; display:flex; gap:14px; width:360px;}.mod-card:nth-child(odd) { align-self:flex-end; }.mod-card:nth-child(even) { align-self:flex-start; }.mc-icon { width:48px; height:48px; background:rgba(106,170,58,.15); flex-shrink:0; display:flex; align-items:center; justify-content:center; font-size:22px; }.mc-info { flex:1; display:flex; flex-direction:column; gap:6px; }.mc-name { font-size:18px; font-weight:700; color:#E8ECF2; font-family:"Schibsted Grotesk",sans-serif; }.mc-meta { font-size:14px; color:#7790A8; font-family:"JetBrains Mono",monospace; }.mc-bar { height:3px; background:rgba(106,170,58,.15); border-radius:2px; overflow:hidden; }.mc-bar-fill { height:100%; background:#6AAA3A; border-radius:2px; width:100%; }/* AI chat */.chat-box { width:440px; background:#141924; border:1px solid rgba(255,255,255,.06); border-radius:6px; overflow:hidden;}.chat-hdr { padding:12px 18px; background:rgba(20,25,36,.95); border-bottom:1px solid rgba(255,255,255,.06); display:flex; align-items:center; gap:8px; }.chat-dot { width:10px; height:10px; border-radius:50%; background:#6AAA3A; box-shadow:0 0 6px rgba(106,170,58,.4); }.chat-title { font-family:"JetBrains Mono",monospace; font-size:14px; color:#7790A8; }.chat-body { padding:16px; display:flex; flex-direction:column; gap:12px; }.chat-msg { max-width:85%; padding:10px 14px; font-size:17px; line-height:1.5; border-radius:5px; color:#E8ECF2; }.chat-msg.usr { align-self:flex-end; background:rgba(106,170,58,.15); border:1px solid rgba(106,170,58,.2); font-size:16px; }.chat-msg.bot { align-self:flex-start; background:rgba(255,179,71,.08); border:1px solid rgba(255,179,71,.15); }.chat-inp { margin:0 16px 14px; padding:10px 14px; background:rgba(255,255,255,.03); border:1px solid rgba(255,255,255,.08); border-radius:5px; font-family:"JetBrains Mono",monospace; font-size:14px; color:#7790A8; }/* Pack cards */.pack-card {  width:380px; background:#141924; border:1px solid rgba(255,255,255,.06); padding:18px; display:flex; gap:14px;}.pack-card:nth-child(odd) { align-self:flex-end; }.pack-card:nth-child(even) { align-self:flex-start; }.pc-thumb { width:56px; height:56px; background:rgba(255,179,71,.1); border:1px solid rgba(255,179,71,.2); flex-shrink:0; display:flex; align-items:center; justify-content:center; font-size:26px; }.pc-info { flex:1; display:flex; flex-direction:column; gap:5px; }.pc-name { font-size:19px; font-weight:700; color:#E8ECF2; font-family:"Schibsted Grotesk",sans-serif; }.pc-desc { font-size:14px; color:#7790A8; }.pc-tag { display:inline-flex; padding:3px 10px; background:rgba(106,170,58,.15); border:1px solid rgba(106,170,58,.25); font-size:12px; color:#6AAA3A; font-family:"JetBrains Mono",monospace; }/* Tool cards */.tool-row { display:flex; flex-direction:column; gap:16px; }
```

```
.tool-row-top { flex:1; display:flex; align-items:flex-start; gap:40px; margin-bottom:20px; }.tool-grid { display:grid; grid-template-columns:repeat(4,1fr); gap:16px; flex:1; }.tool-card {  background:#141924; border:1px solid rgba(255,255,255,.05); padding:18px; display:flex; flex-direction:column; gap:10px;}.tc-icon { font-size:26px; }.tc-title { font-family:"Schibsted Grotesk",sans-serif; font-weight:700; font-size:20px; color:#E8ECF2; }.tc-desc { font-size:14px; color:#7790A8; font-family:"JetBrains Mono",monospace; line-height:1.4; }.tc-bar { height:2px; background:rgba(255,255,255,.06); border-radius:1px; overflow:hidden; margin-top:auto; }.tc-bar-fill { height:100%; background:#6AAA3A; border-radius:1px; }.tool-card:nth-child(2) .tc-bar-fill { background:#FFB347; width:80%; }.tool-card:nth-child(3) .tc-bar-fill { width:65%; }.tool-card:nth-child(4) .tc-bar-fill { background:#FFB347; }/* Version ring */.v-ring-wrap { position:relative; width:340px; height:340px; }.v-ring { position:absolute; inset:0; border:3px solid rgba(106,170,58,.18); border-radius:50%; }.v-ring-i { position:absolute; inset:50px; border:2px dashed rgba(255,179,71,.12); border-radius:50%; }.v-badge {  position:absolute; font-family:"JetBrains Mono",monospace; font-size:26px; font-weight:700;  color:#E8ECF2; background:rgba(10,13,20,.92); border:2px solid rgba(106,170,58,.4); padding:5px 16px; white-space:nowrap;}.v-b1 { top:2%; left:50%; transform:translateX(-50%); }.v-b2 { right:-5px; top:40%; transform:translateY(-50%); }.v-b3 { bottom:2%; left:50%; transform:translateX(-50%); }.v-b4 { left:-5px; top:40%; transform:translateY(-50%); }.v-play {  position:absolute; top:50%; left:50%; transform:translate(-50%,-50%);  width:100px; height:100px; border-radius:50%; background:#6AAA3A;  display:flex; align-items:center; justify-content:center;  box-shadow:0 0 50px rgba(106,170,58,.35);}.v-play::after { content:""; border:solid; border-width:18px 0 18px 34px; border-color:transparent transparent transparent #0A0D14; margin-left:6px; }/* Scene 6 CTA */.s6-center { flex:1; display:flex; flex-direction:column; align-items:center; justify-content:center; gap:28px; padding:100px; }.s6-logo { width:110px; height:110px; border:3px solid #FFB347; background:rgba(20,25,36,.9); display:flex; align-items:center; justify-content:center; box-shadow:0 0 60px rgba(255,179,71,.2); }.s6-logo img { width:60px; height:60px; image-rendering:pixelated; }.s6-head { font-family:"Schibsted Grotesk",sans-serif; font-weight:900; font-size:90px; color:#E8ECF2; text-align:center; line-height:1.1; letter-spacing:.02em; }.s6-head .hl-o { color:#FFB347; }.s6-sub { font-size:28px; color:#7790A8; text-align:center; line-height:1.5; }.cta-btns { display:flex; gap:24px; }.cta-btn { padding:16px 44px; font-family:"Schibsted Grotesk",sans-serif; font-weight:700; font-size:26px; letter-spacing:.04em; }.cta-primary { background:#6AAA3A; color:#0A0D14; box-shadow:0 0 40px rgba(106,170,58,.3); border:none; }.cta-secondary { background:transparent; color:#E8ECF2; border:2px solid rgba(255,255,255,.15); }.s6-foot { font-family:"JetBrains Mono",monospace; font-size:15px; color:rgba(255,255,255,.15); position:absolute; bottom:36px; }/* pixel decor */.px-row { position:absolute; bottom:70px; left:50%; transform:translateX(-50%); display:flex; gap:3px; z-index:2; }.px { width:7px; height:7px; background:rgba(106,170,58,.25); }
```

```
.px:nth-child(3n) { background:rgba(255,179,71,.25); }.scene-num { position:absolute;
top:60px; left:60px; font-family:"JetBrains Mono",monospace; font-size:22px;
color:rgba(255,179,71,.3); z-index:2; }</style></head><body><div data-composition-
id="versepc-promo" data-width="1920" data-height="1080" data-start="0">  <!-- Common BG
layers -->  <div class="bg-grid"></div>  <div class="bg-glow glow-gold"></div>  <div
class="bg-glow glow-green"></div>  <!-- SCENE 0: Title -->  <div class="scene scene-0"
id="s0">    <div class="scene-inner">      <div class="s0-center">        <div class="s0-
logo"><img src="assets/icon.svg" alt="VersePC"></div>        <div class="s0-name"><span
class="hl-o">Verse</span>PC</div>        <div class="s0-tag">新 一 代 Minecraft 启 动 器
</div>      </div>    </div>    <div class="block-dot"></div>    <div class="block-dot">
</div>  </div>  <!-- SCENE 1: Launch -->  <div class="scene" id="s1">    <div class="scene-
inner">      <div class="left">        <div class="scene-label">// 01 — 启动</div>
<div class="scene-title">一键启动<br><span class="hl-g">无限可能</span></div>        <div
class="scene-desc">多版本管理 · 模组加载器自动配置<br>从经典版本到最新快照，一切就绪</div>
<div class="badge-row">          <div class="badge">Forge</div><div
class="badge">Fabric</div>          <div class="badge">NeoForge</div><div
class="badge">OptiFine</div>        </div>      </div>      <div class="right">        <div
class="v-ring-wrap">          <div class="v-ring"></div><div class="v-ring-i"></div>
<div class="v-badge v-b1">1.21.5</div>          <div class="v-badge v-b2">1.20.1</div>
<div class="v-badge v-b3">1.18.2</div>          <div class="v-badge v-b4">1.16.5</div>
<div class="v-play"></div>        </div>      </div>    </div>    <div class="scene-
num">01</div>
```

```
  </div>  <!-- SCENE 2: Mods -->  <div class="scene" id="s2">    <div class="scene-inner">
<div class="left">        <div class="scene-label">// 02 — 模组</div>        <div
class="scene-title">模组管理<br><span class="hl-o">尽在掌握</span></div>        <div
class="scene-desc">支持 Forge · Fabric · NeoForge · OptiFine<br>智能依赖解析，告别模组冲突
</div>      </div>      <div class="right" style="flex-direction:column; gap:14px;">
<div class="mod-card">          <div class="mc-icon">🔮</div>          <div class="mc-
info">            <div class="mc-name">OptiFine</div>            <div class="mc-
meta">v1.21.5 · 已启用</div>            <div class="mc-bar"><div class="mc-bar-fill"></div>
</div>          </div>        </div>        <div class="mod-card">          <div class="mc-
icon">📦</div>          <div class="mc-info">            <div class="mc-name">Just Enough
Items</div>            <div class="mc-meta">v19.21.0 · 已启用</div>            <div
class="mc-bar"><div class="mc-bar-fill"></div></div>          </div>        </div>
<div class="mod-card">          <div class="mc-icon">🌍</div>          <div class="mc-
info">            <div class="mc-name">JourneyMap</div>            <div class="mc-
meta">v6.0.0 · 已启用</div>            <div class="mc-bar"><div class="mc-bar-fill"></div>
</div>          </div>        </div>      </div>    </div>    <div class="scene-
num">02</div>  </div>  <!-- SCENE 3: AI -->  <div class="scene" id="s3">    <div
class="scene-inner">      <div class="left">        <div class="scene-label">// 03 —
AI</div>        <div class="scene-title">内置 AI<br><span class="hl-o">智能问答</span></div>
<div class="scene-desc">崩溃分析 · 模组推荐 · 配置优化<br>你的专属 Minecraft 技术顾问</div>
</div>      <div class="right">        <div class="chat-box">
```

```
          <div class="chat-hdr"><div class="chat-dot"></div><div class="chat-title">verse-
ai</div></div>          <div class="chat-body">            <div class="chat-msg usr">为什么
我的 Forge 启动崩溃？</div>            <div class="chat-msg bot">检测到你的 OptiFine 版本与
<br>Forge 47.3.0 不兼容。<br>推荐升级到 OptiFine HD U I1。</div>            <div class="chat-
msg usr">帮我安装</div>            <div class="chat-msg bot">已自动下载并安装 ✅</div>
</div>          <div class="chat-inp">输入你的问题...</div>        </div>      </div>
</div>    <div class="scene-num">03</div>  </div>  <!-- SCENE 4: Modpacks -->  <div
class="scene" id="s4">    <div class="scene-inner">      <div class="left">        <div
class="scene-label">// 04 — 整合包</div>        <div class="scene-title">整合包<br><span
class="hl-o">一键安装</span></div>        <div class="scene-desc">支持 CurseForge · Modrinth
格式导入<br>自动下载依赖，即装即玩</div>        <div class="stat-row">          <div><div
class="stat-val">50+</div><div class="stat-lbl">内置整合包</div></div>          <div><div
class="stat-val">1-Click</div><div class="stat-lbl">一键导入</div></div>        </div>
</div>      <div class="right" style="flex-direction:column; gap:12px;">        <div
class="pack-card">          <div class="pc-thumb">⚔️</div>          <div class="pc-info">
<div class="pc-name">RLCraft</div>            <div class="pc-desc">高难度生存整合包</div>
<div><span class="pc-tag">CurseForge</span></div>          </div>        </div>        <div
class="pack-card">          <div class="pc-thumb">🏭</div>          <div class="pc-info">
<div class="pc-name">All the Mods 10</div>            <div class="pc-desc">400+ 模组科技魔法
</div>            <div><span class="pc-tag">Modrinth</span></div>          </div>
</div>        <div class="pack-card">          <div class="pc-thumb">🧩</div>          <div
class="pc-info">            <div class="pc-name">Better MC</div>            <div class="pc-
desc">原版增强 · 轻量优化</div>            <div><span class="pc-tag">CurseForge</span></div>
</div>
```

```
        </div>      </div>    </div>    <div class="scene-num">04</div>  </div>  <!-- SCENE
5: Tools -->  <div class="scene" id="s5">    <div class="scene-inner" style="flex-
direction:column; gap:28px;">      <div class="tool-row-top">        <div>          <div
class="scene-label">// 05 — 工具</div>          <div class="scene-title">开发者<br><span
class="hl-g">全能工具箱</span></div>        </div>        <div class="scene-desc"
style="margin-left:auto; align-self:flex-end; text-align:right;">          代码编辑器 · 终端
· 文件管理 · 崩溃分析<br>一切尽在 VersePC        </div>      </div>      <div class="tool-
grid">        <div class="tool-card">          <div class="tc-icon">{ }</div>          <div
class="tc-title">Monaco 编辑器</div>          <div class="tc-desc">VS Code 同款内核<br>语法高
亮 · 自动补全</div>          <div class="tc-bar"><div class="tc-bar-fill" style="width:90%">
</div></div>        </div>        <div class="tool-card">          <div class="tc-
icon">&gt;_</div>          <div class="tc-title">内置终端</div>          <div class="tc-
desc">xterm.js 驱动<br>GPU 加速渲染</div>          <div class="tc-bar"><div class="tc-bar-
fill"></div></div>        </div>        <div class="tool-card">          <div class="tc-
icon">🟨</div>          <div class="tc-title">文件浏览器</div>          <div class="tc-
desc">像 IDE 一样管理<br>实例文件与配置</div>          <div class="tc-bar"><div class="tc-
bar-fill"></div></div>        </div>        <div class="tool-card">          <div
class="tc-icon">🪲</div>          <div class="tc-title">崩溃分析器</div>          <div
class="tc-desc">解析错误日志<br>定位冲突模组</div>          <div class="tc-bar"><div
class="tc-bar-fill"></div></div>        </div>      </div>    </div>    <div class="scene-
num">05</div>  </div>  <!-- SCENE 6: CTA -->  <div class="scene" id="s6">
```

```
    <div class="scene-inner">      <div class="s6-center">        <div class="s6-logo"><img
src="assets/icon.svg" alt="VersePC"></div>        <div class="s6-head">开启你的<br><span
class="hl-o">Minecraft</span> 之旅</div>        <div class="s6-sub">免费 · 开源 · 永久更新
<br>让每一次方块冒险更加精彩</div>        <div class="cta-btns">          <div class="cta-btn
cta-primary">立即下载</div>          <div class="cta-btn cta-secondary">GitHub</div>
</div>      </div>      <div class="s6-foot">VersePC v1.0.0 ·
github.com/doujie081231/versePc</div>    </div>    <div class="px-row">      <div
class="px"></div><div class="px"></div><div class="px"></div>      <div class="px"></div>
<div class="px"></div><div class="px"></div>      <div class="px"></div><div class="px">
</div><div class="px"></div>      <div class="px"></div><div class="px"></div><div
class="px"></div>      <div class="px"></div><div class="px"></div><div class="px">
<div class="px"></div><div class="px"></div><div class="px"></div>    </div>  </div></div>
<script src="https://cdn.jsdelivr.net/npm/gsap@3.14.2/dist/gsap.min.js"></script>
<script>window.__timelines = window.__timelines || {};gsap.defaults({ overwrite:false
});const tl = gsap.timeline({ paused:true });// === COMMON BG ===tl.from(".bg-grid", {
opacity:0, duration:0.5 }, 0);tl.from(".glow-gold", { scale:.5, opacity:0, duration:1.2,
ease:"power2.out" }, 0);tl.from(".glow-green", { scale:.6, opacity:0, duration:1,
ease:"power2.out" }, .2);// === SCENE 0: TITLE (0s-5s) ===tl.set("#s0", { autoAlpha:1 },
.3);tl.from("#s0 .s0-logo",  { scale:.2, opacity:0, duration:.7, ease:"back.out(1.7)" },
.5);tl.from("#s0 .s0-name",  { y:60, opacity:0, duration:.6, ease:"expo.out" },
1.2);tl.from("#s0 .s0-tag",   { y:30, opacity:0, duration:.5, ease:"power3.out" },
1.7);tl.from("#s0 .block-dot",{ scale:0, opacity:0, stagger:.1, duration:.35,
ease:"back.out(2)" }, 1.4);tl.to("#s0", { autoAlpha:0, duration:.35, ease:"power2.in" },
4.7);// === SCENE 1: LAUNCH (5s-10.5s) ===tl.set("#s1", { autoAlpha:1 }, 5);tl.from("#s1
.scene-num", { opacity:0, duration:.3 }, 5.1);tl.from("#s1 .scene-label",{ x:-30,
opacity:0, duration:.4, ease:"power3.out" }, 5.2);tl.from("#s1 .scene-title",{ y:50,
opacity:0, duration:.6, ease:"expo.out" }, 5.4);tl.from("#s1 .scene-desc", { y:30,
opacity:0, duration:.5, ease:"power3.out" }, 5.8);
```

```
tl.from("#s1 .badge",      { y:20, opacity:0, stagger:.08, duration:.4, ease:"power2.out"
}, 6.1);tl.from("#s1 .v-ring",     { scale:.5, opacity:0, duration:.65, ease:"power3.out"
}, 5.5);tl.from("#s1 .v-ring-i",   { scale:0, opacity:0, duration:.45, ease:"power2.out" },
5.8);tl.from("#s1 .v-badge",    { scale:0, opacity:0, stagger:.1, duration:.4,
ease:"back.out(1.5)" }, 6);tl.from("#s1 .v-play",     { scale:0, opacity:0, duration:.5,
ease:"back.out(2)" }, 6.5);tl.to("#s1 .v-play", { boxShadow:"0 0 80px rgba(106,170,58,.5)",
duration:1, repeat:1, yoyo:true, ease:"sine.inOut" }, 7.5);tl.to("#s1", { autoAlpha:0,
duration:.35, ease:"power2.in" }, 10.2);// === SCENE 2: MODS (10.5s-16s) ===tl.set("#s2", {
autoAlpha:1 }, 10.5);tl.from("#s2 .scene-num", { opacity:0, duration:.3 },
10.6);tl.from("#s2 .scene-label",{ x:-30, opacity:0, duration:.4, ease:"power3.out" },
10.7);tl.from("#s2 .scene-title",{ y:50, opacity:0, duration:.6, ease:"expo.out" },
10.9);tl.from("#s2 .scene-desc", { y:30, opacity:0, duration:.5, ease:"power3.out" },
11.3);tl.from("#s2 .mod-card",   { x:60, opacity:0, stagger:.15, duration:.5,
ease:"power3.out" }, 10.8);tl.to("#s2", { autoAlpha:0, duration:.35, ease:"power2.in" },
15.7);// === SCENE 3: AI (16s-21.5s) ===tl.set("#s3", { autoAlpha:1 }, 16);tl.from("#s3
.scene-num", { opacity:0, duration:.3 }, 16.1);tl.from("#s3 .scene-label",{ x:-30,
opacity:0, duration:.4, ease:"power3.out" }, 16.2);tl.from("#s3 .scene-title",{ y:50,
opacity:0, duration:.6, ease:"expo.out" }, 16.4);tl.from("#s3 .scene-desc", { y:30,
opacity:0, duration:.5, ease:"power3.out" }, 16.8);tl.from("#s3 .chat-box",   { scale:.85,
opacity:0, duration:.6, ease:"power3.out" }, 16.5);tl.from("#s3 .chat-msg.usr",{ x:30,
opacity:0, stagger:.25, duration:.4, ease:"power3.out" }, 17.2);tl.from("#s3 .chat-
msg.bot",{ x:-30, opacity:0, stagger:.25, duration:.4, ease:"power3.out" },
17.4);tl.from("#s3 .chat-inp",   { opacity:0, y:8, duration:.3 }, 18.5);tl.to("#s3", {
autoAlpha:0, duration:.35, ease:"power2.in" }, 21.2);// === SCENE 4: MODPACKS (21.5s-27s)
===tl.set("#s4", { autoAlpha:1 }, 21.5);tl.from("#s4 .scene-num", { opacity:0, duration:.3
}, 21.6);tl.from("#s4 .scene-label",{ x:-30, opacity:0, duration:.4, ease:"power3.out" },
21.7);tl.from("#s4 .scene-title",{ y:50, opacity:0, duration:.6, ease:"expo.out" },
21.9);tl.from("#s4 .scene-desc", { y:30, opacity:0, duration:.5, ease:"power3.out" },
22.3);tl.from("#s4 .pack-card",  { x:60, opacity:0, stagger:.15, duration:.5,
ease:"power3.out" }, 21.9);tl.from("#s4 .stat-val",   { y:30, opacity:0, stagger:.15,
duration:.5, ease:"back.out(1.5)" }, 23);tl.from("#s4 .stat-lbl",   { opacity:0,
stagger:.15, duration:.3 }, 23.2);tl.to("#s4", { autoAlpha:0, duration:.35,
ease:"power2.in" }, 26.7);// === SCENE 5: TOOLS (27s-32.5s) ===tl.set("#s5", { autoAlpha:1
}, 27);tl.from("#s5 .scene-num",   { opacity:0, duration:.3 }, 27.1);tl.from("#s5 .scene-
label", { x:-30, opacity:0, duration:.4, ease:"power3.out" }, 27.2);tl.from("#s5 .scene-
title", { y:50, opacity:0, duration:.6, ease:"expo.out" }, 27.4);tl.from("#s5 .scene-desc",
{ x:40, opacity:0, duration:.5, ease:"power3.out" }, 27.7);tl.from("#s5 .tool-card",   {
y:50, opacity:0, stagger:.1, duration:.5, ease:"power3.out" }, 28);tl.from("#s5 .tc-bar-
fill", { scaleX:0, stagger:.1, duration:.5, ease:"power3.out", transformOrigin:"left" },
28.7);tl.to("#s5", { autoAlpha:0, duration:.35, ease:"power2.in" }, 32.2);
```

```
// === SCENE 6: CTA (32.5s-38s) ===tl.set("#s6", { autoAlpha:1 }, 32.5);tl.from("#s6 .s6-
logo",   { scale:.2, opacity:0, duration:.8, ease:"back.out(1.7)" }, 32.8);tl.from("#s6
.s6-head",   { y:60, opacity:0, duration:.7, ease:"expo.out" }, 33.5);tl.from("#s6 .s6-
sub",    { y:30, opacity:0, duration:.5, ease:"power3.out" }, 34);tl.from("#s6 .cta-
primary", { x:-50, opacity:0, duration:.5, ease:"back.out(1.5)" }, 34.5);tl.from("#s6 .cta-
secondary",{ x:50, opacity:0, duration:.5, ease:"back.out(1.5)" }, 34.65);tl.from("#s6
.px",           { scale:0, opacity:0, stagger:.03, duration:.3, ease:"back.out(1.5)" },
35.2);tl.from("#s6 .s6-foot",      { opacity:0, duration:.4 }, 35.8);tl.to("#s6 .cta-
primary", { boxShadow:"0 0 60px rgba(106,170,58,.5)", duration:1, repeat:1, yoyo:true,
ease:"sine.inOut" }, 36);// Registerwindow.__timelines["versepc-promo"] = tl;</script>
</body></html>
============================================================文件: versepc-promo\promo-
fastcut.html============================================================
<!DOCTYPE html><html lang="zh-CN"><head><meta charset="UTF-8"><meta name="viewport"
content="width=device-width, initial-scale=1.0"><title>VersePC FastCut</title><link
href="https://fonts.googleapis.com/css2?
family=Inter:wght@300;400;500;600;700;800&display=swap" rel="stylesheet">
<style>*,*::before,*::after{margin:0;padding:0;box-sizing:border-
box}body{background:#fff;overflow:hidden;font-family:"Inter",-apple-
system,BlinkMacSystemFont,sans-serif}[data-composition-id="versepc-fastcut"]{
width:1920px;height:1080px;position:relative;overflow:hidden;background:#fff;}.scene{
position:absolute;inset:0;  display:flex;align-items:center;justify-content:center;
visibility:hidden;opacity:0;}/* ===== Act 1: Logo ===== */.a1-center{  display:flex;flex-
direction:column;align-items:center;gap:28px;}.a1-logo-wrap{  width:120px;height:120px;
border:1.5px solid rgba(0,0,0,.08);  display:flex;align-items:center;justify-
content:center;  box-shadow:0 4px 24px rgba(0,0,0,.04);}.a1-logo-wrap
img{width:64px;height:64px;image-rendering:pixelated;}
```

```
.a1-title{  font-size:96px;font-weight:700;color:#1D1D1F;  letter-spacing:-.03em;line-
height:1;}.a1-title .hl{color:#6AAA3A;}.a1-sub{  font-size:24px;font-
weight:400;color:#86868B;  letter-spacing:.08em;}/* ===== Bottom tag line (shared) =====
*/.tagline{  position:absolute;bottom:80px;left:50%;transform:translateX(-50%);  font-
size:18px;font-weight:400;color:#86868B;  letter-spacing:.06em;white-space:nowrap;}/* =====
Act 2: Version ===== */.a2-wrap{  width:100%;padding:0 140px;  display:flex;align-
items:center;justify-content:space-between;}.a2-left{display:flex;flex-
direction:column;align-items:center;gap:36px;}.a2-ring-
wrap{position:relative;width:300px;height:300px;}.a2-ring{  position:absolute;inset:0;
border:2px solid rgba(0,0,0,.06);border-radius:50%;}.a2-ring-i{
position:absolute;inset:48px;  border:1.5px dashed rgba(0,0,0,.04);border-radius:50%;}.a2-
badge{  position:absolute;font-size:14px;font-weight:500;color:#1D1D1F;
background:#fff;border:1px solid rgba(0,0,0,.06);  padding:4px 14px;white-space:nowrap;
box-shadow:0 1px 6px rgba(0,0,0,.04);}.a2-
b1{top:0;left:50%;transform:translateX(-50%);}.a2-b2{right:-10px;top:38%;}.a2-
b3{bottom:0;left:50%;transform:translateX(-50%);}.a2-b4{left:-10px;top:38%;}.a2-play{
position:absolute;top:50%;left:50%;transform:translate(-50%,-50%);
width:72px;height:72px;border-radius:50%;  background:#6AAA3A;  display:flex;align-
items:center;justify-content:center;}.a2-play::after{  content:"";border:solid;border-
width:14px 0 14px 26px;
```

```
  border-color:transparent transparent transparent #fff;margin-left:5px;}.a2-
right{display:flex;flex-direction:column;gap:16px;}.a2-loader{  display:flex;align-
items:center;gap:12px;  padding:10px 20px;border:1px solid rgba(0,0,0,.06);
background:#fff;box-shadow:0 1px 8px rgba(0,0,0,.03);}.a2-loader-
dot{width:8px;height:8px;border-radius:50%;flex-shrink:0;}.ldot-
f{background:#FFB347;}.ldot-a{background:#6AAA3A;}.ldot-n{background:#7C4DFF;}.a2-loader
span{font-size:15px;font-weight:500;color:#1D1D1F;letter-spacing:.02em;}/* ===== Act 3:
Mods ===== */.a3-wrap{  width:100%;padding:0 140px;  display:flex;align-
items:center;justify-content:flex-end;}.a3-cards{display:flex;flex-
direction:column;gap:14px;}.a3-card{  display:flex;align-items:center;gap:16px;
width:420px;padding:16px 20px;  background:#fff;border:1px solid rgba(0,0,0,.06);  box-
shadow:0 2px 12px rgba(0,0,0,.04);}.a3-icon{  width:44px;height:44px;flex-shrink:0;
background:rgba(106,170,58,.08);border:1px solid rgba(106,170,58,.15);  display:flex;align-
items:center;justify-content:center;  font-size:20px;}.a3-info{flex:1;display:flex;flex-
direction:column;gap:4px;}.a3-name{font-size:16px;font-weight:600;color:#1D1D1F;}.a3-
meta{font-size:13px;color:#86868B;}.a3-bar{height:3px;background:rgba(0,0,0,.04);border-
radius:2px;overflow:hidden;margin-top:2px;}.a3-bar-
fill{height:100%;background:#6AAA3A;border-radius:2px;width:0%;}/* ===== Act 4: Batch
Download ===== */.a4-wrap{  width:100%;padding:0 200px;  display:flex;flex-
direction:column;gap:20px;}.a4-row{  display:flex;align-items:center;gap:16px;}.a4-check{
width:22px;height:22px;flex-shrink:0;  border:1.5px solid rgba(0,0,0,.12);
background:#fff;display:flex;align-items:center;justify-content:center;
```

```
  font-size:14px;color:#fff;}.a4-check.done{background:#6AAA3A;border-color:#6AAA3A;}.a4-
check.done::after{content:"✓";color:#fff;font-weight:700;font-size:13px;}.a4-row-name{
font-size:15px;font-weight:500;color:#1D1D1F;width:100px;}.a4-bar-
wrap{flex:1;height:6px;background:rgba(0,0,0,.04);border-radius:3px;overflow:hidden;}.a4-
bar-fill{height:100%;border-radius:3px;width:0%;}.a4-bar-fill.g{background:#6AAA3A;}.a4-
bar-fill.o{background:#FFB347;}.a4-btn{  align-self:center;  padding:10px 36px;margin-
top:12px;  background:#1D1D1F;color:#fff;border:none;  font-family:"Inter",sans-serif;font-
size:15px;font-weight:500;  letter-spacing:.02em;cursor:default;}.a4-done-row{
display:flex;align-items:center;gap:10px;}.a4-done-row .check-icon{
width:28px;height:28px;border-radius:50%;background:#6AAA3A;  display:flex;align-
items:center;justify-content:center;  color:#fff;font-weight:700;font-size:15px;}/* =====
Act 5: Mystery ===== */.a5-wrap{  width:100%;padding:0 200px;  display:flex;flex-
direction:column;align-items:center;gap:32px;}.a5-blur-area{  position:relative;
width:500px;height:340px;  overflow:hidden;}.a5-blur-content{  position:absolute;inset:0;
display:flex;flex-direction:column;gap:12px;padding:30px;}.a5-blur-bubble{  max-
width:80%;padding:12px 16px;border-radius:8px;  font-size:15px;line-
height:1.5;color:#1D1D1F;  background:rgba(0,0,0,.03);border:1px solid rgba(0,0,0,.04);
backdrop-filter:blur(18px);-webkit-backdrop-filter:blur(18px);}.a5-blur-
bubble.bot{background:rgba(106,170,58,.04);border-color:rgba(106,170,58,.1);align-
self:flex-start;}.a5-blur-bubble.usr{background:rgba(0,0,0,.02);align-self:flex-end;}.a5-
glow{
```

```
  position:absolute;bottom:-80px;left:50%;transform:translateX(-50%);
width:300px;height:120px;  background:radial-gradient(ellipse, rgba(106,170,58,.15) 0%,
transparent 70%);}.a5-flash-code{  position:absolute;top:30px;right:20px;  font-family:"SF
Mono","Menlo","Consolas",monospace;font-size:12px;  color:rgba(0,0,0,.2);line-height:1.6;
opacity:0;}.a5-tagline{  font-size:28px;font-weight:500;color:#1D1D1F;  letter-
spacing:.12em;}/* ===== Act 6: Modpacks ===== */.a6-wrap{  width:100%;padding:0 140px;
display:flex;align-items:center;gap:60px;}.a6-cards{display:flex;flex-
direction:column;gap:14px;flex:1;}.a6-card{  display:flex;align-items:center;gap:16px;
padding:16px 20px;  background:#fff;border:1px solid rgba(0,0,0,.06);  box-shadow:0 2px
12px rgba(0,0,0,.04);}.a6-thumb{  width:48px;height:48px;flex-shrink:0;
background:rgba(255,179,71,.08);border:1px solid rgba(255,179,71,.15);  display:flex;align-
items:center;justify-content:center;font-size:22px;}.a6-info{flex:1;display:flex;flex-
direction:column;gap:4px;}.a6-name{font-size:16px;font-weight:600;color:#1D1D1F;}.a6-
desc{font-size:13px;color:#86868B;}.a6-tag{  display:inline-flex;padding:2px 10px;font-
size:11px;font-weight:500;  background:rgba(106,170,58,.06);border:1px solid
rgba(106,170,58,.12);  color:#6AAA3A;}.a6-stats{display:flex;flex-
direction:column;gap:20px;flex-shrink:0;}.a6-stat{text-align:center;}.a6-stat-val{font-
size:48px;font-weight:700;color:#1D1D1F;line-height:1;}.a6-stat-lbl{font-
size:14px;color:#86868B;margin-top:4px;}/* ===== Act 7: Finale ===== */.a7-wrap{
display:flex;flex-direction:column;align-items:center;gap:20px;}.a7-line{font-
size:40px;font-weight:500;color:#1D1D1F;letter-spacing:.15em;}
```

```
.a7-line.thin{font-weight:300;color:#86868B;}.a7-
divider{width:40px;height:1px;background:rgba(0,0,0,.08);margin:8px 0;}.a7-logo-small{
width:56px;height:56px;  border:1px solid rgba(0,0,0,.06);  display:flex;align-
items:center;justify-content:center;}.a7-logo-small img{width:32px;height:32px;image-
rendering:pixelated;}.a7-date{  font-size:72px;font-weight:700;color:#1D1D1F;  letter-
spacing:.04em;line-height:1.1;  margin-top:8px;}.a7-date .hl{color:#6AAA3A;}</style></head>
<body><div data-composition-id="versepc-fastcut" data-width="1920" data-height="1080" data-
start="0">  <!-- ===== ACT 1: Logo 浮现 (0-3s) ===== -->  <div class="scene" id="s1">
<div class="a1-center">      <div class="a1-logo-wrap"><img src="assets/icon.svg"
alt="VersePC"></div>      <div class="a1-title"><span class="hl">Verse</span>PC</div>
<div class="a1-sub">新一代 Minecraft 启动器</div>    </div>  </div>  <!-- ===== ACT 2: 版本启
动 (3-6.5s) ===== -->  <div class="scene" id="s2">    <div class="a2-wrap">      <div
class="a2-left">        <div class="a2-ring-wrap">          <div class="a2-ring"></div><div
class="a2-ring-i"></div>          <div class="a2-badge a2-b1">1.21.5</div>          <div
class="a2-badge a2-b2">1.20.1</div>          <div class="a2-badge a2-b3">1.18.2</div>
<div class="a2-badge a2-b4">1.16.5</div>          <div class="a2-play"></div>        </div>
</div>      <div class="a2-right">        <div class="a2-loader"><div class="a2-loader-dot
ldot-f"></div><span>Forge</span></div>        <div class="a2-loader"><div class="a2-loader-
dot ldot-a"></div><span>Fabric</span></div>        <div class="a2-loader"><div class="a2-
loader-dot ldot-n"></div><span>NeoForge</span></div>      </div>    </div>    <div
class="tagline">一键启动 · 无限可能</div>  </div>
```

```
  <!-- ===== ACT 3: 模组管理 (6.5-10s) ===== -->  <div class="scene" id="s3">    <div
class="a3-wrap">      <div class="a3-cards">        <div class="a3-card">          <div
class="a3-icon">🔮</div>          <div class="a3-info">            <div class="a3-
name">OptiFine</div>            <div class="a3-meta">v1.21.5 · 性能优化</div>
<div class="a3-bar"><div class="a3-bar-fill"></div></div>          </div>        </div>
<div class="a3-card">          <div class="a3-icon">📦</div>          <div class="a3-info">
<div class="a3-name">Just Enough Items</div>            <div class="a3-meta">v19.21.0 · 物品
查询</div>            <div class="a3-bar"><div class="a3-bar-fill"></div></div>
</div>        </div>        <div class="a3-card">          <div class="a3-icon">🌍</div>
<div class="a3-info">            <div class="a3-name">JourneyMap</div>            <div
class="a3-meta">v6.0.0 · 小地图</div>            <div class="a3-bar"><div class="a3-bar-
fill"></div></div>          </div>        </div>      </div>    </div>    <div
class="tagline">模组管理 · 尽在掌握</div>  </div>  <!-- ===== ACT 4: 批量下载 (10-14s) =====
-->  <div class="scene" id="s4">    <div class="a4-wrap">      <div class="a4-row"><div
class="a4-check"></div><span class="a4-row-name">OptiFine</span><div class="a4-bar-wrap">
<div class="a4-bar-fill g"></div></div></div>      <div class="a4-row"><div class="a4-
check"></div><span class="a4-row-name">JEI</span><div class="a4-bar-wrap"><div class="a4-
bar-fill o"></div></div></div>      <div class="a4-row"><div class="a4-check"></div><span
class="a4-row-name">JourneyMap</span><div class="a4-bar-wrap"><div class="a4-bar-fill g">
</div></div></div>      <div class="a4-row"><div class="a4-check"></div><span class="a4-
row-name">Inventory Profiler</span><div class="a4-bar-wrap"><div class="a4-bar-fill o">
</div></div></div>      <div class="a4-row"><div class="a4-check"></div><span class="a4-
row-name">Sodium</span><div class="a4-bar-wrap"><div class="a4-bar-fill g"></div></div>
</div>      <div class="a4-btn">批量下载（5）</div>      <div class="a4-done-row"
style="opacity:0"><div class="check-icon">✓</div><span style="font-size:16px;font-
weight:500;color:#1D1D1F">5 个模组已安装</span></div>    </div>    <div class="tagline">批量
下载 · 一键安装</div>  </div>  <!-- ===== ACT 5: ??? 神秘功能 (14-18s) ===== -->  <div
class="scene" id="s5">
```

```
    <div class="a5-wrap">      <div class="a5-blur-area">        <div class="a5-blur-
content" style="filter:blur(14px);-webkit-filter:blur(14px);">          <div class="a5-
blur-bubble usr" style="max-width:75%">为什么我的 Forge 启动崩溃了？</div>          <div
class="a5-blur-bubble bot" style="max-width:80%">检测到模组冲突，正在分析日志…</div>
<div class="a5-blur-bubble usr" style="max-width:60%">帮我修复</div>          <div
class="a5-blur-bubble bot" style="max-width:65%">已自动修复 ✅</div>        </div>
<div class="a5-glow"></div>        <div class="a5-flash-code">[17:32:41] Detected mod
conflict: OptiFine × Sodium<br>  → Auto-resolving... Done.</div>      </div>      <div
class="a5-tagline">还有更多…</div>    </div>  </div>  <!-- ===== ACT 6: 整合包 (18-21.5s)
===== -->  <div class="scene" id="s6">    <div class="a6-wrap">      <div class="a6-cards">
<div class="a6-card">          <div class="a6-thumb">⚔️</div>          <div class="a6-
info">            <div class="a6-name">RLCraft</div>            <div class="a6-desc">高难度
生存整合包</div>            <div><span class="a6-tag">CurseForge</span></div>
</div>        </div>        <div class="a6-card">          <div class="a6-thumb">🏭</div>
<div class="a6-info">            <div class="a6-name">All the Mods 10</div>            <div
class="a6-desc">400+ 模组科技魔法</div>            <div><span class="a6-tag">Modrinth</span>
</div>          </div>        </div>        <div class="a6-card">          <div class="a6-
thumb">🧩</div>          <div class="a6-info">            <div class="a6-name">Better
MC</div>            <div class="a6-desc">原版增强 · 轻量优化</div>            <div><span
class="a6-tag">CurseForge</span></div>          </div>        </div>      </div>      <div
class="a6-stats">        <div class="a6-stat"><div class="a6-stat-val">50+</div><div
class="a6-stat-lbl">整合包</div></div>        <div class="a6-stat"><div class="a6-stat-
val">1-Click</div><div class="a6-stat-lbl">一键导入</div></div>      </div>    </div>
<div class="tagline">整合包 · 一键安装</div>
```

```
  </div>  <!-- ===== ACT 7: 收官 (21.5-30s) ===== -->  <div class="scene" id="s7">    <div
class="a7-wrap">      <div class="a7-line">免费</div>      <div class="a7-line">开源</div>
<div class="a7-line thin">永久更新</div>      <div class="a7-divider"></div>      <div
class="a7-logo-small"><img src="assets/icon.svg" alt="VersePC"></div>      <div class="a7-
date"><span class="hl">6 月 21 日</span> 见</div>    </div>  </div></div><script
src="https://cdn.jsdelivr.net/npm/gsap@3.14.2/dist/gsap.min.js"></script>
<script>gsap.defaults({overwrite:false, ease:"sine.inOut"});const tl =
gsap.timeline({paused:true});// ================================================// ACT 1:
Logo (0s - 3s)// ================================================tl.set("#s1",
{autoAlpha:1},0);tl.from("#s1 .a1-logo-wrap",
{scale:.85,opacity:0,duration:.8,ease:"sine.inOut"},.2);tl.from("#s1 .a1-title",
{y:30,opacity:0,duration:.7,ease:"power3.out"},.8);tl.from("#s1 .a1-sub",
{y:15,opacity:0,duration:.5,ease:"power3.out"},1.4);tl.to("#s1",
{autoAlpha:0,duration:.35,ease:"power2.in"},2.6);//
================================================// ACT 2: Version (3s - 6.5s)//
================================================tl.set("#s2",{autoAlpha:1},3);tl.from("#s2
.a2-ring",{scale:.9,opacity:0,duration:.55,ease:"sine.inOut"},3.1);tl.from("#s2 .a2-ring-
i",{scale:.8,opacity:0,duration:.4,ease:"sine.inOut"},3.3);tl.from("#s2 .a2-badge",
{scale:.8,opacity:0,stagger:.08,duration:.35,ease:"power3.out"},3.3);tl.from("#s2 .a2-
play",{scale:.8,opacity:0,duration:.4,ease:"power3.out"},3.8);tl.from("#s2 .a2-loader",
{x:30,opacity:0,stagger:.1,duration:.45,ease:"power3.out"},3.3);tl.from("#s2 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},4.5);// pulse the play buttontl.to("#s2 .a2-
play",{boxShadow:"0 0 40px
rgba(106,170,58,.25)",duration:.8,yoyo:true,repeat:1,ease:"sine.inOut"},4.2);tl.to("#s2",
{autoAlpha:0,duration:.3,ease:"power2.in"},6.2);//
================================================// ACT 3: Mods (6.5s - 10s)//
================================================tl.set("#s3",
{autoAlpha:1},6.5);tl.from("#s3 .a3-card",
{x:60,opacity:0,stagger:.12,duration:.5,ease:"power3.out"},6.6);tl.to("#s3 .a3-bar-fill",
{width:"100%",stagger:.12,duration:.5,ease:"power3.out",transformOrigin:"left"},7.3);
```

```
tl.from("#s3 .tagline",{y:12,opacity:0,duration:.4,ease:"power3.out"},7.8);tl.to("#s3",
{autoAlpha:0,duration:.3,ease:"power2.in"},9.7);//
================================================// ACT 4: Batch Download (10s - 14s)//
================================================tl.set("#s4",{autoAlpha:1},10);// stagger
checkbox checkstl.to("#s4 .a4-check",{
className:"+=done",stagger:.08,duration:.15,ease:"none"},10.2);// button
appearstl.from("#s4 .a4-btn",
{scale:.92,opacity:0,duration:.4,ease:"sine.inOut"},10.8);tl.to("#s4 .a4-btn",
{opacity:.4,duration:.2},11.1); // click feedback// progress bars all at once with
staggered delaystl.to("#s4 .a4-bar-fill",{
width:"100%",stagger:.15,duration:.6,ease:"sine.inOut",transformOrigin:"left"},11.2);//
done rowtl.to("#s4 .a4-done-row",
{autoAlpha:1,duration:.4,ease:"sine.inOut"},12.2);tl.from("#s4 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},13);tl.to("#s4",
{autoAlpha:0,duration:.3,ease:"power2.in"},13.7);//
================================================// ACT 5: Mystery (14s - 18s)//
================================================tl.set("#s5",{autoAlpha:1},14);tl.from("#s5
.a5-blur-area",{scale:.92,opacity:0,duration:.8,ease:"sine.inOut"},14.1);// green glow
pulsetl.to("#s5 .a5-glow",
{scale:1.15,opacity:.6,duration:1.2,yoyo:true,repeat:1,ease:"sine.inOut"},14.6);// flash
code snippettl.to("#s5 .a5-flash-code",{autoAlpha:1,duration:.3},15.2);tl.to("#s5 .a5-
flash-code",{autoAlpha:0,duration:.3},15.9);tl.to("#s5 .a5-flash-code",
{autoAlpha:1,duration:.2},16.4);tl.to("#s5 .a5-flash-code",
{autoAlpha:0,duration:.3},16.9);tl.from("#s5 .a5-tagline",
{y:15,opacity:0,duration:.6,ease:"sine.inOut"},15.5);tl.to("#s5",
{autoAlpha:0,duration:.35,ease:"power2.in"},17.6);//
================================================// ACT 6: Modpacks (18s - 21.5s)//
================================================tl.set("#s6",{autoAlpha:1},18);tl.from("#s6
.a6-card",{y:30,opacity:0,stagger:.15,duration:.5,ease:"power3.out"},18.1);tl.from("#s6
.a6-tag",{opacity:0,stagger:.15,duration:.3,ease:"sine.inOut"},18.6);tl.from("#s6 .a6-stat-
val",{scale:.8,opacity:0,stagger:.15,duration:.5,ease:"back.out(1.4)"},18.8);tl.from("#s6
.a6-stat-lbl",{opacity:0,stagger:.15,duration:.3},19.1);tl.from("#s6 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},19.5);tl.to("#s6",
{autoAlpha:0,duration:.3,ease:"power2.in"},21.2);//
================================================
```

```
// ACT 7: Finale (21.5s - 30s)//
================================================tl.set("#s7",
{autoAlpha:1},21.5);tl.from("#s7 .a7-line",
{y:25,opacity:0,stagger:.25,duration:.6,ease:"power3.out"},21.6);tl.from("#s7 .a7-divider",
{scaleX:0,opacity:0,duration:.4,ease:"power3.out",transformOrigin:"center"},23.6);tl.from("
#s7 .a7-logo-small",{scale:.8,opacity:0,duration:.5,ease:"sine.inOut"},24.2);tl.from("#s7
.a7-date",{y:30,opacity:0,duration:.7,ease:"power3.out"},25);// keep finale on screen until
end// total timeline ~30s// Registerwindow.__timelines = window.__timelines ||
{};window.__timelines["versepc-fastcut"] = tl;</script></body></html>
============================================================文件: 文档鉴别材
料.html============================================================
<!DOCTYPE html><html lang="zh-CN"><head>    <meta charset="UTF-8">    <title>软件说明书 -
VersePC</title>    <style>        @page { size: A4; margin: 2cm; }        body { font-
family: "Microsoft YaHei", SimSun, sans-serif; font-size: 12pt; line-height: 1.8; color:
#333; }        .cover { text-align: center; padding-top: 150px; page-break-after: always; }
.cover h1 { font-size: 28pt; margin-bottom: 20px; }        .cover h2 { font-size: 18pt;
font-weight: normal; color: #666; }        .cover .version { font-size: 14pt; margin-top:
30px; }        h1 { font-size: 20pt; text-align: center; margin: 30px 0; }        h2 {
font-size: 16pt; margin: 25px 0 15px 0; border-bottom: 1px solid #ccc; padding-bottom: 5px;
}        p { text-indent: 2em; margin: 10px 0; }        ul { margin: 10px 0 10px 2em; }
li { margin: 5px 0; }        .section { page-break-before: always; }        .feature-box {
background-color: #f5f5f5; padding: 15px; margin: 15px 0; border-left: 4px solid #007bff; }
table { width: 100%; border-collapse: collapse; margin: 15px 0; }        table, th, td {
border: 1px solid #ddd; }        th, td { padding: 10px; text-align: left; }        th {
background-color: #f0f0f0; }    </style></head><body>    <div class="cover">
<h1>VersePC</h1>        <h2>软件说明书</h2>        <div class="version">
```

```
            <p>版本：V1.0</p>            <p>日期：2026年06月</p>        </div>    </div>
<div class="section">        <h1>一、软件概述</h1>        <h2>1.1 软件简介</h2>
<p>VersePC是一款基于Electron框架开发的Minecraft游戏启动器，为玩家提供一站式的Minecraft游戏管理体验。软件集成了游戏启动、版本管理、模组管理、整合包管理、账户管理等核心功能，旨在为Minecraft玩家提供便捷、高效、美观的游戏启动与管理工具。</p>        <h2>1.2 开发目的</h2>        <p>随着Minecraft游戏的不断发展，玩家对于游戏启动器的需求日益增长。现有的启动器产品在用户体验、功能完整性、界面美观度等方面存在不足。VersePC的开发旨在解决这些问题，为玩家提供一款功能丰富、界面精美、操作便捷的Minecraft启动器。</p>        <h2>1.3 适用范围</h2>        <p>本软件适用于所有Minecraft玩家，特别是对游戏管理有较高要求的用户。支持Windows操作系统，兼容Minecraft Java Edition的各个版本。</p>
</div>        <div class="section">        <h1>二、系统要求</h1>        <h2>2.1 硬件要求</h2>
<table>            <tr><th>项目</th><th>最低配置</th><th>推荐配置</th></tr>            <tr>
<td>处理器</td><td>Intel Core i3 或同等性能</td><td>Intel Core i5-10400F 或更高</td></tr>
<tr><td>内存</td><td>4GB RAM</td><td>8GB RAM 或更高</td></tr>            <tr><td>显卡</td>
<td>支持OpenGL 2.1</td><td>GTX 1660 或更高</td></tr>            <tr><td>存储空间</td>
<td>500MB 可用空间</td><td>1GB 或更多</td></tr>        </table>        <h2>2.2 软件要求</h2>
<ul>            <li>操作系统：Windows 10/11 64位</li>            <li>运行时：Java Runtime
Environment 8+（软件可自动检测和管理）</li>            <li>网络：需要互联网连接以下载游戏版本和模组</li>        </ul>    </div>        <div class="section">        <h1>三、功能模块</h1>
<h2>3.1 游戏启动模块</h2>        <div class="feature-box">            <p><strong>功能描述：
</strong>提供Minecraft游戏的启动功能，支持一键启动游戏。</p>            <ul>
<li>支持多种Minecraft版本的启动</li>                <li>自动检测和配置Java环境</li>
<li>游戏启动前的依赖检查</li>                <li>启动进度实时显示</li>            </ul>
</div>        <h2>3.2 版本管理模块</h2>        <div class="feature-box">            <p>
<strong>功能描述：</strong>管理本地安装的Minecraft版本，支持版本的安装、删除和切换。</p>
<ul>                <li>版本列表展示与搜索</li>
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                <li>一键安装新版本</li>                <li>版本删除与清理</li>
<li>版本信息查看</li>            </ul>        </div>        <h2>3.3 模组管理模块</h2>
<div class="feature-box">            <p><strong>功能描述：</strong>管理Minecraft模组，支持从
Modrinth等平台下载和安装模组。</p>            <ul>                <li>模组搜索与浏览</li>
<li>一键安装模组</li>                <li>模组启用/禁用管理</li>                <li>模组冲突检测
</li>            </ul>        </div>        <h2>3.4 整合包管理模块</h2>        <div
class="feature-box">            <p><strong>功能描述：</strong>支持Minecraft整合包的导入和管
理。</p>            <ul>                <li>支持多种整合包格式导入</li>                <li>整
合包版本管理</li>                <li>整合包配置文件管理</li>            </ul>        </div>
<h2>3.5 账户管理模块</h2>        <div class="feature-box">            <p><strong>功能描述：
</strong>管理Minecraft账户，支持微软账号登录。</p>            <ul>                <li>微软账号
OAuth登录</li>                <li>账户信息显示</li>                <li>皮肤预览</li>
<li>多账户切换</li>            </ul>        </div>        <h2>3.6 设置模块</h2>        <div
class="feature-box">            <p><strong>功能描述：</strong>软件各项设置的配置。</p>
<ul>                <li>主题切换（支持深色/浅色主题）</li>                <li>Java路径配置
</li>                <li>游戏目录设置</li>                <li>网络代理设置</li>
<li>语言设置</li>            </ul>        </div>        <h2>3.7 Java管理模块</h2>        <div
class="feature-box">            <p><strong>功能描述：</strong>自动检测和管理系统中的Java环境。
</p>            <ul>                <li>Java版本自动检测</li>
```

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                <li>Java路径配置</li>                <li>Java下载与安装</li>            </ul>
</div>        <h2>3.8 文件浏览器模块</h2>        <div class="feature-box">            <p>
<strong>功能描述：</strong>内置文件浏览器，方便用户管理游戏文件。</p>            <ul>
<li>目录浏览与导航</li>                <li>文件预览</li>                <li>文件编辑（集成
Monaco Editor）</li>            </ul>        </div>        <h2>3.9 AI助手模块</h2>
<div class="feature-box">            <p><strong>功能描述：</strong>集成AI助手，提供智能问答和
帮助。</p>            <ul>                <li>自然语言问答</li>                <li>游戏问题诊
断</li>                <li>配置建议</li>            </ul>        </div>    </div>
<div class="section">        <h1>四、操作指南</h1>        <h2>4.1 安装与启动</h2>
<p>1. 下载VersePC安装包</p>        <p>2. 运行安装程序，选择安装目录</p>        <p>3. 完成安装
后，双击桌面图标启动软件</p>        <h2>4.2 账户登录</h2>        <p>1. 点击左侧导航栏的"账户"选
项</p>        <p>2. 选择"添加账户"</p>        <p>3. 按照提示完成微软账号授权登录</p>
<h2>4.3 安装游戏版本</h2>        <p>1. 点击左侧导航栏的"版本管理"</p>        <p>2. 选择需要安装
的版本</p>        <p>3. 点击"安装"按钮，等待下载完成</p>        <h2>4.4 安装模组</h2>
<p>1. 点击左侧导航栏的"模组管理"</p>        <p>2. 在搜索框输入模组名称</p>        <p>3. 选择目标
模组，点击"安装"</p>        <h2>4.5 启动游戏</h2>        <p>1. 在首页选择要启动的版本</p>
<p>2. 点击"启动游戏"按钮</p>        <p>3. 等待游戏加载完成</p>    </div>        <div
class="section">        <h1>五、技术特点</h1>
```

```
        <h2>5.1 架构设计</h2>        <p>采用Electron框架，实现跨平台桌面应用开发。使用自定义协议替代传统HTTP服务器，消除端口冲突，提升安全性。</p>        <h2>5.2 前端技术</h2>        <ul>
<li>HTML5 + CSS3 + JavaScript</li>            <li>Monaco Editor（代码编辑器）</li>
<li>Three.js（3D渲染）</li>            <li>xterm.js（终端模拟）</li>        </ul>
<h2>5.3 后端技术</h2>        <ul>            <li>Node.js + Express</li>
<li>WebSocket（实时通信）</li>            <li>自定义协议处理</li>        </ul>        <h2>5.4 安全特性</h2>        <ul>            <li>代码混淆保护</li>            <li>所有权验证机制</li>
<li>AI训练防护</li>        </ul>    </div>        <div class="section">        <h1>六、版本信息</h1>        <table>            <tr><th>项目</th><th>信息</th></tr>            <tr><td>软件名称</td><td>VersePC</td></tr>            <tr><td>版本号</td><td>V1.0</td></tr>
<tr><td>开发语言</td><td>JavaScript</td></tr>            <tr><td>运行平台</td><td>Windows 10/11</td></tr>            <tr><td>发布日期</td><td>2026年06月</td></tr>            <tr><td>版权所有</td><td>豆杰</td></tr>        </table>    </div></body></html>
============================================================文件: 程序鉴别材料.html============================================================
<!DOCTYPE html><html lang="zh-CN"><head>    <meta charset="UTF-8">    <title>程序鉴别材料 - VersePC</title>    <style>        @page { size: A4; margin: 2cm; }        body { font-family: Consolas, monospace; font-size: 10pt; line-height: 1.4; color: #333; }
.page-header { text-align: center; font-size: 14pt; font-weight: bold; margin-bottom: 20px; padding-bottom: 10px; border-bottom: 2px solid #333; }        .section-title { font-size: 12pt; font-weight: bold; margin: 20px 0 10px 0; padding: 5px; background-color: #f0f0f0; }
```

.code-block { font-family: Consolas, monospace; font-size: 9pt; white-space: pre-wrap; word-wrap: break-word; margin: 10px 0; padding: 10px; background-color: #f8f8f8; border: 1px solid #ddd; } .page-number { text-align: center; font-size: 9pt; color: #666; margin-top: 10px; } .separator { page-break-after: always; } </style></head><body> 程序鉴别材料 软件名称：VersePC 版本号：V1.0 前1500行代码（第1-30页） ==文件: .source-backup\agent-engine.js==

```
/**
 * VersePC Agent Engine
 *
 * 核心设计：
 * 1. 状态机驱动：IDLE → RUNNING → STREAMING → ACTING → OBSERVING → REFLECTING → DONE
 * 2. 增量流式输出：仅发送新字符，消息去重 (OutputManager)
 * 3. 事件驱动：统一 say/ask 消息格式
 * 4. 全功能保留：意图检测、计划生成、卡死检测、反思、被动检测、并行工具、进度轮询
 */

const https = require('https');
const http = require('http');
const { URL } = require('url');
const { getPluginManager } = require('./plugin-manager');

// ============================================================================
// Agent State Machine
// ============================================================================

const AgentState = {
    IDLE: 'idle',
    RUNNING: 'running',
    STREAMING: 'streaming',
    ACTING: 'acting',
    OBSERVING: 'observing',
    REFLECTING: 'reflecting',
    RESPONDING: 'responding',
    WAITING_FOR_INPUT: 'waiting_for_input',
    DONE: 'done',
    STUCK: 'stuck',
    ERROR: 'error'
};

const SayType = {
    TEXT: 'text',
    REASONING: 'reasoning',
    TOOL_START: 'tool_start',
    TOOL_RESULT: 'tool_result',
    TOOL_END: 'tool_end',
    ERROR: 'error',
    COMPLETION: 'completion',
    API_STARTED: 'api_req_started',
    API_FINISHED: 'api_req_finished',
    FOLLOWUP: 'followup',
    PLAN_CREATED: 'plan_created',
    PLAN_STEP_UPDATE: 'plan_step_update',
    PLAN_DONE: 'plan_done',
```

```
</div><div class="page-number">第 1 页</div><div class="separator"></div><div class="code-
block">    THINKING_STEP: 'thinking_step',
    REFLECTION: 'reflection',
    PROGRESS: 'progress'
};

const AskType = {
    TOOL_APPROVAL: 'tool_approval',
    FOLLOWUP: 'followup'
};

// ============================================================================
// Output Manager (增量输出 + 去重)
// ============================================================================

class OutputManager {
    constructor() {
        this.displayedMessages = new Map();
        this.streamedContent = new Map();
        this.currentlyStreamingTs = null;
        this.tsCounter = 0;
    }

    _nextTs() { return ++this.tsCounter; }

    _streamDelta(ts, text) {
        const previous = this.streamedContent.get(ts);
        if (!previous) {
            this.streamedContent.set(ts, { text, headerShown: true });
            this.currentlyStreamingTs = ts;
            return { action: 'full', text };
        }
        if (text.length &gt; previous.text.length &amp;&amp;
text.startsWith(previous.text)) {
            const delta = text.slice(previous.text.length);
            this.streamedContent.set(ts, { text, headerShown: true });
            return { action: 'delta', text: delta };
        }
        return { action: 'skip' };
    }

    _finishStream(ts) {
        if (this.currentlyStreamingTs === ts) {
            this.currentlyStreamingTs = null;
        }
    }

    processMessage(msg) {
        const text = msg.text || '';
        const isPartial = msg.partial === true;
        let ts;

</div><div class="page-number">第 2 页</div><div class="separator"></div><div class="code-
block">        if (isPartial &amp;&amp; msg.type === 'say' &amp;&amp; (msg.say ===
SayType.TEXT || msg.say === SayType.REASONING || msg.say === SayType.COMPLETION)) {
            if (this.currentlyStreamingTs) {
                const activeMsg = this.displayedMessages.get(this.currentlyStreamingTs);
```

```
                if (activeMsg &amp;&amp; activeMsg.say === msg.say &amp;&amp;
activeMsg.partial) {
                    ts = this.currentlyStreamingTs;
                } else {
                    ts = this._nextTs();
                }
            } else {
                ts = this._nextTs();
            }
        } else {
            if (!isPartial &amp;&amp; msg.type === 'say' &amp;&amp;
this.currentlyStreamingTs) {
                const activeMsg = this.displayedMessages.get(this.currentlyStreamingTs);
                if (activeMsg &amp;&amp; activeMsg.say === msg.say) {
                    ts = this.currentlyStreamingTs;
                } else {
                    ts = msg.ts || this._nextTs();
                }
            } else {
                ts = msg.ts || this._nextTs();
            }
        }

        const previous = this.displayedMessages.get(ts);
        const alreadyComplete = previous &amp;&amp; !previous.partial;

        if (msg.type === 'say') {
            return this._processSay(ts, msg.say, text, isPartial, alreadyComplete, msg);
        }
        if (msg.type === 'ask') {
            return this._processAsk(ts, msg.ask, text, isPartial, alreadyComplete, msg);
        }
        return null;
    }

    _processSay(ts, say, text, isPartial, alreadyComplete, msg) {
        switch (say) {
            case SayType.TEXT:
            case SayType.REASONING:
            case SayType.COMPLETION:
                if (isPartial &amp;&amp; text) {
                    const delta = this._streamDelta(ts, text);
                    this.displayedMessages.set(ts, { ts, say, text, partial: true });
                    return { type: 'say', say, ts, text: delta.text, partial: true, delta:
delta.action === 'delta' };
                }
                if (!isPartial &amp;&amp; !alreadyComplete) {
                    const streamed = this.streamedContent.get(ts);
                    if (streamed &amp;&amp; text &amp;&amp; text.length &gt;
streamed.text.length &amp;&amp; text.startsWith(streamed.text)) {
                        const remaining = text.slice(streamed.text.length);
</div><div class="page-number">第 3 页</div><div class="separator"></div><div class="code-
block">                        this._finishStream(ts);
                        this.displayedMessages.set(ts, { ts, say, text, partial: false });
                        return { type: 'say', say, ts, text: remaining, partial: false,
trailing: true };
                    }
```

```
                    this._finishStream(ts);
                    this.displayedMessages.set(ts, { ts, say, text, partial: false });
                    return { type: 'say', say, ts, text: text || '', partial: false };
                }
                break;

            case SayType.TOOL_START:
            case SayType.TOOL_RESULT:
            case SayType.TOOL_END:
            case SayType.ERROR:
            case SayType.PLAN_CREATED:
            case SayType.PLAN_STEP_UPDATE:
            case SayType.PLAN_DONE:
            case SayType.THINKING_STEP:
            case SayType.REFLECTION:
            case SayType.PROGRESS:
                this.displayedMessages.set(ts, { ts, text, partial: false });
                return { type: 'say', say, ts, text, partial: false };

            case SayType.API_STARTED:
                this.displayedMessages.set(ts, { ts, text, partial: true });
                return { type: 'say', say, ts, text, partial: false };

            case SayType.API_FINISHED:
            case SayType.FOLLOWUP:
                this.displayedMessages.set(ts, { ts, text, partial: false });
                return { type: 'say', say, ts, text, partial: false };
        }
        return null;
    }

    _processAsk(ts, ask, text, isPartial, alreadyComplete, fullMsg) {
        this.displayedMessages.set(ts, { ts, text, partial: false });
        return { type: 'ask', ask, ts, text, partial: false, args: fullMsg.args, toolName:
fullMsg.toolName, risk: fullMsg.risk };
    }

    clear() {
        this.displayedMessages.clear();
        this.streamedContent.clear();
        this.currentlyStreamingTs = null;
        this.tsCounter = 0;
    }
}

// ============================================================================
// AI Platform &amp;amp; Provider 配置（所有平台）
</div><div class="page-number">第 4 页</div><div class="separator"></div><div class="code-
block">// ============================================================================
// 结构：
//   PLATFORMS: providerKey → { name, baseUrl, authType, apiFormat, thinkingParams }
//   MODELS: modelId → { provider, name, free }
//
// apiFormat 可选值：'openai'(默认) | 'anthropic' | 'google'
// authType 可选值：'bearer'(默认) | 'x-api-key' | 'url_key'

const PLATFORMS = {
```

```
    zhipu: {
        name: '智谱 GLM',
        baseUrl: 'https://open.bigmodel.cn/api/paas/v4',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    deepseek: {
        name: 'DeepSeek',
        baseUrl: 'https://api.deepseek.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    qwen: {
        name: '通义千问',
        baseUrl: 'https://dashscope.aliyuncs.com/compatible-mode/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    moonshot: {
        name: 'Moonshot Kimi',
        baseUrl: 'https://api.moonshot.cn/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    yi: {
        name: '零一万物',
        baseUrl: 'https://api.lingyiwanwu.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    baichuan: {
        name: '百川智能',
        baseUrl: 'https://api.baichuan-ai.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
</div><div class="page-number">第 5 页</div><div class="separator"></div><div class="code-
block">    },
    minimax: {
        name: 'MiniMax',
        baseUrl: 'https://api.minimax.chat/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    stepfun: {
        name: '阶跃星辰',
        baseUrl: 'https://api.stepfun.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
```

```
    siliconflow: {
        name: 'SiliconFlow',
        baseUrl: 'https://api.siliconflow.cn/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    openrouter: {
        name: 'OpenRouter',
        baseUrl: 'https://openrouter.ai/api/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: { enable_thinking: true }
    },
    groq: {
        name: 'Groq',
        baseUrl: 'https://api.groq.com/openai/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    openai: {
        name: 'OpenAI',
        baseUrl: 'https://api.openai.com/v1',
        authType: 'bearer',
        apiFormat: 'openai',
        thinkingParams: {}
    },
    anthropic: {
        name: 'Anthropic',
        baseUrl: 'https://api.anthropic.com',
        authType: 'x-api-key',
        apiFormat: 'anthropic',
        thinkingParams: {}
    },
</div><div class="page-number">第 6 页</div><div class="separator"></div><div class="code-
block">    google: {
        name: 'Google Gemini',
        baseUrl: 'https://generativelanguage.googleapis.com/v1beta',
        authType: 'url_key',
        apiFormat: 'google',
        thinkingParams: {}
    }
};

const MODELS = {
    // 智谱 GLM
    'glm-5.1': { provider: 'zhipu', name: 'GLM-5.1' },
    'glm-5': { provider: 'zhipu', name: 'GLM-5' },
    'glm-5-plus': { provider: 'zhipu', name: 'GLM-5-Plus' },
    'glm-5-air': { provider: 'zhipu', name: 'GLM-5-Air' },
    'glm-5-flash': { provider: 'zhipu', name: 'GLM-5-Flash', free: true },
    'glm-4.7': { provider: 'zhipu', name: 'GLM-4.7' },
    // DeepSeek
    'deepseek-v4-pro': { provider: 'deepseek', name: 'DeepSeek-V4-Pro' },
    'deepseek-v4-flash': { provider: 'deepseek', name: 'DeepSeek-V4-Flash' },
    'deepseek-chat': { provider: 'deepseek', name: 'DeepSeek-V3.2' },
```

```
    'deepseek-reasoner': { provider: 'deepseek', name: 'DeepSeek-R1' },
    // 通义千问
    'qwen3.6-max-preview': { provider: 'qwen', name: 'Qwen3.6-Max' },
    'qwen3.6-plus': { provider: 'qwen', name: 'Qwen3.6-Plus' },
    'qwen3.6-flash': { provider: 'qwen', name: 'Qwen3.6-Flash', free: true },
    'qwen3-235b-a22b': { provider: 'qwen', name: 'Qwen3-235B' },
    'qwen3-30b-a3b': { provider: 'qwen', name: 'Qwen3-30B', free: true },
    'qwq-plus': { provider: 'qwen', name: 'QwQ-Plus' },
    // Moonshot Kimi
    'kimi-k2.6': { provider: 'moonshot', name: 'Kimi-K2.6' },
    'kimi-k2.5': { provider: 'moonshot', name: 'Kimi-K2.5' },
    'moonshot-v1-128k': { provider: 'moonshot', name: 'Moonshot-v1-128k' },
    'moonshot-v1-32k': { provider: 'moonshot', name: 'Moonshot-v1-32k' },
    // 零一万物 Yi
    'yi-lightning': { provider: 'yi', name: 'Yi-Lightning', free: true },
    'yi-large': { provider: 'yi', name: 'Yi-Large' },
    'yi-large-turbo': { provider: 'yi', name: 'Yi-Large-Turbo' },
    'yi-medium': { provider: 'yi', name: 'Yi-Medium' },
    // 百川
    'Baichuan4-Turbo': { provider: 'baichuan', name: 'Baichuan4-Turbo' },
    'Baichuan4-Air': { provider: 'baichuan', name: 'Baichuan4-Air' },
    'Baichuan4': { provider: 'baichuan', name: 'Baichuan4' },
    'Baichuan3-Turbo': { provider: 'baichuan', name: 'Baichuan3-Turbo' },
    // MiniMax
    'MiniMax-M2.7': { provider: 'minimax', name: 'MiniMax-M2.7' },
    'MiniMax-M2.5': { provider: 'minimax', name: 'MiniMax-M2.5' },
    'MiniMax-M2.1': { provider: 'minimax', name: 'MiniMax-M2.1' },
    // 阶跃星辰
    'step-2-16k': { provider: 'stepfun', name: 'Step-2-16K' },
</div><div class="page-number">第 7 页</div><div class="separator"></div><div class="code-
block">    'step-1-8k': { provider: 'stepfun', name: 'Step-1-8K', free: true },
    // SiliconFlow
    'Pro/deepseek-ai/DeepSeek-V4-Pro': { provider: 'siliconflow', name: 'DeepSeek-V4-Pro'
},
    'deepseek-ai/DeepSeek-V4-Flash': { provider: 'siliconflow', name: 'DeepSeek-V4-Flash',
free: true },
    'Qwen/Qwen3-235B-A22B': { provider: 'siliconflow', name: 'Qwen3-235B', free: true },
    'Qwen/Qwen3-30B-A3B': { provider: 'siliconflow', name: 'Qwen3-30B', free: true },
    'Qwen/Qwen3.6-Flash': { provider: 'siliconflow', name: 'Qwen3.6-Flash', free: true },
    'Pro/zai-org/GLM-5.1': { provider: 'siliconflow', name: 'GLM-5.1', free: true },
    // OpenRouter
    'deepseek/deepseek-v4-pro': { provider: 'openrouter', name: 'DeepSeek V4 Pro' },
    'deepseek/deepseek-v4-flash:free': { provider: 'openrouter', name: 'DeepSeek V4 Flash',
free: true },
    'qwen/qwen3-235b-a22b': { provider: 'openrouter', name: 'Qwen3 235B' },
    'google/gemini-2.5-flash': { provider: 'openrouter', name: 'Gemini 2.5 Flash', free:
true },
    'anthropic/claude-sonnet-4-6': { provider: 'openrouter', name: 'Claude Sonnet 4.6' },
    'anthropic/claude-3.5-sonnet': { provider: 'openrouter', name: 'Claude 3.5 Sonnet',
free: true },
    // Groq
    'llama-3.3-70b-versatile': { provider: 'groq', name: 'Llama 3.3 70B', free: true },
    'qwen/qwen3-32b': { provider: 'groq', name: 'Qwen3 32B', free: true },
    'deepseek-r1-distill-llama-70b': { provider: 'groq', name: 'DeepSeek R1 70B', free:
true },
    // OpenAI
    'gpt-4.1': { provider: 'openai', name: 'GPT-4.1' },
```

```
    'gpt-4.1-mini': { provider: 'openai', name: 'GPT-4.1-mini' },
    'gpt-4.1-nano': { provider: 'openai', name: 'GPT-4.1-nano' },
    'gpt-4o': { provider: 'openai', name: 'GPT-4o' },
    'gpt-4o-mini': { provider: 'openai', name: 'GPT-4o-mini' },
    'o3-mini': { provider: 'openai', name: 'o3-mini' },
    'o3': { provider: 'openai', name: 'o3' },
    // Anthropic Claude
    'claude-sonnet-4-6-20250217': { provider: 'anthropic', name: 'Claude Sonnet 4.6' },
    'claude-opus-4-7-20260416': { provider: 'anthropic', name: 'Claude Opus 4.7' },
    'claude-sonnet-4-5-20250929': { provider: 'anthropic', name: 'Claude Sonnet 4.5' },
    'claude-opus-4-5-20251124': { provider: 'anthropic', name: 'Claude Opus 4.5' },
    'claude-haiku-4-5-20251001': { provider: 'anthropic', name: 'Claude Haiku 4.5' },
    // Google Gemini
    'gemini-3.5-flash': { provider: 'google', name: 'Gemini 3.5 Flash' },
    'gemini-2.5-pro': { provider: 'google', name: 'Gemini 2.5 Pro' },
    'gemini-2.5-flash': { provider: 'google', name: 'Gemini 2.5 Flash', free: true },
    'gemini-2.5-flash-lite': { provider: 'google', name: 'Gemini 2.5 Flash-Lite', free:
true },
    'gemini-2.0-flash': { provider: 'google', name: 'Gemini 2.0 Flash', free: true }
};

function getProviderInfo(modelId) {
    const modelInfo = MODELS[modelId];
    if (!modelInfo) return { platform: PLATFORMS.zhipu, modelInfo: { name: modelId } };
    return { platform: PLATFORMS[modelInfo.provider] || PLATFORMS.zhipu, modelInfo };
}

function getProviderForModel(modelId) {
    return getProviderInfo(modelId).platform;
}
</div><div class="page-number">第 8 页</div><div class="separator"></div><div class="code-
block">
function buildApiHeaders(platform, apiKey) {
    const headers = { 'Content-Type': 'application/json' };
    switch (platform.authType) {
        case 'x-api-key':
            headers['x-api-key'] = apiKey;
            headers['anthropic-version'] = '2023-06-01';
            break;
        case 'url_key':
            headers['x-goog-api-key'] = apiKey;
            break;
        case 'bearer':
        default:
            headers['Authorization'] = `Bearer ${apiKey}`;
            break;
    }
    if (platform.apiFormat === 'openrouter') {
        headers['HTTP-Referer'] = 'https://versepc.app';
        headers['X-Title'] = 'VersePC';
    }
    return headers;
}

function buildChatEndpoint(platform, modelId, apiKey) {
    const base = platform.baseUrl;
    switch (platform.apiFormat) {
```

```
        case 'anthropic':
            return { url: base + '/v1/messages', method: 'POST' };
        case 'google':
            return { url: base + '/models/' + modelId + ':streamGenerateContent?
alt=sse&amp;key=' + encodeURIComponent(apiKey), method: 'POST' };
        case 'openai':
        default:
            return { url: base + '/chat/completions', method: 'POST' };
    }
}

function buildNonStreamingEndpoint(platform, modelId, apiKey) {
    const base = platform.baseUrl;
    switch (platform.apiFormat) {
        case 'google':
            return { url: base + '/models/' + modelId + ':generateContent?key=' +
encodeURIComponent(apiKey), method: 'POST' };
        default:
            return buildChatEndpoint(platform, modelId, apiKey);
    }
}

// Anthropic 消息格式转换
function _toAnthropicMessages(messages) {
    const system = messages.find(m =&gt; m.role === 'system');
    const others = messages.filter(m =&gt; m.role !== 'system');
</div><div class="page-number">第 9 页</div><div class="separator"></div><div class="code-
block">    const result = { messages: [] };
    if (system) result.system = system.content;

    let i = 0;
    while (i &lt; others.length) {
        const m = others[i];

        if (m.role === 'tool') {
            const toolResults = [];
            while (i &lt; others.length &amp;&amp; others[i].role === 'tool') {
                const toolContent = typeof others[i].content === 'string' ?
others[i].content : '';
                toolResults.push({ type: 'tool_result', tool_use_id: others[i].tool_call_id
|| 'unknown', content: toolContent });
                i++;
            }
            if (toolResults.length &gt; 0) {
                result.messages.push({ role: 'user', content: toolResults });
            }
            continue;
        }

        const content = [];
        if (m.content) content.push({ type: 'text', text: m.content });
        if (m.tool_calls) {
            for (const tc of m.tool_calls) {
                let input = {};
                try { input = JSON.parse(tc.function.arguments); } catch (e) {}
                content.push({ type: 'tool_use', id: tc.id, name: tc.function.name, input
});
```

```
            }
        }

        if (content.length &gt; 0) {
            result.messages.push({ role: m.role === 'assistant' ? 'assistant' : 'user',
content });
        }
        i++;
    }

    if (result.messages.length === 0 &amp;&amp; others.length &gt; 0) {
        result.messages = others.map(m =&gt; ({ role: m.role === 'assistant' ? 'assistant'
: 'user', content: m.content || '...' }));
    }
    return result;
}

function _toAnthropicTools(tools) {
    if (!tools || !tools.length) return undefined;
    return tools.map(t =&gt; ({ name: t.function.name, description: t.function.description,
input_schema: t.function.parameters }));
}

// Google Gemini 消息格式转换
function _toGoogleMessages(messages) {
    const systemMsg = messages.find(m =&gt; m.role === 'system');
</div><div class="page-number">第 10 页</div><div class="separator"></div><div class="code-
block">    const nonSystem = messages.filter(m =&gt; m.role !== 'system');
    const result = { contents: [] };
    if (systemMsg) result.system_instruction = { parts: [{ text: systemMsg.content }] };

    for (const m of nonSystem) {
        const parts = [];
        if (m.content) parts.push({ text: m.content });
        if (m.tool_calls) {
            for (const tc of m.tool_calls) {
                let args = {};
                try { args = JSON.parse(tc.function.arguments); } catch (e) {}
                parts.push({ functionCall: { name: tc.function.name, args } });
            }
        }
        if (m.role === 'tool') {
            try {
                const parsed = JSON.parse(m.content);
                parts.push({ functionResponse: { name: m.name || m.tool_name || '',
response: parsed } });
            } catch (e) {
                parts.push({ functionResponse: { name: m.name || m.tool_name || '',
response: { result: m.content || '' } } });
            }
        }
        if (parts.length &gt; 0) {
            result.contents.push({
                role: m.role === 'assistant' ? 'model' : (m.role === 'tool' ? 'function' :
'user'),
                parts
            });
```

```
        }
    }
    return result;
}

function _toGoogleTools(tools) {
    if (!tools || !tools.length) return undefined;
    return [{ functionDeclarations: tools.map(t =&gt; ({ name: t.function.name,
description: t.function.description, parameters: t.function.parameters })) }];
}

// ============================================================================
// Tool 定义
// ============================================================================

const AI_TOOLS = [
    { type: 'function', function: { name: 'bash', description: 'Execute a shell command.
Supports any CLI tool (git, npm, node, python, pip, etc.). For long-running processes (dev
servers, watchers), use background=true. Use manage_processes(action="list") to see running
processes, manage_processes(action="output", pid=N) to read output,
manage_processes(action="stop", pid=N) to stop.', parameters: { type: 'object', properties:
{ command: { type: 'string', description: 'The command to execute' }, cwd: { type:
'string', description: 'Working directory (absolute path). Defaults to previous cwd or
project root.' }, timeout: { type: 'number', description: 'Timeout in seconds (default 120,
max 600). For long-running commands use background=true instead.' }, background: { type:
'boolean', description: 'Run in background (for dev servers, watchers). Returns immediately
with process info. Use bash(command="taskkill /PID &lt;pid&gt; /F") to stop.' }, restart: {
type: 'boolean', description: 'Reset shell session state (cwd, env)' } }, required:
['command'] } } },
    { type: 'function', function: { name: 'str_replace_based_edit_tool', description: 'File
editing tool: view, create, and edit files. Commands: view (read file), create (new file),
str_replace (exact string replacement), insert (insert at line). create cannot be used on
existing files.', parameters: { type: 'object', properties: { command: { type: 'string',
enum: ['view', 'create', 'str_replace', 'insert'], description: 'Command to execute' },
path: { type: 'string', description: 'Absolute path to file or directory' }, file_text: {
type: 'string', description: 'Required for create: file content' }, old_str: { type:
'string', description: 'Required for str_replace: exact string to replace (must be unique)'
}, new_str: { type: 'string', description: 'New string for str_replace/insert' },
insert_line: { type: 'integer', description: 'Required for insert: line number to insert
after' }, view_range: { type: 'array', items: { type: 'integer' }, description: 'Optional
for view: line range [start, end]' } }, required: ['command', 'path'] } } },
    { type: 'function', function: { name: 'json_edit_tool', description: 'JSON file editing
tool using JSONPath expressions. Operations: view, set, add, remove. Examples:
$.users[0].name, $.config.database', parameters: { type: 'object', properties: { operation:
{ type: 'string', enum: ['view', 'set', 'add', 'remove'], description: 'Operation to
perform' }, file_path: { type: 'string', description: 'Absolute path to JSON file' },
json_path: { type: 'string', description: 'JSONPath expression' }, value: { type: 'object',
description: 'Value to set or add (required for set/add)' }, pretty_print: { type:
'boolean', description: 'Format output (default true)' } }, required: ['operation',
'file_path'] } } },
    { type: 'function', function: { name: 'sequential_thinking', description: 'Break down
complex problems into sequential thinking steps. Each step produces a conclusion. Supports
revising previous steps. Use when deep analysis is needed.', parameters: { type: 'object',
properties: { thought: { type: 'string', description: 'Thinking content for current step'
}, thought_number: { type: 'number', description: 'Current step number' }, total_thoughts:
{ type: 'number', description: 'Estimated total steps' }, next_thought_needed: { type:
'boolean', description: 'Whether another step is needed' }, is_revision: { type: 'boolean',
```

```
description: 'Whether revising a previous step' }, revises_thought: { type: 'number',
description: 'Step number being revised (when is_revision=true)' }, branch_from_thought: {
type: 'number', description: 'Branch from this step (optional)' }, branch_id: { type:
'string', description: 'Branch identifier (optional)' } }, required: ['thought',
'thought_number', 'total_thoughts', 'next_thought_needed'] } } },
    { type: 'function', function: { name: 'attempt_completion', description: 'Report task
completion. Only call after verifying the task is done. The result will be presented to the
user for confirmation.', parameters: { type: 'object', properties: { result: { type:
'string', description: 'Final result message with summary of completed work' } }, required:
['result'] } } },
    { type: 'function', function: { name: 'ckg', description: 'Code Knowledge Graph: search
for functions, classes, and class methods in the codebase.', parameters: { type: 'object',
properties: { command: { type: 'string', enum: ['search_function', 'search_class',
'search_class_method'], description: 'Search command' }, path: { type: 'string',
description: 'Codebase path' }, identifier: { type: 'string', description:
'Function/class/method name to search' }, print_body: { type: 'boolean', description:
'Print function/class body (default true)' } }, required: ['command', 'path', 'identifier']
} } },
    { type: 'function', function: { name: 'update_todo_list', description: 'Create or
update a task plan. MUST be called as the FIRST action for any task requiring tools.
Decomposes the user request into concrete, actionable tasks and tracks progress. Format: [
] pending, [-] in progress, [x] completed.', parameters: { type: 'object', properties: {
todos: { type: 'string', description: 'Complete task list in Markdown format. Example:\n- [
] Task 1: Analyze code structure\n- [-] Task 2: Modify CSS styles (currently executing)\n-
[x] Task 3: Bug fix completed' } }, required: ['todos'] } } },
    {
</div><div class="page-number">第 11 页</div><div class="separator"></div><div class="code-
block">        type: 'function',
        function: {
            name: 'sub_agent_dispatch',
            description: '派遣子代理执行特定任务。file_search: 在代码库中搜索文件和目录,
code_analysis: 分析代码结构和调用链, resource_download: 搜索Minecraft资源, crash_analysis: 分
析崩溃日志, code_completion: 代码补全和优化, auto: 自动选择最合适的子代理类型',
            parameters: {
                type: 'object',
                properties: {
                    agent_type: {
                        type: 'string',
                        enum: ['file_search', 'code_analysis', 'resource_download',
'crash_analysis', 'code_completion', 'auto'],
                        description: '子代理类型。使用 auto 可自动根据任务描述选择最合适的子代
理'
                    },
                    task: {
                        type: 'string',
                        description: '子代理要执行的具体任务描述'
                    }
                },
                required: ['agent_type', 'task']
            }
        }
    },
    { type: 'function', function: { name: 'start_preview', description: 'Start a local HTTP
server to preview web files (HTML/CSS/JS). Opens a browser preview panel in the UI. Use
this when the user wants to see how their web code looks. Use command="stop" to stop the
server.', parameters: { type: 'object', properties: { command: { type: 'string', enum:
['start', 'stop'], description: 'start to begin preview, stop to end it' }, root: { type:
```

```
'string', description: 'Absolute path to the directory to serve (required for start)' },
port: { type: 'number', description: 'Port number (default 8080)' } }, required:
['command'] } } },
    { type: 'function', function: { name: 'manage_processes', description: 'List and manage
background processes started by bash. Use to check dev server output, list running
processes, or stop processes.', parameters: { type: 'object', properties: { action: { type:
'string', enum: ['list', 'output', 'stop', 'stop_all'], description: 'list: show running
processes. output: get stdout/stderr of a process. stop: kill a process by PID. stop_all:
kill all background processes.' }, pid: { type: 'number', description: 'Process ID
(required for output and stop actions)' }, tail: { type: 'number', description: 'Number of
last lines to return for output (default 50)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'ask_user', description: 'Ask the user a question
to get clarification or make a decision. Use when you need user input to proceed. Supports
free text or multiple choice.', parameters: { type: 'object', properties: { question: {
type: 'string', description: 'The question to ask the user' }, options: { type: 'array',
items: { type: 'string' }, description: 'Optional multiple choice options. If provided,
user can select one. If omitted, user can type free text.' }, context: { type: 'string',
description: 'Optional context or explanation for why you are asking' } }, required:
['question'] } } },
    { type: 'function', function: { name: 'undo_edit', description: 'Manage file edit
backups. List recent AI file changes, view diffs, and restore previous versions. Use
action="list" to see history, action="restore" to rollback, action="diff" to compare
versions.', parameters: { type: 'object', properties: { action: { type: 'string', enum:
['list', 'restore', 'diff', 'session'], description: 'list: show backup history. restore:
rollback to a backup. diff: compare backup with current. session: show session summary.' },
backup_id: { type: 'string', description: 'Backup ID (required for restore and diff
actions)' }, file_path: { type: 'string', description: 'Filter backups by file path
(optional for list)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'view_history', description: 'View change history
and audit log. Track all AI file modifications, tool executions, and session activity.',
parameters: { type: 'object', properties: { action: { type: 'string', enum: ['changes',
'audit', 'summary'], description: 'changes: file modification history. audit: tool
execution log. summary: session statistics.' }, file_path: { type: 'string', description:
'Filter changes by file path (optional)' }, tool_name: { type: 'string', description:
'Filter by tool name (optional)' }, limit: { type: 'number', description: 'Max results to
return (default 20)' } }, required: ['action'] } } },
    { type: 'function', function: { name: 'validate_code', description: 'Validate code
syntax before writing. Checks JS/TS/JSON/CSS/HTML for syntax errors. Use before large file
writes to catch errors early.', parameters: { type: 'object', properties: { content: {
type: 'string', description: 'The code content to validate' }, file_path: { type: 'string',
description: 'File path (used to determine language for validation)' } }, required:
['content', 'file_path'] } } },
    { type: 'function', function: { name: 'build_index', description: 'Build a searchable
index of the codebase. Must be called before semantic_search. Indexes all code files in the
directory for fast semantic search.', parameters: { type: 'object', properties: { root_dir:
{ type: 'string', description: 'Absolute path to the root directory to index' } },
required: ['root_dir'] } } },
    { type: 'function', function: { name: 'semantic_search', description: 'Search the
codebase using natural language queries. Finds files by meaning, not just keywords.
Requires build_index to be called first. Use for queries like "find authentication code",
"where is the database connection", "error handling logic".', parameters: { type: 'object',
properties: { query: { type: 'string', description: 'Natural language search query
describing what you are looking for' }, root_dir: { type: 'string', description: 'Limit
search to this directory (optional)' }, max_results: { type: 'number', description:
'Maximum results to return (default 10)' } }, required: ['query'] } } },
    { type: 'function', function: { name: 'index_stats', description: 'Get statistics about
the current code index (number of files, tokens, memory usage).', parameters: { type:
```

```
'object', properties: {} } } },
];

let _pluginManager = null;
function _getPluginTools() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getTools();
    } catch (e) { return []; }
}

function _getPluginDisplayNames() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getToolDisplayNames();
    } catch (e) { return {}; }
}

function _getPluginRisks() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
</div><div class="page-number">第 12 页</div><div class="separator"></div>
```

```
<div class="code-block">        return _pluginManager.getToolRisks();
    } catch (e) { return {}; }
}

function _getPluginPromptExtensions() {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.getPromptExtensions();
    } catch (e) { return []; }
}

function _isPluginTool(name) {
    try {
        if (!_pluginManager) _pluginManager = getPluginManager();
        return _pluginManager.isPluginTool(name);
    } catch (e) { return false; }
}

function _getAllTools() {
    return [...AI_TOOLS, ..._getPluginTools()];
}

const toolDescriptions = AI_TOOLS.map(t =&gt; `- ${t.function.name}:
${t.function.description}`).join('\n');

const TOOL_RISK = {
    str_replace_based_edit_tool: 'safe', json_edit_tool: 'safe', ckg: 'safe',
    sequential_thinking: 'safe', attempt_completion: 'safe',
    bash: 'safe', update_todo_list: 'safe',
    sub_agent_dispatch: 'safe', start_preview: 'safe', manage_processes: 'safe',
    ask_user: 'safe',
    undo_edit: 'safe',
    view_history: 'safe',
    validate_code: 'safe',
    build_index: 'safe',
    semantic_search: 'safe',
    index_stats: 'safe'
};

const TOOL_DISPLAY_NAMES = {
    bash: '执行命令', str_replace_based_edit_tool: '编辑文件',
    json_edit_tool: '编辑JSON', sequential_thinking: '分步思考',
    attempt_completion: '完成任务', ckg: '代码图谱',
    update_todo_list: '更新计划',
    sub_agent_dispatch: '派遣子代理',
    start_preview: '启动预览',
    manage_processes: '进程管理',
    ask_user: '询问用户',
    undo_edit: '撤销编辑',
    view_history: '查看历史',
    validate_code: '验证代码',
</div><div class="page-number">第 13 页</div><div class="separator"></div><div class="code-
block">    build_index: '构建索引',
    semantic_search: '语义搜索',
    index_stats: '索引统计'
};
```

```
const AGENT_META = {
    file_search: {
        name: '文件搜索代理', role: 'File Search', avatar: 'robot', color: '#4caf50',
        systemPrompt: `你是 VersePC 项目的文件搜索代理。你的任务是在代码库中快速定位文件和目录。

## 工作流程
1. 首先理解搜索目标（文件名、关键词、文件类型等）
2. 使用 bash 工具执行搜索命令：
   - 按文件名搜索：find . -name "pattern" -type f
   - 按内容搜索：grep -r "pattern" --include="*.js" --include="*.css" --include="*.html" -l
   - 查看目录结构：ls -la，tree（限制深度）
   - 查看文件信息：wc -l，file，stat
3. 对搜索结果进行筛选和排序
4. 输出结构化结果

## 输出格式
搜索完成后，用以下格式总结：
- 搜索目标：xxx
- 找到 N 个文件
- 关键文件列表（路径 + 用途说明）
- 相关代码片段（如有）

## 注意事项
- 优先搜索项目根目录下的 js/、css/、agent-engine.js 等核心文件
- 搜索时排除 node_modules、dist、.git 目录
- 如果搜索结果过多，按相关性排序，只展示最相关的
- 用中文输出结果`
    },
    code_analysis: {
        name: '代码分析代理', role: 'Code Analysis', avatar: 'robot', color: '#9c27b0',
        systemPrompt: `你是 VersePC 项目的代码分析代理。你的任务是分析代码结构、理解项目架构、追踪函数调用链。

## 工作流程
1. 首先确定分析目标（文件、函数、模块）
2. 使用 bash 工具阅读代码：
   - cat 查看文件内容
   - grep 搜索函数调用
   - find 查找相关文件
3. 分析代码结构和依赖关系
4. 输出分析报告

## 分析维度
- 文件职责：该文件的主要功能
- 依赖关系：依赖了哪些模块，被哪些模块依赖
- 函数调用链：关键函数的调用路径
</div><div class="page-number">第 14 页</div><div class="separator"></div><div class="code-block">- 数据流：数据如何在模块间传递
- 潜在问题：发现的 bug、性能问题、代码异味

## 输出格式
分析完成后，用以下格式总结：
- 分析目标：xxx
- 文件结构：列出相关文件及其职责
- 核心逻辑：关键函数和调用链
- 发现的问题（如有）：问题描述 + 建议修复方案
```

```
## 注意事项
- 用中文输出分析结果
- 代码片段保留原始格式
- 行号引用格式：文件名:行号`
    },
    resource_download: {
        name: '资源搜索代理', role: 'Resource Search', avatar: 'robot', color: '#8d6e63',
        systemPrompt: `你是 Minecraft 资源搜索代理。你的任务是搜索和推荐 Minecraft 相关资源。

## 支持的资源类型
- Mod（模组）
- 整合包（Modpack）
- 材质包（Texture Pack）
- 光影包（Shader Pack）

## 工作流程
1. 理解用户需求（版本、类型、功能偏好）
2. 搜索 CurseForge 和 Modrinth 上的资源
3. 筛选和推荐

## 推荐标准
- 版本兼容性：优先推荐与用户 Minecraft 版本兼容的资源
- 稳定性：优先推荐更新频繁、bug 少的资源
- 下载量和评分：作为参考指标
- 兼容性：推荐的资源之间不要有冲突

## 输出格式
- 资源名称 + 简介
- 版本兼容信息
- 下载量/评分
- 安装建议
- 注意事项（如有）

## 注意事项
- 用中文输出推荐结果
- 如果资源需要前置依赖，一并说明`
    },
    crash_analysis: {
        name: '崩溃分析代理', role: 'Crash Analysis', avatar: 'robot', color: '#9e9e9e',
        systemPrompt: `你是 Minecraft 崩溃分析代理。你的任务是分析崩溃日志，定位错误原因并提供修复建议。
</div><div class="page-number">第 15 页</div><div class="separator"></div><div class="code-block">
## 工作流程
1. 定位日志文件（crash-reports/、logs/ 目录）
2. 使用 bash 工具读取和分析日志
3. 识别崩溃类型和原因
4. 提供修复建议

## 崩溃类型识别
- Mod 冲突：检查多个 Mod 的兼容性
- 内存不足：检查 JVM 参数和内存分配
- 配置错误：检查配置文件格式和值
- 版本不兼容：检查 Mod 与 Minecraft 版本的兼容性
- 驱动问题：检查显卡驱动版本
- Java 版本：检查 Java 版本是否匹配
```

```
## 输出格式
- 崩溃类型：xxx
- 错误原因：详细描述
- 涉及文件：相关日志文件和配置文件
- 修复步骤：按优先级排列的修复方案
- 预防建议：避免再次发生的方法

## 注意事项
- 用中文输出分析结果
- 引用关键日志行（文件:行号）
- 修复步骤要具体可操作`
    },
    code_completion: {
        name: '代码补全代理', role: 'Code Completion', avatar: 'robot', color: '#00bcd4',
        systemPrompt: `你是 VersePC 项目的代码补全代理。你的任务是代码重写、优化、补全和重构。

## 工作流程
1. 阅读并理解目标代码的上下文和意图
2. 分析代码结构、变量作用域、依赖关系
3. 生成高质量的补全/优化代码
4. 验证生成的代码语法正确性

## 支持的任务类型
- 代码补全：根据上下文补全未完成的代码
- 代码重写：重构现有代码以提升可读性或性能
- 代码优化：优化算法、减少冗余、提升效率
- 模式匹配：按照项目现有代码风格生成新代码

## 输出格式
- 原始代码（简要引用）
- 修改后的完整代码
- 修改说明：改了什么、为什么改
- 影响范围：可能受影响的文件和函数

## 注意事项
</div><div class="page-number">第 16 页</div><div class="separator"></div><div class="code-block">- 用中文输出说明，代码本身保持原语言
- 保持与项目现有代码风格一致
- 不要引入项目中未使用的新依赖
- 代码片段保留原始缩进格式`
    }
};



// ============================================================================
// HTTP API 工具
// ============================================================================

function makeApiStreamRequest(apiUrl, bodyStr, headers) {
    const options = {
        hostname: apiUrl.hostname,
        path: apiUrl.pathname + (apiUrl.search || ''),
        method: 'POST',
        headers: { ...headers, 'Content-Length': Buffer.byteLength(bodyStr), 'Connection':
'close' },
```

```
        agent: false
    };
    const proto = apiUrl.protocol === 'https:' ? https : http;
    return new Promise((resolve, reject) =&gt; {
        const req = proto.request(options, (res) =&gt; {
            if (res.statusCode &gt;= 400) {
                let errData = '';
                res.on('data', chunk =&gt; errData += chunk);
                res.on('end', () =&gt; {
                    let errMsg = `HTTP ${res.statusCode}`;
                    try {
                        const parsed = JSON.parse(errData);
                        errMsg = parsed.error?.message || parsed.error?.code ||
parsed.message || errMsg;
                    } catch (e) {}
                    reject(new Error(errMsg));
                });
                return;
            }
            resolve(res);
        });
        req.on('error', reject);
        req.setTimeout(60000, () =&gt; { req.destroy(); reject(new Error('API连接超时
(60s)')); });
        req.write(bodyStr);
        req.end();
    });
}

function makeApiRequest(apiUrl, bodyStr, headers) {
    const options = {
        hostname: apiUrl.hostname,
        path: apiUrl.pathname + (apiUrl.search || ''),
</div><div class="page-number">第 17 页</div><div class="separator"></div><div class="code-
block">        method: 'POST',
        headers: { ...headers, 'Content-Length': Buffer.byteLength(bodyStr), 'Connection':
'close' },
        agent: false
    };
    const proto = apiUrl.protocol === 'https:' ? https : http;
    return new Promise((resolve, reject) =&gt; {
        const req = proto.request(options, (res) =&gt; {
            let data = '';
            res.on('data', chunk =&gt; data += chunk);
            res.on('end', () =&gt; {
                if (res.statusCode &gt;= 400) {
                    let errMsg = `HTTP ${res.statusCode}`;
                    try {
                        const parsed = JSON.parse(data);
                        errMsg = parsed.error?.message || parsed.error?.code ||
parsed.message || errMsg;
                    } catch (e) {}
                    reject(new Error(errMsg));
                    return;
                }
                try {
                    const parsed = JSON.parse(data);
```

```
                    if (parsed.error) {
                        reject(new Error(parsed.error.message || parsed.error.code ||
JSON.stringify(parsed.error)));
                        return;
                    }
                    resolve(parsed.choices?.[0]?.message?.content || '');
                } catch (e) {
                    reject(e);
                }
            });
        });
        req.on('error', reject);
        req.setTimeout(30000, () =&gt; { req.destroy(); reject(new Error('API请求超时
(30s)')); });
        req.write(bodyStr);
        req.end();
    });
}

// ============================================================================
// Agent Engine (全功能)
// ============================================================================

class AgentEngine {
    constructor(options = {}) {
        this.onChunk = options.onChunk || (() =&gt; {});
        this.onRequestApproval = options.onRequestApproval || null;
        this.onAskUser = options.onAskUser || null;
        this.executeTool = options.executeTool || null;
        this.output = new OutputManager();

</div><div class="page-number">第 18 页</div><div class="separator"></div><div class="code-
block">        this.enablePlanning = options.enablePlanning !== false;
        this.enableReflection = options.enableReflection !== false;
        this.enableStuckDetection = options.enableStuckDetection !== false;
        this.enablePassiveDetection = options.enablePassiveDetection !== false;
        this.maxRounds = options.maxRounds || 24;
        this.maxConsecutiveFailures = options.maxConsecutiveFailures || 3;
        this.maxPassiveDetections = options.maxPassiveDetections || 2;
        this.maxRepeatText = options.maxRepeatText || 2;

        this._aborted = false;
        this._actionHistory = [];
        this._lastTextContent = '';
        this._repeatTextCount = 0;
        this._consecutiveFailures = 0;
        this._consecutiveMistakes = 0;
        this._passiveDetectionCount = 0;
        this._activePlan = null;
        this._planStepResults = {};
        this._provider = options.provider || null;
        this._apiUrl = options.apiUrl || null;
        this._apiHeaders = options.apiHeaders || null;
        this._model = options.model || null;
    }

    abort() {
```

```
        this._aborted = true;
    }

    _send(msg) {
        const passthroughTypes = ['subagent_start', 'subagent_chunk', 'subagent_end'];
        if (passthroughTypes.includes(msg.type)) {
            this.onChunk(msg);
            return;
        }
        const processed = this.output.processMessage(msg);
        if (processed) {
            this.onChunk(processed);
        }
    }

    _initReasoningThrottle() {
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        this._reasoningBuffer = null;
        this._reasoningStarted = false;
    }

    _sendReasoningDelta(delta, fullReasoning) {
</div><div class="page-number">第 19 页</div><div class="separator"></div><div class="code-
block">        if (!this._reasoningStarted) {
            this._reasoningStarted = true;
            this.onChunk({ type: 'reasoning_start', content: '' });
        }
        this._reasoningBuffer = (this._reasoningBuffer || '') + delta;
        if (!this._reasoningFlushTimer) {
            this._reasoningFlushTimer = setTimeout(() =&gt; {
                this._reasoningFlushTimer = null;
                if (this._reasoningBuffer) {
                    const buf = this._reasoningBuffer;
                    this._reasoningBuffer = null;
                    this.onChunk({ type: 'reasoning_content', content: buf, partial: true
});
                }
            }, 80);
        }
    }

    _flushReasoningThrottle() {
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        if (this._reasoningStarted &amp;amp;&amp;amp; this._reasoningBuffer) {
            const buf = this._reasoningBuffer;
            this._reasoningBuffer = null;
            this.onChunk({ type: 'reasoning_content', content: buf, partial: true });
        }
    }

    _finishReasoningThrottle() {
```

```
        if (this._reasoningFlushTimer) {
            clearTimeout(this._reasoningFlushTimer);
            this._reasoningFlushTimer = null;
        }
        this._reasoningBuffer = null;
        this._reasoningStarted = false;
    }

    /**
     * 主入口：处理聊天
     */
    async processChat({ apiKey, model, messages, temperature, enableTools, apiFormat:
customApiFormat, baseUrl: customBaseUrl, language, maxRounds }) {
        this._aborted = false;
        this.output.clear();
        this._actionHistory = [];
        this._lastTextContent = '';
        this._repeatTextCount = 0;
        this._consecutiveFailures = 0;
        this._consecutiveMistakes = 0;
        this._passiveDetectionCount = 0;
</div><div class="page-number">第 20 页</div><div class="separator"></div><div class="code-
block">        this._activePlan = null;
        this._planStepResults = {};
        this._thinkingSteps = [];
        this._apiKey = apiKey;
        this._model = model || 'glm-5-flash';

        if (!apiKey) {
            this._send({ type: 'say', say: SayType.ERROR, text: '未配置 API Key，请在设置中填
写' });
            return;
        }
        if (maxRounds) this.maxRounds = maxRounds;

        let apiFormat;
        let requestModel = this._model;
        if (this._apiUrl &amp;&amp; this._apiHeaders &amp;&amp; !customBaseUrl) {
            apiFormat = this._provider?.apiFormat || 'openai';
        } else if (customBaseUrl &amp;&amp; customApiFormat) {
            const authType = customApiFormat === 'anthropic' ? 'x-api-key' : 'bearer';
            this._provider = { name: 'Custom', baseUrl: customBaseUrl, authType, apiFormat:
customApiFormat, thinkingParams: {} };
            const endpoint = buildChatEndpoint(this._provider, this._model, apiKey);
            this._apiUrl = new URL(endpoint.url);
            this._apiHeaders = buildApiHeaders(this._provider, apiKey);
            apiFormat = customApiFormat;
            const customMatch = this._model.match(/^custom:(https?:\/\/.+):(.+)$/);
            if (customMatch) requestModel = customMatch[2];
        } else {
            this._provider = getProviderForModel(this._model);
            const endpoint = buildChatEndpoint(this._provider, this._model, apiKey);
            this._apiUrl = new URL(endpoint.url);
            this._apiHeaders = buildApiHeaders(this._provider, apiKey);
            apiFormat = this._provider.apiFormat || 'openai';
        }
        this._requestModel = requestModel;
```

```
        const tools = enableTools !== false ? _getAllTools() : undefined;

        let conversation = [...messages];        this.conversation = conversation;
        const lastUserMsg = [...messages].reverse().find(m =&gt; m.role === 'user');

        const contextSummary = this._buildContextSummary();
        if (contextSummary) conversation.push({ role: 'system', content: contextSummary });

        conversation.push({ role: 'system', content: `OS: Windows. You can use both CMD and
Unix-style commands (git, npm, node, python, pip, npx, etc. all work in the bash tool).
Common commands: dir/ls, type/cat, findstr/grep, copy/cp, move/mv, del/rm, mkdir. Use bash
tool for all CLI operations including git, npm, python, testing, and formatting.` });

        const pluginPrompts = _getPluginPromptExtensions();
        if (pluginPrompts.length &gt; 0) {
            conversation.push({ role: 'system', content: '## Plugin Capabilities\n\n' +
pluginPrompts.join('\n\n') });
        }

        conversation.push({
</div><div class="page-number">第 21 页</div><div class="separator"></div><div class="code-
block">            role: 'system',
            content: `## 全栈开发能力

你具备完整的全栈开发能力。所有 CLI 操作通过 bash 工具执行。

### 文件操作
- 查看文件：str_replace_based_edit_tool(command="view", path="绝对路径")
- 创建文件：str_replace_based_edit_tool(command="create", path="绝对路径", file_text="内容")
- 编辑文件：str_replace_based_edit_tool(command="str_replace", path="绝对路径", old_str="原
文", new_str="新文")
- 写入/覆盖文件：str_replace_based_edit_tool(command="create", ...) 或通过 bash 使用 echo/重定
向
- 搜索文件：bash(command="dir /s /b *.js") 或 bash(command="findstr /s /n \"pattern\" *.js")
- 搜索文件内容：bash(command="findstr /s /n \"keyword\" *.*")

### 版本控制 (Git)
- 查看状态：bash(command="git status")
- 查看差异：bash(command="git diff") 或 bash(command="git diff --cached")
- 提交更改：bash(command="git add . &amp;&amp; git commit -m \"message\"")
- 查看日志：bash(command="git log --oneline -20")
- 分支操作：bash(command="git branch") / bash(command="git checkout -b new-branch")
- 合并分支：bash(command="git merge branch-name")

### 包管理
- npm: bash(command="npm install package-name") / bash(command="npm run build") /
bash(command="npm test")
- pip: bash(command="pip install package-name") / bash(command="pip list")
- npx: bash(command="npx create-react-app my-app")

### 开发服务器
- 启动开发服务器（后台运行）：bash(command="npm run dev", background=true)
- 启动 HTTP 预览：start_preview(command="start", root="项目目录路径")
- 停止预览：start_preview(command="stop")

### 代码质量
- 运行测试：bash(command="npm test") 或 bash(command="npx jest --verbose")
```

- 代码检查：bash(command="npx eslint src/") 或 bash(command="npx eslint --fix src/")
- 代码格式化：bash(command="npx prettier --write \"src/**/*.{js,ts,css,json}\"")
- TypeScript 检查：bash(command="npx tsc --noEmit")

构建与部署
- 构建项目：bash(command="npm run build")
- 查看构建输出：bash(command="dir dist") 或 bash(command="ls dist")
- 启动生产服务器：bash(command="node server.js", background=true)

工作流最佳实践
1. 收到开发任务后，先用 str_replace_based_edit_tool(view) 了解项目结构
2. 使用 update_todo_list 创建任务计划
3. 逐步执行：编辑文件 → 运行测试 → 修复错误 → 提交代码
4. 对于需要长时间运行的服务（dev server、watcher），使用 background=true
5. 完成后用 attempt_completion 提交总结`
        });

第 22 页   if (this.enablePlanning && lastUserMsg && tools) {
            const intent = this._detectIntent(lastUserMsg.content);
            if (intent.intent === 'complex') {
                const stepHint = intent.steps_needed >= 3
                    ? `这是一个复杂的多步骤任务（检测到 ${intent.steps_needed}+ 个步骤）。`
                    : '这是一个需要多步骤执行的任务。';
                conversation.push({
                    role: 'system',
                    content: `${stepHint}

任务工作流

请使用 update_todo_list 工具创建任务计划，将用户的请求分解为具体的、可执行的任务。

执行流程
1. 调用 update_todo_list 创建任务列表
   - [ ] 任务1：具体描述
   - [ ] 任务2：具体描述
   - [ ] 任务3：具体描述
2. 逐个执行任务：
   a. 将当前任务标记为 [-] 进行中（调用 update_todo_list）
   b. 执行该任务所需的所有工具调用
   c. 任务完成后，将任务标记为 [x] 已完成（调用 update_todo_list）
3. 所有任务完成后，调用 attempt_completion 提交最终总结

关键规则
- 每个任务应该是自包含的，将相关的工具调用归组到一起
- attempt_completion 的 result 字段必须包含清晰的中文总结
- 如果某个任务失败，标记为 [x] 并注明错误，然后继续下一个任务
- 永远不要在回复中使用 emoji

子代理派遣规则
当你需要搜索文件、分析代码、搜索资源或分析崩溃日志时，你必须调用 sub_agent_dispatch 工具，而不是自己执行这些任务。
- 搜索文件/目录 → sub_agent_dispatch(agent_type="file_search", task="具体任务描述")
- 分析代码结构 → sub_agent_dispatch(agent_type="code_analysis", task="具体任务描述")
- 搜索Minecraft资源 → sub_agent_dispatch(agent_type="resource_download", task="具体任务描述")
- 分析崩溃日志 → sub_agent_dispatch(agent_type="crash_analysis", task="具体任务描述")
- 代码补全/优化 → sub_agent_dispatch(agent_type="code_completion", task="具体任务描述")

```
**推荐：使用 agent_type="auto" 可自动选择最合适的子代理类型**，系统会根据任务描述智能匹配。例如：
- sub_agent_dispatch(agent_type="auto", task="在项目中搜索所有配置文件")
- sub_agent_dispatch(agent_type="auto", task="分析这个崩溃日志的错误原因")

绝对不要在文本中写"[执行] 调用子代理"，而是必须实际调用 sub_agent_dispatch 工具。

### 子代理使用场景指南
- **复杂搜索任务** → file_search：查找文件、搜索代码、定位资源
- **代码理解任务** → code_analysis：分析函数调用链、理解模块依赖、代码审查
- **Minecraft资源任务** → resource_download：搜索模组、整合包、材质包、光影包
- **崩溃诊断任务** → crash_analysis：分析崩溃日志、定位错误原因、提供修复建议
</div><div class="page-number">第 23 页</div><div class="separator"></div><div class="code-block">- **代码补全任务** → code_completion：代码重写、优化、补全建议

**最佳实践**：对于大多数任务，推荐使用 agent_type="auto"，系统会根据任务描述自动选择最合适的子代理类型。`
                });
            } else {
                conversation.push({
                    role: 'system',
                    content: `## 执行指南

这是一个简短的问候或确认，直接回复即可。

### 执行流程
1. 直接回复用户
2. 如果需要执行操作，使用工具完成

### 关键规则
- 用中文回复
- 如果执行了操作，调用 attempt_completion 提交结果总结
- 永远不要在回复中使用 emoji`
                });
            }
        }

        if (language &amp;&amp; language !== 'en') {
            const _langNames = { 'zh-CN': '简体中文', 'zh-TW': '繁體中文', 'ja': '日本語',
'ko': '한국어' };
            const _langName = _langNames[language] || '简体中文';
            conversation.push({
                role: 'system',
                content: `FINAL REMINDER — Language: You MUST respond entirely in
${_langName}. All explanations, todo descriptions, plan text, and completion summaries must
be in ${_langName}. Only code, file paths, and technical identifiers stay in English.`
            });
        }

        let totalUsage = { prompt_tokens: 0, completion_tokens: 0, total_tokens: 0, rounds:
0 };        let toolResults = [];        for (let round = 0; round &lt; this.maxRounds;
round++) {
            if (this._aborted) break;
            await new Promise(resolve =&gt; setImmediate(resolve));

            if (apiFormat === 'openai') {
                const toRemove = new Set();
```

```
                for (let i = 0; i &lt; conversation.length; i++) {
                    const msg = conversation[i];
                    if (msg.role === 'assistant' &amp;&amp; msg.tool_calls &amp;&amp;
msg.tool_calls.length &gt; 0) {
                        const expectedIds = new Set(msg.tool_calls.map(tc =&gt; tc.id));
                        const foundIds = new Set();
                        for (let j = i + 1; j &lt; conversation.length; j++) {
                            const next = conversation[j];
                            if (next.role === 'assistant') break;
                            if (next.role === 'tool' &amp;&amp;
expectedIds.has(next.tool_call_id)) {
</div><div class="page-number">第 24 页</div><div class="separator"></div><div class="code-
block">                                foundIds.add(next.tool_call_id);
                            }
                        }
                        if (foundIds.size &lt; expectedIds.size) {
                            toRemove.add(i);
                            for (let j = i + 1; j &lt; conversation.length; j++) {
                                if (conversation[j].role === 'assistant') break;
                                if (conversation[j].role === 'tool') toRemove.add(j);
                            }
                        }
                    }
                }
                for (let i = conversation.length - 1; i &gt;= 0; i--) {
                    if (toRemove.has(i)) conversation.splice(i, 1);
                }
                for (let i = conversation.length - 1; i &gt;= 0; i--) {
                    if (conversation[i].role === 'tool') {
                        let hasAssistantBefore = false;
                        for (let j = i - 1; j &gt;= 0; j--) {
                            if (conversation[j].role === 'assistant') {
                                hasAssistantBefore = conversation[j].tool_calls &amp;&amp;
conversation[j].tool_calls.length &gt; 0;
                                break;
                            }
                        }
                        if (!hasAssistantBefore) conversation.splice(i, 1);
                    }
                }
            }

            let bodyStr;
            const hasTools = tools &amp;&amp; tools.length &gt; 0;

            if (apiFormat === 'anthropic') {
                const anthropicMessages = _toAnthropicMessages(conversation);
                const reqBody = {
                    model: this._requestModel,
                    max_tokens: 8192,
                    stream: true,
                    ...(anthropicMessages.system ? { system: anthropicMessages.system } :
{}),
                    messages: anthropicMessages.messages
                };
                if (hasTools) reqBody.tools = _toAnthropicTools(tools);
                bodyStr = JSON.stringify(reqBody);
```

```
            } else if (apiFormat === 'google') {
                const googleMessages = _toGoogleMessages(conversation);
                const reqBody = {
                    ...(googleMessages.system_instruction ? { system_instruction:
googleMessages.system_instruction } : {}),
                    contents: googleMessages.contents,
                    generationConfig: { temperature: temperature != null ? temperature :
0.7 }
                };
</div><div class="page-number">第 25 页</div><div class="separator"></div><div class="code-
block">                if (hasTools) reqBody.tools = _toGoogleTools(tools);
                bodyStr = JSON.stringify(reqBody);
            } else {
                const reqBody = {
                    model: this._requestModel,
                    messages: conversation,
                    temperature: temperature != null ? temperature : 0.7,
                    stream: true,
                    stream_options: { include_usage: true }
                };
                if (hasTools) { reqBody.tools = tools; reqBody.tool_choice = 'auto'; }
                if (this._provider.thinkingParams &amp;&amp;
Object.keys(this._provider.thinkingParams).length &gt; 0) {
                    Object.assign(reqBody, this._provider.thinkingParams);
                }
                if (reqBody.enable_thinking === true) delete reqBody.temperature;
                bodyStr = JSON.stringify(reqBody);
            }

            let res;
            const MAX_RETRIES = 2;
            let lastError = null;
            for (let attempt = 0; attempt &lt;= MAX_RETRIES; attempt++) {
                try {
                    this._send({ type: 'say', say: SayType.API_STARTED, text: '' });
                    res = await makeApiStreamRequest(this._apiUrl, bodyStr,
this._apiHeaders);
                    lastError = null;
                    break;
                } catch (e) {
                    lastError = e;
                    const isRetryable = e.message &amp;&amp;
(e.message.includes('ECONNRESET') || e.message.includes('ECONNREFUSED') ||
e.message.includes('ETIMEDOUT') || e.message.includes('socket hang up'));
                    if (isRetryable &amp;&amp; attempt &lt; MAX_RETRIES) {
                        this._send({ type: 'say', say: SayType.HEARTBEAT, text: `连接中断，
正在重试 (${attempt + 1}/${MAX_RETRIES})...` });
                        await new Promise(r =&gt; setTimeout(r, 2000 * (attempt + 1)));
                        continue;
                    }
                    this._send({ type: 'say', say: SayType.ERROR, text: e.message });
                    this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
                    return;
                }
            }
            if (lastError) {
                this._send({ type: 'say', say: SayType.ERROR, text: lastError.message });
```

```
                this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
                return;
            }

            const roundData = await this._processStream(res, apiFormat);
            if (this._aborted) break;

            if (roundData.usage) {
</div><div class="page-number">第 26 页</div><div class="separator"></div><div class="code-
block">                totalUsage.prompt_tokens += roundData.usage.prompt_tokens || 0;
                totalUsage.completion_tokens += roundData.usage.completion_tokens || 0;
                totalUsage.total_tokens += roundData.usage.total_tokens || 0;
            }
            totalUsage.rounds++;

            if (roundData.toolCalls.length &gt; 0 &amp;&amp; roundData.finishReason ===
'tool_calls') {
                const assistantMsg = {
                    role: 'assistant',
                    content: roundData.fullContent || '',
                    tool_calls: roundData.toolCalls.map(tc =&gt; ({
                        id: tc.id, type: 'function',
                        function: { name: tc.name, arguments: tc.argsStr }
                    }))
                };
                if (this._provider.thinkingParams &amp;&amp;
this._provider.thinkingParams.enable_thinking) {
                    assistantMsg.reasoning_content = roundData.fullReasoning || '';
                } else if (roundData.fullReasoning) {
                    assistantMsg.reasoning_content = roundData.fullReasoning;
                }
                conversation.push(assistantMsg);

                const stuckDetected = this.enableStuckDetection &amp;&amp;
this._detectStuck();
                if (stuckDetected) {
                    this._send({ type: 'say', say: SayType.ERROR, text: '检测到循环调用，已自
动中断' });
                    this._send({
                        type: 'say', say: SayType.TOOL_START,
                        text: JSON.stringify(roundData.toolCalls.map(tc =&gt; ({
                            id: tc.id, name: tc.name,
                            displayName: TOOL_DISPLAY_NAMES[tc.name] ||
_getPluginDisplayNames()[tc.name] || tc.name,
                            args: tc.argsStr
                        })))
                    });
                    for (const tc of roundData.toolCalls) {
                        this._send({
                            type: 'say', say: SayType.TOOL_RESULT,
                            text: JSON.stringify({ id: tc.id, name: tc.name, error: '检测到
循环调用，已自动中断' })
                        });
                    }
                    this._send({ type: 'say', say: SayType.TOOL_END, text: '' });
                    conversation.push({
                        role: 'system',
```

```
                        content: 'AGENT STATE: STUCK — Loop detected. You must try a
completely different approach, or call attempt_completion to report current progress and
difficulties.'
                    });
                    continue;
                }

                toolResults = await this._executeTools(roundData.toolCalls, lastUserMsg);
                if (this._aborted) break;

</div><div class="page-number">第 27 页</div><div class="separator"></div><div class="code-
block">                for (const tr of toolResults) {
                    conversation.push({
                        role: 'tool',
                        tool_call_id: tr.id,
                        name: tr.name,
                        content: tr.result
                    });
                }

                await this._evaluateAndGuide(toolResults, conversation, lastUserMsg,
round);

                if (toolResults.some(r =&gt; r.isCompletion)) break;

                this._compressIfNeeded(conversation);
                continue;
            }

            if (roundData.fullContent &amp;&amp; roundData.toolCalls.length === 0) {
                if (this.enablePassiveDetection) {
                    const isPassive = this._detectPassive(roundData.fullContent,
lastUserMsg);
                    if (isPassive &amp;&amp; this._passiveDetectionCount &lt;
this.maxPassiveDetections) {
                        this._passiveDetectionCount++;
                        conversation.push({
                            role: 'system',
                            content: `Your response is too passive. You have tools to
autonomously gather information and execute actions. NEVER ask the user for information you
can obtain yourself.

Review the user's request. Think about what information you need and which tool can provide
it. Call the tool immediately.

Available tools: bash, str_replace_based_edit_tool, json_edit_tool, ckg,
sequential_thinking, attempt_completion, update_todo_list.
Take action now. Do not explain your limitations.`
                        });
                        continue;
                    }
                }

                const similarity = this._computeTextSimilarity(roundData.fullContent,
this._lastTextContent);
                if (similarity &gt; 0.4 &amp;&amp; this._lastTextContent.length &gt; 10) {
                    this._repeatTextCount++;
```

```
                    if (this._repeatTextCount &gt;= this.maxRepeatText) {
                        this._send({ type: 'say', say: SayType.COMPLETION, text: '' });
                        return;
                    }
                } else if (this._detectInternalRepetition(roundData.fullContent)) {
                    this._repeatTextCount++;
                    if (this._repeatTextCount &gt;= this.maxRepeatText) {
                        this._send({ type: 'say', say: SayType.COMPLETION, text: '' });
                        return;
                    }
                } else {
                    this._repeatTextCount = 0;
</div><div class="page-number">第 28 页</div><div class="separator"></div><div class="code-
block">                }
                this._lastTextContent = roundData.fullContent;
            }

            if (roundData.fullContent || roundData.fullReasoning) {
                const finalAssistantMsg = { role: 'assistant', content:
roundData.fullContent || '' };
                if (roundData.fullReasoning) finalAssistantMsg.reasoning_content =
roundData.fullReasoning;
                conversation.push(finalAssistantMsg);
            }

            if (roundData.fullContent) {                conversation.push({ role:
'assistant', content: roundData.fullContent });            }            const
completionFromTool = toolResults.find(r =&gt; r.isCompletion);
            const completionText = completionFromTool ? completionFromTool.completionText :
'';
            this._send({ type: 'say', say: SayType.COMPLETION, text: roundData.fullContent
|| completionText || '' });
            if (totalUsage.total_tokens &gt; 0) {
                this._send({ type: 'usage', usage: totalUsage });
            }
            this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
            return;
        }

        const finalCompletion = toolResults.find(r =&gt; r.isCompletion);
        this._send({ type: 'say', say: SayType.COMPLETION, text: finalCompletion ?
finalCompletion.completionText : '' });
        if (totalUsage.total_tokens &gt; 0) {
            this._send({ type: 'usage', usage: totalUsage });
        }
        this._send({ type: 'say', say: SayType.API_FINISHED, text: '' });
    }
    // Context Builder
    // ========================================================================

    _buildContextSummary() {
        try {
            const os = require('os');
            const path = require('path');
            const fs = require('fs');
            const settingsPath = path.join(os.homedir(), '.versepc', 'settings.json');
            if (fs.existsSync(settingsPath)) {
```

```
                const settings = JSON.parse(fs.readFileSync(settingsPath, 'utf-8'));
                let s = '## 当前工作区状态（自动获取，无需询问用户）\n';
                if (settings.selectedVersion) s += `- 当前版本:
${settings.selectedVersion}\n`;
                if (settings.javaPath) s += `- Java路径: ${settings.javaPath}\n`;
                if (settings.maxMemory) s += `- 最大内存: ${settings.maxMemory}\n`;
                if (settings.gameDir) s += `- 游戏目录: ${settings.gameDir}\n`;
                s += '\n以上信息已自动获取。如需更详细的信息（模组列表、版本列表等），请使用工具获
取。';
                return s;
            }
</div><div class="page-number">第 29 页</div>
```

```
<div class="separator"></div><div class="code-block">        } catch (e) {}
        return null;
    }

    // ========================================================================
    // Intent Detection
    // ========================================================================

    _detectIntent(userMessage) {
        const lower = userMessage.toLowerCase();
        const greetingOnly = /^(你好|hi|hello|hey|谢谢|感谢|ok|好的|知道了|你是谁|你叫什么|你是|说说|聊聊|讲讲)\s*[!! ?? 。.]*$/i;
        if (greetingOnly.test(lower.trim())) return { intent: 'simple', reason: 'greeting', steps_needed: 0 };
        if (userMessage.trim().length &lt;= 6 &amp;&amp; !/[安装|创建|修改|删除|修复|配置|写|做|运行|执行|添加|更新]/.test(lower)) return { intent: 'simple', reason: 'short', steps_needed: 0 };

        const explicitMultiStep = /然后|接着|之后再|先.*再|先.*然后|第一步.*第二步|首先.*然后.*最后/;
        if (explicitMultiStep.test(lower)) {
            const conjunctionCount = (lower.match(/然后|接着|之后再|再(?!次)/g) || []).length;
            return { intent: 'complex', reason: 'explicit_multi_step', steps_needed: conjunctionCount + 1 };
        }

        const actionVerbs = /安装|卸载|创建|删除|修改|编辑|配置|部署|构建|编译|调试|修复|优化|迁移|升级|写|做|运行|执行|添加|更新|实现|重构|迁移|集成|生成|重写/;
        const actionMatches = lower.match(new RegExp(actionVerbs.source, 'g'));
        if (actionMatches &amp;&amp; actionMatches.length &gt;= 2) {
            return { intent: 'complex', reason: 'multi_action', steps_needed: actionMatches.length };
        }

        return { intent: 'complex', reason: 'general_task', steps_needed: 2 };
    }

    // ========================================================================
    // Stream Processing
    // ========================================================================

    async _processStream(res, apiFormat) {
        if (apiFormat === 'anthropic') return this._processAnthropicStream(res);
        if (apiFormat === 'google') return this._processGoogleStream(res);
        return this._processOpenAIStream(res);
    }

    // OpenAI-compatible SSE
    async _processOpenAIStream(res) {
        let buffer = '';
        let fullContent = '';
        let prevContentLen = 0;
        let fullReasoning = '';
        let prevReasoningLen = 0;
        let toolCalls = [];
```

```
        let finishReason = null;
        let reasoningStarted = false;
        let doneReceived = false;
</div><div class="page-number">第 30 页</div><div class="section-title">后1500行代码（第31-60页）</div><div class="code-block">                const { targetPath } = oieBody;
if (!targetPath || !fs.existsSync(targetPath)) {                    sendError('Invalid or non-existent path', 400);                    break;                }
try {                    const { shell } = require('electron');                    await shell.openPath(targetPath);                    sendJSON({ success: true });
} catch (e) {                    sendError(e.message);                }
break;            }            case '/api/create-shortcut': {                const scBody = await readBody();                const shortcutType = scBody.type || 'desktop';
const shortcutVersion = scBody.versionId || '';                const exePath = process.execPath;                const shortcutName = shortcutVersion ? `VersePC - ${shortcutVersion}` : 'VersePC';                let shortcutDir;                if (shortcutType === 'desktop') {                  shortcutDir = path.join(os.homedir(), 'Desktop');                } else {                    shortcutDir = path.join(os.homedir(), 'AppData', 'Roaming', 'Microsoft', 'Windows', 'Start Menu', 'Programs');                }                const shortcutPath = path.join(shortcutDir, `${shortcutName.replace(/["$`]/g, '')}.lnk`);                const psScript = `$ws = New-Object -ComObject WScript.Shell$sc = $ws.CreateShortcut("${shortcutPath.replace(/"/g, '`"')}")$sc.TargetPath = "${exePath.replace(/"/g, '`"')}"$sc.WorkingDirectory = "${path.dirname(exePath).replace(/"/g, '`"')}"$sc.Description = "VersePC Minecraft Launcher"$sc.Save()`.trim();                const tmpScript = path.join(os.tmpdir(), 'versepc_shortcut.ps1');                fs.writeFileSync(tmpScript, psScript, 'utf8');
exec(`powershell -NoProfile -ExecutionPolicy Bypass -File "${tmpScript}"`, { timeout: 10000 }, (err) =&gt; {                    try { fs.unlinkSync(tmpScript); } catch(e) {}
if (err) { sendError(err.message); return; }                    sendJSON({ success: true, path: shortcutPath });                });</div><div class="page-number">第 31 页</div><div class="separator"></div><div class="code-block">                break;            }
case '/api/screenshots': {                const ssVersionId = parsedUrl.query.versionId || '';                const ssDir = resolveSavesDir(ssVersionId).replace('saves', 'screenshots');                const globalSsDir = path.join(DATA_DIR, 'screenshots');
const screenshots = [];                for (const dir of [ssDir, globalSsDir]) {
if (!fs.existsSync(dir)) continue;                    try {
fs.readdirSync(dir).forEach(f =&gt; {                            if (/\.(png|jpg|jpeg|bmp)$/i.test(f)) {                                const filePath = path.join(dir, f);                                const stat = fs.statSync(filePath);
screenshots.push({                                    name: f,
path: filePath,                                    size: stat.size,
time: stat.mtimeMs,                                    url: `/api/screenshot?path=${encodeURIComponent(filePath)}`                                });
}                        });                    } catch(e) {}                }
screenshots.sort((a, b) =&gt; b.time - a.time);                sendJSON({ screenshots });
break;            }            case '/api/screenshot': {                const SCREENSHOT_ALLOWED_BASES = [DATA_DIR, ...Object.values(VERSIONS_DIR)].map(d =&gt; path.resolve(d));                function isScreenshotPathAllowed(p) {
const resolved = path.resolve(p);                    return SCREENSHOT_ALLOWED_BASES.some(base =&gt; resolved.toLowerCase().startsWith(base.toLowerCase()));                }                if (method === 'DELETE') {                    const delPath = decodeURIComponent(parsedUrl.query.path || '');                    if (!delPath) {
sendError('Missing path', 400); break; }                    if (!isScreenshotPathAllowed(delPath)) { sendError('Forbidden', 403); break; }
try {                        fs.unlinkSync(delPath);                        sendJSON({ success: true });                    } catch(e) {
sendError(e.message);                    }                    break;</div><div class="page-
```

```
number">第 32 页</div><div class="separator"></div><div class="code-block">                }
const ssPath = decodeURIComponent(parsedUrl.query.path || '');                if (!ssPath
|| !fs.existsSync(ssPath)) {                    sendError('Not found', 404);
break;                }                if (!isScreenshotPathAllowed(ssPath)) {
sendError('Forbidden', 403); break; }                const ssExt =
path.extname(ssPath).toLowerCase();                const ssMime = ssExt === '.png' ?
'image/png' : ssExt === '.jpg' || ssExt === '.jpeg' ? 'image/jpeg' : 'image/bmp';
const ssData = fs.readFileSync(ssPath);                res.writeHead(200, { 'Content-Type':
ssMime, 'Cache-Control': 'max-age=3600' });                res.end(ssData);
break;            }            case '/api/server/ping': {                const pingHost =
parsedUrl.query.host;                const pingPort = parseInt(parsedUrl.query.port) ||
25565;                if (!pingHost) { sendJSON({ error: 'host required' }, 400); break; }
const pingResult = await mcPing(pingHost, pingPort);                sendJSON(pingResult);
break;            }            default:                sendError('Not found', 404);
}    } catch (e) {        console.error('API Error:', e.message);
sendError(e.message);    }}function fetchJSONWithMethod(urlStr, method, body, headers) {
return new Promise((resolve, reject) =&gt; {        const urlObj = new URL(urlStr);
const options = {            hostname: urlObj.hostname,            path: urlObj.pathname +
urlObj.search,            method: method,            headers: headers || {}        };
const req = https.request(options, (res) =&gt; {            let data = '';
res.on('data', chunk =&gt; data += chunk);            res.on('end', () =&gt; {
try { resolve(JSON.parse(data)); }                catch (e) { reject(e); }            });
</div><div class="page-number">第 33 页</div><div class="separator"></div><div class="code-
block">        });        req.on('error', reject);        req.setTimeout(15000, () =&gt; {
req.destroy(); reject(new Error('Request timeout')); });        if (body) req.write(body);
req.end();    });}function fetchJSONWithAuth(urlStr, token) {    return new
Promise((resolve, reject) =&gt; {        const req = https.get(urlStr, {
headers: { 'Authorization': `Bearer ${token}`, 'User-Agent': 'VersePC/1.0' }        },
(res) =&gt; {            let data = '';            res.on('data', chunk =&gt; data +=
chunk);            res.on('end', () =&gt; {                try { resolve(JSON.parse(data));
}                catch (e) { reject(e); }            });        });        req.on('error',
reject);        req.setTimeout(15000, () =&gt; { req.destroy(); reject(new Error('Request
timeout')); });    });}function getDirSize(dirPath, depth = 0) {    if (depth &gt; 20)
return 0;    let size = 0;    try {        fs.readdirSync(dirPath).forEach(file =&gt; {
const filePath = path.join(dirPath, file);            const stat = fs.statSync(filePath);
if (stat.isDirectory()) size += getDirSize(filePath, depth + 1);            else size +=
stat.size;        });    } catch (e) {}    return size;}function analyzeExitCode(code,
versionId) {    const analysis = { code, reason: '', suggestion: '', isCrash: false };
if (code === 0) {        analysis.reason = '正常退出';        analysis.suggestion = '';
return analysis;    }    if (code === 1) {        analysis.isCrash = true;</div><div
class="page-number">第 34 页</div><div class="separator"></div><div class="code-block">
analysis.reason = '游戏异常退出（通用错误）';        analysis.suggestion = '可能是模组冲突或
Java参数问题，请查看崩溃日志';    } else if (code === -1) {        analysis.isCrash = true;
analysis.reason = '游戏进程被强制终止';        analysis.suggestion = '可能是内存不足或用户手动
结束进程';    } else if (code === 137) {        analysis.isCrash = true;
analysis.reason = '内存不足（OOM Killer）';        analysis.suggestion = '请增加分配内存或减少
模组数量';    } else if (code === 134) {        analysis.isCrash = true;
analysis.reason = '程序异常终止（SIGABRT）';        analysis.suggestion = '可能是JVM内部错误，
尝试更新Java版本';    } else if (code === 139) {        analysis.isCrash = true;
analysis.reason = '段错误（SIGSEGV）';        analysis.suggestion = '可能是JVM崩溃或原生库问
题，尝试更新显卡驱动和Java';    } else if (code === -7 || code === -1073741819) {
analysis.isCrash = true;        analysis.reason = 'JVM 崩溃（访问违规）';
analysis.suggestion = '可能是显卡驱动不兼容或内存损坏，请更新显卡驱动和Java版本，尝试减少分配内
存';    } else {        analysis.isCrash = true;        analysis.reason = `异常退出（退出码:
${code}）`;        analysis.suggestion = '请查看崩溃日志获取更多信息';    }    const settings
= loadSettingsCached();    const searchDirs = [];    if (settings.versionIsolation
&amp;&amp; versionId) {        searchDirs.push(path.join(VERSIONS_DIR, versionId, 'crash-
```

```
reports'));    }    searchDirs.push(path.join(DATA_DIR, 'crash-reports'));
searchDirs.push(path.join(settings.gameDir || DATA_DIR, 'crash-reports'));    for (const
dir of searchDirs) {        if (!fs.existsSync(dir)) continue;        const files =
fs.readdirSync(dir).filter(f =&gt; f.endsWith('.txt')).sort().reverse();        if
(files.length &gt; 0) {            try {                const content =
fs.readFileSync(path.join(dir, files[0]), 'utf8');                if
(content.includes('java.lang.OutOfMemoryError')) {                    analysis.reason = '内
存不足（OutOfMemoryError）';                    analysis.suggestion = '请在设置中增加最大内存
分配';                } else if (content.includes('UnsupportedClassVersionError') ||
content.includes('Unsupported major.minor version')) {                    analysis.reason =
'Java版本不兼容';                    analysis.suggestion = '游戏需要更高版本的Java，请在设置中
更换Java版本';                } else if (content.includes('java.lang.NoSuchMethodError') ||
content.includes('NoClassDefFoundError')) {                    analysis.reason = '模组版本不
兼容';</div><div class="page-number">第 35 页</div><div class="separator"></div><div
class="code-block">                    analysis.suggestion = '请检查模组是否与当前游戏版本和加
载器版本匹配';                } else if (content.includes('Unable to make protected final')
|| content.includes('does not export')) {                    analysis.reason = 'Java版本过高
导致模块访问限制';                    analysis.suggestion = '请降级Java版本或使用Java 8/17启
动';                } else if (content.includes('FMLCommonSetupEvent') ||
content.includes('fml')) {                    analysis.reason = 'Forge/Fabric初始化失败';
analysis.suggestion = '请检查模组兼容性，尝试移除最近添加的模组';                } else if
(content.includes('ShaderCompilationException') || content.includes('shader')) {
analysis.reason = '着色器编译失败';                    analysis.suggestion = '可能是光影模组问
题，尝试移除光影模组';                } else if (content.includes('Mixin') ||
content.includes('mixin')) {                    analysis.reason = 'Mixin注入失败';
analysis.suggestion = '可能是模组与当前版本不兼容，检查Mixin相关模组';                } else if
(content.includes('OpenGL') || content.includes('GLFW')) {
analysis.reason = '图形驱动问题';                    analysis.suggestion = '请更新显卡驱动或检
查OpenGL支持';                } else if (content.includes('Invalid paths argument') ||
content.includes('contained no existing paths')) {                    analysis.reason =
'Forge核心库文件缺失（Invalid paths argument）';                    analysis.suggestion =
'Forge安装不完整，请前往版本设置使用"文件修复"功能，或重新安装该Forge版本';                }
analysis.crashLogFile = path.join(dir, files[0]);                break;            } catch
(e) {}        }    }    return analysis;}function logStartupInfo() {
validateInstalledVersions();    console.log(`\n  VersePC - Minecraft Launcher`);
console.log(`  ──────────────────────────────────────────`);    console.log(`  Data:
${DATA_DIR}`);    console.log(`  Platform: ${getPlatformKey()}`);    console.log(`
──────────────────────────────────────────`);    const installed = getInstalledVersions();
if (installed.length &gt; 0) {        console.log(`  Installed:       ${installed.map(v
=&gt; v.id).join(', ')}`);    }    console.log('');    setTimeout(() =&gt; {        const
javaList = [...detectBundledJava(), ...detectSystemJava()];        if (javaList.length &gt;
0) {            console.log(`  Java:            ${javaList[0].version}
(${javaList[0].path})`);        } else {            console.log('  Java:            未检测
到');        }</div><div class="page-number">第 36 页</div><div class="separator"></div><div
class="code-block">        getVersionManifest().then(manifest =&gt; {
console.log(`  Latest Release:  ${manifest.latest.release}`);            console.log(`
Latest Snapshot: ${manifest.latest.snapshot}`);            console.log(`  Total Versions:
${manifest.versions.length}`);        }).catch(() =&gt; {            console.log('
Warning: Could not fetch version manifest (will retry)');        });    }, 100);}/*
@versepc-protected: anti-ai-plagiarism-v1.0
*/============================================================文件: sse-
server.js============================================================/**
 * ==========================================================================
 *  VersePC - Minecraft Launcher
 *  Copyright (c) 2026 豆杰. All Rights Reserved.
 * ==========================================================================
 *
```

```
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 * ============================================================================
 */

/**
 * VersePC SSE 服务器 - 零外部依赖，纯 Node.js http 模块
 * 绕过 Electron IPC 序列化瓶颈，直接通过 HTTP+SSE 流式传输 AI 响应
 */

const http = require('http');
</div><div class="page-number">第 37 页</div><div class="separator"></div><div class="code-
block">
function createSSEServer(mainExports = {}) {
    const { executeTool = null } = mainExports;
    const BASE_PORT = 3001;
    const approvalPendingMap = {};

    // JSON 解析请求体
    function parseBody(req) {
        return new Promise((resolve) =&gt; {
            let body = '';
            req.on('data', c =&gt; { body += c; if (body.length &gt; 10 * 1024 * 1024)
req.destroy(); });
            req.on('end', () =&gt; { try { resolve(JSON.parse(body)); } catch (e) {
resolve({}); } });
        });
    }

    // CORS 头
    function setCORS(res) {
        res.setHeader('Access-Control-Allow-Origin', '*');
        res.setHeader('Access-Control-Allow-Methods', 'GET, POST, OPTIONS');
        res.setHeader('Access-Control-Allow-Headers', 'Content-Type');
    }

    const server = http.createServer(async (req, res) =&gt; {
        setCORS(res);
```

```
        if (req.method === 'OPTIONS') {
            res.writeHead(204); res.end(); return;
        }

        const url = new URL(req.url, `http://localhost:${BASE_PORT}`);
        const pathname = url.pathname;

        // 健康检查
        if (req.method === 'GET' &amp;&amp; pathname === '/api/health') {
            res.writeHead(200, { 'Content-Type': 'application/json' });
            res.end(JSON.stringify({ status: 'ok', port: BASE_PORT }));
            return;
        }

        // 工具审批
        if (req.method === 'POST' &amp;&amp; pathname === '/api/chat/approve') {
            const body = await parseBody(req);
            const p = approvalPendingMap[body.approvalId];
            if (p) {
                clearTimeout(p.timeout);
                delete approvalPendingMap[body.approvalId];
                p.resolve({ approved: !!body.approved });
            }
            res.writeHead(200, { 'Content-Type': 'application/json' });
            res.end(JSON.stringify({ success: !!p }));
</div><div class="page-number">第 38 页</div><div class="separator"></div><div class="code-
block">            return;
        }

        // SSE 聊天流
        if (req.method === 'POST' &amp;&amp; pathname === '/api/chat') {
            const body = await parseBody(req);
            const { apiKey, model = 'glm-5-flash', messages = [], temperature = 0.7,
enableTools = true } = body;
            const chatId = `chat_${Date.now()}_${Math.random().toString(36).substr(2, 9)}`;

            if (!apiKey) {
                res.writeHead(400, { 'Content-Type': 'application/json' });
                res.end(JSON.stringify({ error: 'API Key 未配置' }));
                return;
            }

            res.writeHead(200, {
                'Content-Type': 'text/event-stream',
                'Cache-Control': 'no-cache, no-transform',
                'Connection': 'keep-alive',
                'X-Accel-Buffering': 'no',
                'Access-Control-Allow-Origin': '*',
            });

            const heartbeat = setInterval(() =&gt; { res.write(': hb\n\n'); }, 30000);

            const cleanup = () =&gt; {
                clearInterval(heartbeat);
                Object.keys(approvalPendingMap).forEach(k =&gt; {
                    if (approvalPendingMap[k].chatId === chatId) {
                        approvalPendingMap[k].resolve({ approved: false, timeout: true });
```

```
                        delete approvalPendingMap[k];
                    }
                });
            };

            res.on('close', cleanup);

            let _resClosed = false;
            res.on('close', () =&gt; { _resClosed = true; });
            const sendChunk = (data) =&gt; {
                if (_resClosed) return;
                try { res.write(`data: ${JSON.stringify(data)}\n\n`); } catch (e) {
_resClosed = true; }
            };

            const onChunk = (processed) =&gt; {
                if (!processed) return;
                if (processed.type &amp;amp;&amp;amp; processed.type !== 'say') {
                    sendChunk(processed);
                    return;
                }
</div><div class="page-number">第 39 页</div><div class="separator"></div><div class="code-
block">                sendChunk(processed);
            };

            const TOOL_RISK_MAP = {
                write_to_file: 'dangerous', replace_in_file: 'dangerous', delete_file:
'dangerous',
                execute_command: 'moderate', spawn_process: 'moderate',
                browser_action: 'moderate', mcp_tool: 'moderate',
                search_files: 'safe', list_files: 'safe', list_code_definition_names:
'safe',
                read_file: 'safe', ask_followup_question: 'safe', attempt_completion:
'safe',
                update_todo_list: 'safe', sequential_thinking: 'safe',
            };
            const onRequestApproval = (toolName, argsStr) =&gt; {
                const risk = TOOL_RISK_MAP[toolName] || 'moderate';
                const aid =
`apv_${Date.now().toString(36)}_${Math.random().toString(36).substr(2, 6)}`;
                sendChunk({ type: 'approval_requested', approvalId: aid, toolName, risk,
args: argsStr });
                return new Promise((resolve) =&gt; {
                    const t = setTimeout(() =&gt; {
                        if (approvalPendingMap[aid]) { delete approvalPendingMap[aid];
resolve({ approved: false, toolName, timeout: true }); }
                    }, 60000);
                    approvalPendingMap[aid] = { resolve, timeout: t, chatId };
                });
            };

            try {
                const { AgentEngine: EngineClass } = require('./agent-engine');
                const engine = new EngineClass({ executeTool, onRequestApproval, onChunk,
logger: console });
                await engine.processChat({ apiKey, model, messages, temperature,
enableTools, maxRounds: 24 });
```

```
            } catch (e) {
                sendChunk({ type: 'error', error: e.message });
            } finally {
                cleanup();
                try { res.end(); } catch (e) {}
            }
            return;
        }

        res.writeHead(404, { 'Content-Type': 'application/json' });
        res.end(JSON.stringify({ error: 'not found' }));
    });

    function tryListen(port) {
        server.removeAllListeners('error');
        server.on('error', (err) =&gt; {
            if (err.code === 'EADDRINUSE' &amp;&amp; port &lt; BASE_PORT + 100) {
                console.log(`[SSE] 端口 ${port} 被占用，尝试 ${port + 1}`);
                tryListen(port + 1);
            } else {
                console.error(`[SSE] 启动失败:`, err.message);
            }
        });
</div><div class="page-number">第 40 页</div><div class="separator"></div><div class="code-
block">        server.listen(port, () =&gt; {
            server._actualPort = port;
            console.log(`[SSE] 启动: http://localhost:${port}`);
        });
    }
    tryListen(BASE_PORT);
    return { server, get PORT() { return server._actualPort || BASE_PORT; } };
}

module.exports = { createSSEServer
};============================================================文件: versepc-
promo\index.html============================================================&lt;!DOCTYPE
html&gt;&lt;html lang="zh-CN"&gt;&lt;head&gt;&lt;meta charset="UTF-8"&gt;&lt;meta
name="viewport" content="width=device-width, initial-scale=1.0"&gt;&lt;title&gt;VersePC
Promo&lt;/title&gt;&lt;link href="https://fonts.googleapis.com/css2?
family=Schibsted+Grotesk:wght@700;900&amp;family=JetBrains+Mono:wght@400;700&amp;display=sw
ap" rel="stylesheet"&gt;&lt;style&gt;*, *::before, *::after { margin:0; padding:0; box-
sizing:border-box; }body { background:#0A0D14; overflow:hidden; }[data-composition-
id="versepc-promo"] {  width:1920px; height:1080px; position:relative; overflow:hidden;
background:#0A0D14;}.scene {  position:absolute; inset:0;  display:flex; align-
items:center; justify-content:center;  visibility:hidden; opacity:0;}.scene-inner {
width:100%; height:100%;  display:flex; padding:100px 120px;  gap:40px; position:relative;
z-index:1;}.left { flex:1; display:flex; flex-direction:column; justify-content:center;
gap:16px; }.right { flex:1; display:flex; align-items:center; justify-content:center;
}.scene-label {  font-family:"JetBrains Mono",monospace; font-size:16px;  color:#FFB347;
letter-spacing:.1em; margin-bottom:8px;}.scene-title {  font-family:"Schibsted
Grotesk",sans-serif; font-weight:900;  font-size:80px; color:#E8ECF2; line-height:1.1;
letter-spacing:-.02em;}.scene-title .hl-g { color:#6AAA3A; }.scene-title .hl-o {
color:#FFB347; }.scene-desc {  font-size:26px; color:#7790A8; line-height:1.5;</div><div
class="page-number">第 41 页</div><div class="separator"></div><div class="code-
block">}.badge-row { display:flex; gap:12px; margin-top:8px; }.badge {  padding:8px 18px;
border:1.5px solid rgba(255,255,255,.1);  background:rgba(20,25,36,.8); font-
family:"JetBrains Mono",monospace;  font-size:16px; color:#FFB347; letter-
```

```
spacing:.04em;}.stat-row { display:flex; gap:48px; margin-top:8px; }.stat-val {  font-family:"Schibsted Grotesk",sans-serif; font-weight:900;  font-size:56px; color:#FFB347; line-height:1;}.stat-lbl { font-size:15px; color:#7790A8; font-family:"JetBrains Mono",monospace; margin-top:4px; }.bg-grid {  position:absolute; inset:0; z-index:0; background-image:    linear-gradient(rgba(255,255,255,.015) 1px,transparent 1px), linear-gradient(90deg,rgba(255,255,255,.015) 1px,transparent 1px);  background-size:80px 80px;}.bg-glow {  position:absolute; border-radius:50%; filter:blur(120px); z-index:0;}.glow-gold { background:#FFB347; width:600px; height:600px; top:50%; right:-100px; transform:translateY(-50%); opacity:.08; }.glow-green { background:#6AAA3A; width:500px; height:500px; bottom:-100px; left:20%; opacity:.08; }.glow-mid { background:#FFB347; width:700px; height:700px; top:50%; left:50%; transform:translate(-50%,-50%); opacity:.06; }/* Scene 0 - Title */.scene-0 .s0-center { flex:1; display:flex; flex-direction:column; align-items:center; justify-content:center; gap:32px; padding:100px; }.s0-logo { width:130px; height:130px; border:3px solid #FFB347; background:rgba(20,25,36,.9); display:flex; align-items:center; justify-content:center; box-shadow:0 0 60px rgba(255,179,71,.2); }.s0-logo img { width:70px; height:70px; image-rendering:pixelated; }.s0-name { font-family:"Schibsted Grotesk",sans-serif; font-weight:900; font-size:130px; color:#E8ECF2; letter-spacing:.04em; line-height:1; text-align:center; }.s0-name .hl-o { color:#FFB347; }.s0-tag { font-size:34px; color:#7790A8; letter-spacing:.08em; text-align:center; }.block-dot {  position:absolute; z-index:2;  width:18px; height:18px; background:rgba(106,170,58,.3);  border:1px solid rgba(106,170,58,.4);}.block-dot:nth-child(1) { right:120px; top:140px; }.block-dot:nth-child(2) { left:140px; bottom:180px; width:24px; height:24px; background:rgba(255,179,71,.3); border-color:rgba(255,179,71,.4); }/* Mod cards */.mod-card {  background:#141924; border:1px solid rgba(255,255,255,.06); padding:18px; display:flex; gap:14px; width:360px;}.mod-card:nth-child(odd) { align-self:flex-end; }.mod-card:nth-child(even) { align-self:flex-start; }.mc-icon { width:48px; height:48px; background:rgba(106,170,58,.15); flex-shrink:0; display:flex; align-items:center; justify-content:center; font-size:22px; }</div><div class="page-number">第 42 页</div><div class="separator"></div><div class="code-block">.mc-info { flex:1; display:flex; flex-direction:column; gap:6px; }.mc-name { font-size:18px; font-weight:700; color:#E8ECF2; font-family:"Schibsted Grotesk",sans-serif; }.mc-meta { font-size:14px; color:#7790A8; font-family:"JetBrains Mono",monospace; }.mc-bar { height:3px; background:rgba(106,170,58,.15); border-radius:2px; overflow:hidden; }.mc-bar-fill { height:100%; background:#6AAA3A; border-radius:2px; width:100%; }/* AI chat */.chat-box { width:440px; background:#141924; border:1px solid rgba(255,255,255,.06); border-radius:6px; overflow:hidden;}.chat-hdr { padding:12px 18px; background:rgba(20,25,36,.95); border-bottom:1px solid rgba(255,255,255,.06); display:flex; align-items:center; gap:8px; }.chat-dot { width:10px; height:10px; border-radius:50%; background:#6AAA3A; box-shadow:0 0 6px rgba(106,170,58,.4); }.chat-title { font-family:"JetBrains Mono",monospace; font-size:14px; color:#7790A8; }.chat-body { padding:16px; display:flex; flex-direction:column; gap:12px; }.chat-msg { max-width:85%; padding:10px 14px; font-size:17px; line-height:1.5; border-radius:5px; color:#E8ECF2; }.chat-msg.usr { align-self:flex-end; background:rgba(106,170,58,.15); border:1px solid rgba(106,170,58,.2); font-size:16px; }.chat-msg.bot { align-self:flex-start; background:rgba(255,179,71,.08); border:1px solid rgba(255,179,71,.15); }.chat-inp { margin:0 16px 14px; padding:10px 14px; background:rgba(255,255,255,.03); border:1px solid rgba(255,255,255,.08); border-radius:5px; font-family:"JetBrains Mono",monospace; font-size:14px; color:#7790A8; }/* Pack cards */.pack-card {  width:380px; background:#141924; border:1px solid rgba(255,255,255,.06); padding:18px; display:flex; gap:14px;}.pack-card:nth-child(odd) { align-self:flex-end; }.pack-card:nth-child(even) { align-self:flex-start; }.pc-thumb { width:56px; height:56px; background:rgba(255,179,71,.1); border:1px solid rgba(255,179,71,.2); flex-shrink:0; display:flex; align-items:center; justify-content:center; font-size:26px; }.pc-info { flex:1; display:flex; flex-direction:column; gap:5px; }.pc-name { font-size:19px; font-weight:700; color:#E8ECF2; font-family:"Schibsted Grotesk",sans-serif; }.pc-desc { font-size:14px; color:#7790A8; }.pc-tag { display:inline-flex; padding:3px 10px; background:rgba(106,170,58,.15); border:1px solid rgba(106,170,58,.25); font-size:12px; color:#6AAA3A; font-family:"JetBrains
```

Mono",monospace; }/* Tool cards */.tool-row { display:flex; flex-direction:column; gap:16px; }.tool-row-top { flex:1; display:flex; align-items:flex-start; gap:40px; margin-bottom:20px; }.tool-grid { display:grid; grid-template-columns:repeat(4,1fr); gap:16px; flex:1; }.tool-card { background:#141924; border:1px solid rgba(255,255,255,.05); padding:18px; display:flex; flex-direction:column; gap:10px;}.tc-icon { font-size:26px; }.tc-title { font-family:"Schibsted Grotesk",sans-serif; font-weight:700; font-size:20px; color:#E8ECF2; }.tc-desc { font-size:14px; color:#7790A8; font-family:"JetBrains Mono",monospace; line-height:1.4; }.tc-bar { height:2px; background:rgba(255,255,255,.06); border-radius:1px; overflow:hidden; margin-top:auto; }.tc-bar-fill { height:100%; background:#6AAA3A; border-radius:1px; }.tool-card:nth-child(2) .tc-bar-fill { background:#FFB347; width:80%; }.tool-card:nth-child(3) .tc-bar-fill { width:65%; }.tool-card:nth-child(4) .tc-bar-fill { background:#FFB347; }/* Version ring */.v-ring-wrap { position:relative; width:340px; height:340px; }.v-ring { position:absolute; inset:0; border:3px solid rgba(106,170,58,.18); border-radius:50%; } 第 43 页 .v-ring-i { position:absolute; inset:50px; border:2px dashed rgba(255,179,71,.12); border-radius:50%; }.v-badge { position:absolute; font-family:"JetBrains Mono",monospace; font-size:26px; font-weight:700; color:#E8ECF2; background:rgba(10,13,20,.92); border:2px solid rgba(106,170,58,.4); padding:5px 16px; white-space:nowrap;}.v-b1 { top:2%; left:50%; transform:translateX(-50%); }.v-b2 { right:-5px; top:40%; transform:translateY(-50%); }.v-b3 { bottom:2%; left:50%; transform:translateX(-50%); }.v-b4 { left:-5px; top:40%; transform:translateY(-50%); }.v-play { position:absolute; top:50%; left:50%; transform:translate(-50%,-50%); width:100px; height:100px; border-radius:50%; background:#6AAA3A; display:flex; align-items:center; justify-content:center; box-shadow:0 0 50px rgba(106,170,58,.35);}.v-play::after { content:""; border:solid; border-width:18px 0 18px 34px; border-color:transparent transparent transparent #0A0D14; margin-left:6px; }/* Scene 6 CTA */.s6-center { flex:1; display:flex; flex-direction:column; align-items:center; justify-content:center; gap:28px; padding:100px; }.s6-logo { width:110px; height:110px; border:3px solid #FFB347; background:rgba(20,25,36,.9); display:flex; align-items:center; justify-content:center; box-shadow:0 0 60px rgba(255,179,71,.2); }.s6-logo img { width:60px; height:60px; image-rendering:pixelated; }.s6-head { font-family:"Schibsted Grotesk",sans-serif; font-weight:900; font-size:90px; color:#E8ECF2; text-align:center; line-height:1.1; letter-spacing:.02em; }.s6-head .hl-o { color:#FFB347; }.s6-sub { font-size:28px; color:#7790A8; text-align:center; line-height:1.5; }.cta-btns { display:flex; gap:24px; }.cta-btn { padding:16px 44px; font-family:"Schibsted Grotesk",sans-serif; font-weight:700; font-size:26px; letter-spacing:.04em; }.cta-primary { background:#6AAA3A; color:#0A0D14; box-shadow:0 0 40px rgba(106,170,58,.3); border:none; }.cta-secondary { background:transparent; color:#E8ECF2; border:2px solid rgba(255,255,255,.15); }.s6-foot { font-family:"JetBrains Mono",monospace; font-size:15px; color:rgba(255,255,255,.15); position:absolute; bottom:36px; }/* pixel decor */.px-row { position:absolute; bottom:70px; left:50%; transform:translateX(-50%); display:flex; gap:3px; z-index:2; }.px { width:7px; height:7px; background:rgba(106,170,58,.25); }.px:nth-child(3n) { background:rgba(255,179,71,.25); }.scene-num { position:absolute; top:60px; left:60px; font-family:"JetBrains Mono",monospace; font-size:22px; color:rgba(255,179,71,.3); z-index:2; }</style></head><body><div data-composition-id="versepc-promo" data-width="1920" data-height="1080" data-start="0"> <!-- Common BG layers --> <div class="bg-grid"></div> <div class="bg-glow glow-gold"></div> <div class="bg-glow glow-green"></div> <!-- SCENE 0: Title --> <div class="scene scene-0" id="s0">   <div class="scene-inner">   <div class="s0-center"> 第 44 页

  <div class="s0-logo"><img src="assets/icon.svg" alt="VersePC"></div> <div class="s0-name">VersePC</div> <div class="s0-tag">新 一 代 Minecraft 启 动 器</div> </div> </div> <div class="block-dot"></div> <div class="block-dot"></div> </div> <!-- SCENE 1: Launch --> <div class="scene" id="s1"> <div class="scene-inner"> <div class="left"> <div class="scene-label">// 01 — 启动</div> <div class="scene-title">一键启动
无限可能</div> <div class="scene-desc">多版本管理 · 模组加载器自动配置
从经典版本到最新快照，一切就绪</div> <div class="badge-row"> <div class="badge">Forge</div><div class="badge">Fabric</div> <div class="badge">NeoForge</div><div class="badge">OptiFine</div></div> </div> <div class="right"> <div class="v-ring-wrap"> <div class="v-ring"></div><div class="v-ring-i"></div> <div class="v-badge v-b1">1.21.5</div><div class="v-badge v-b2">1.20.1</div> <div class="v-badge v-b3">1.18.2</div> <div class="v-badge v-b4">1.16.5</div><div class="v-play"></div> </div> </div></div> <div class="scene-num">01</div> </div> <!-- SCENE 2: Mods --> <div class="scene" id="s2"> <div class="scene-inner"> <div class="left"> <div class="scene-label">// 02 — 模组</div> <div class="scene-title">模组管理
尽在掌握</div> <div class="scene-desc">支持 Forge · Fabric · NeoForge · OptiFine
智能依赖解析，告别模组冲突</div> </div> <div class="right" style="flex-direction:column; gap:14px;"> <div class="mod-card"> <div class="mc-icon">🔮</div> <div class="mc-info"> <div class="mc-name">OptiFine</div> <div class="mc-meta">v1.21.5 · 已启用</div> <div class="mc-bar"><div class="mc-bar-fill"></div></div> 第 45 页 </div> </div> <div class="mod-card"> <div class="mc-icon">📦</div> <div class="mc-info"> <div class="mc-name">Just Enough Items</div> <div class="mc-meta">v19.21.0 · 已启用</div> <div class="mc-bar"><div class="mc-bar-fill"></div></div> </div> </div> <div class="mod-card"> <div class="mc-icon">🌍</div> <div class="mc-info"> <div class="mc-name">JourneyMap</div> <div class="mc-meta">v6.0.0 · 已启用</div> <div class="mc-bar"><div class="mc-bar-fill"></div></div> </div> </div> </div> </div> <div class="scene-num">02</div> </div> <!-- SCENE 3: AI --> <div class="scene" id="s3"> <div class="scene-inner"> <div class="left"> <div class="scene-label">// 03 — AI</div> <div class="scene-title">内置 AI
智能问答</div> <div class="scene-desc">崩溃分析 · 模组推荐 · 配置优化
你的专属 Minecraft 技术顾问</div> </div> </div> <div class="right"> <div class="chat-box"> <div class="chat-hdr"><div class="chat-dot"></div><div class="chat-title">verse-ai</div></div> <div class="chat-body"> <div class="chat-msg usr">为什么我的 Forge 启动崩溃？</div> <div class="chat-msg bot">检测到你的 OptiFine 版本与
Forge 47.3.0 不兼容。
推荐升级到 OptiFine HD U I1。</div> <div class="chat-msg usr">帮我安装</div> <div class="chat-msg bot">已自动下载并安装 ✅</div> </div> <div class="chat-inp">输入你的问题...</div> </div> </div> </div> <div class="scene-num">03</div> </div> <!-- SCENE 4: Modpacks -->

<div class="scene" id="s4"> <div class="scene-inner"> 第 46 页 <div class="left"> <div class="scene-label">// 04 — 整合包</div> <div class="scene-title">整合包
一键安装 </div> <div class="scene-desc">支持 CurseForge · Modrinth 格式导入
自动下载依赖，即装即玩</div> <div class="stat-row"> <div><div class="stat-val">50+</div><div class="stat-lbl">内置整合包 </div></div> <div><div class="stat-val">1-Click</div><div class="stat-lbl">一键导入</div></div> </div> </div> <div class="right" style="flex-direction:column; gap:12px;"> <div class="pack-card"> <div class="pc-thumb">⚔️</div> <div class="pc-info"> <div class="pc-name">RLCraft</div> <div class="pc-desc">高难度生存整合包 </div> <div>CurseForge</div> </div> </div> <div class="pack-card"> <div class="pc-thumb">🏭</div> <div class="pc-info"> <div class="pc-name">All the Mods 10</div> <div class="pc-desc">400+ 模组科技魔法</div> <div>Modrinth</div> </div> </div> <div class="pack-card"> <div class="pc-thumb">🧩</div> <div class="pc-info"> <div class="pc-name">Better MC</div> <div class="pc-desc">原版增强 · 轻量优化</div> <div>CurseForge</div> </div> </div> </div> <div class="scene-num">04</div> </div> <!-- SCENE 5: Tools --> <div class="scene" id="s5"> <div class="scene-inner" style="flex-direction:column; gap:28px;"> <div class="tool-row-top"> <div> <div class="scene-label">// 05 — 工具</div> <div class="scene-title">开发者
全能工具箱 </div> </div> <div class="scene-desc" style="margin-left:auto; align-self:flex-end; text-align:right;"> 代码编辑器 · 终端 · 文件管理 · 崩溃分析
一切尽在 VersePC </div> 第 47 页 </div> <div class="tool-grid"> <div class="tool-card"> <div class="tc-icon">{ }</div> <div class="tc-title">Monaco 编辑器</div> <div class="tc-desc">VS Code 同款内核
语法高亮 · 自动补全</div> <div class="tc-bar"><div class="tc-bar-fill" style="width:90%"></div></div> </div> <div class="tool-card"> <div class="tc-icon">>_</div> <div class="tc-title">内置终端</div> <div class="tc-desc">xterm.js 驱动
GPU 加速渲染</div> <div class="tc-bar"><div class="tc-bar-fill"></div></div> </div> <div class="tool-card"> <div class="tc-icon">🗂️</div> <div class="tc-title">文件浏览器</div> <div class="tc-desc">像 IDE 一样管理
实例文件与配置</div> <div class="tc-bar"><div class="tc-bar-fill"></div></div> </div> <div class="tool-card"> <div class="tc-icon">🦎</div> <div class="tc-title">崩溃分析器</div> <div class="tc-desc">解析错误日志
定位冲突模组</div> <div class="tc-bar"><div class="tc-bar-fill"></div></div> </div> </div> </div> <div class="scene-num">05</div> </div> <!-- SCENE 6: CTA --> <div class="scene" id="s6"> <div class="scene-inner"> <div class="s6-center"> <div class="s6-logo"><img src="assets/icon.svg" alt="VersePC"></div> <div class="s6-head">开启你的
Minecraft 之旅</div> <div class="s6-sub">免费 · 开源 · 永久更新
让每一次方块冒险更加精彩</div> <div class="cta-btns"> <div class="cta-btn cta-primary">立即下载</div>

<div class="cta-btn cta-secondary">GitHub</div> </div>
</div> <div class="s6-foot">VersePC v1.0.0 ·
github.com/doujie081231/versePc</div> </div> <div class="px-row">
<div class="px"></div><div class="px"></div><div
class="px"></div> <div class="px"></div><div
class="px"></div><div class="px"></div> <div
class="px"></div><div class="px"></div><div
class="px"></div> <div class="px"></div><div
class="px"></div><div class="px"></div>
第 48 页 <div
class="px"></div><div class="px"></div><div
class="px"></div> <div class="px"></div><div
class="px"></div><div class="px"></div> </div>
</div></div><script
src="https://cdn.jsdelivr.net/npm/gsap@3.14.2/dist/gsap.min.js"></script><scrip
t>window.__timelines = window.__timelines || {};gsap.defaults({ overwrite:false });const
tl = gsap.timeline({ paused:true });// === COMMON BG ===tl.from(".bg-grid", { opacity:0,
duration:0.5 }, 0);tl.from(".glow-gold", { scale:.5, opacity:0, duration:1.2,
ease:"power2.out" }, 0);tl.from(".glow-green", { scale:.6, opacity:0, duration:1,
ease:"power2.out" }, .2);// === SCENE 0: TITLE (0s-5s) ===tl.set("#s0", { autoAlpha:1 },
.3);tl.from("#s0 .s0-logo", { scale:.2, opacity:0, duration:.7, ease:"back.out(1.7)" },
.5);tl.from("#s0 .s0-name", { y:60, opacity:0, duration:.6, ease:"expo.out" },
1.2);tl.from("#s0 .s0-tag", { y:30, opacity:0, duration:.5, ease:"power3.out" },
1.7);tl.from("#s0 .block-dot",{ scale:0, opacity:0, stagger:.1, duration:.35,
ease:"back.out(2)" }, 1.4);tl.to("#s0", { autoAlpha:0, duration:.35, ease:"power2.in" },
4.7);// === SCENE 1: LAUNCH (5s-10.5s) ===tl.set("#s1", { autoAlpha:1 }, 5);tl.from("#s1
.scene-num", { opacity:0, duration:.3 }, 5.1);tl.from("#s1 .scene-label",{ x:-30,
opacity:0, duration:.4, ease:"power3.out" }, 5.2);tl.from("#s1 .scene-title",{ y:50,
opacity:0, duration:.6, ease:"expo.out" }, 5.4);tl.from("#s1 .scene-desc", { y:30,
opacity:0, duration:.5, ease:"power3.out" }, 5.8);tl.from("#s1 .badge", { y:20,
opacity:0, stagger:.08, duration:.4, ease:"power2.out" }, 6.1);tl.from("#s1 .v-ring", {
scale:.5, opacity:0, duration:.65, ease:"power3.out" }, 5.5);tl.from("#s1 .v-ring-i", {
scale:0, opacity:0, duration:.45, ease:"power2.out" }, 5.8);tl.from("#s1 .v-badge", {
scale:0, opacity:0, stagger:.1, duration:.4, ease:"back.out(1.5)" }, 6);tl.from("#s1 .v-
play", { scale:0, opacity:0, duration:.5, ease:"back.out(2)" }, 6.5);tl.to("#s1 .v-
play", { boxShadow:"0 0 80px rgba(106,170,58,.5)", duration:1, repeat:1, yoyo:true,
ease:"sine.inOut" }, 7.5);tl.to("#s1", { autoAlpha:0, duration:.35, ease:"power2.in" },
10.2);// === SCENE 2: MODS (10.5s-16s) ===tl.set("#s2", { autoAlpha:1 }, 10.5);tl.from("#s2
.scene-num", { opacity:0, duration:.3 }, 10.6);tl.from("#s2 .scene-label",{ x:-30,
opacity:0, duration:.4, ease:"power3.out" }, 10.7);tl.from("#s2 .scene-title",{ y:50,
opacity:0, duration:.6, ease:"expo.out" }, 10.9);tl.from("#s2 .scene-desc", { y:30,
opacity:0, duration:.5, ease:"power3.out" }, 11.3);tl.from("#s2 .mod-card", { x:60,
opacity:0, stagger:.15, duration:.5, ease:"power3.out" }, 10.8);tl.to("#s2", { autoAlpha:0,
duration:.35, ease:"power2.in" }, 15.7); 第 49 页 <div
class="separator"> // === SCENE 3: AI (16s-21.5s)
===tl.set("#s3", { autoAlpha:1 }, 16);tl.from("#s3 .scene-num", { opacity:0, duration:.3 },
16.1);tl.from("#s3 .scene-label",{ x:-30, opacity:0, duration:.4, ease:"power3.out" },
16.2);tl.from("#s3 .scene-title",{ y:50, opacity:0, duration:.6, ease:"expo.out" },
16.4);tl.from("#s3 .scene-desc", { y:30, opacity:0, duration:.5, ease:"power3.out" },
16.8);tl.from("#s3 .chat-box", { scale:.85, opacity:0, duration:.6, ease:"power3.out" },
16.5);tl.from("#s3 .chat-msg.usr",{ x:30, opacity:0, stagger:.25, duration:.4,
ease:"power3.out" }, 17.2);tl.from("#s3 .chat-msg.bot",{ x:-30, opacity:0, stagger:.25,
duration:.4, ease:"power3.out" }, 17.4);tl.from("#s3 .chat-inp", { opacity:0, y:8,
duration:.3 }, 18.5);tl.to("#s3", { autoAlpha:0, duration:.35, ease:"power2.in" }, 21.2);//
=== SCENE 4: MODPACKS (21.5s-27s) ===tl.set("#s4", { autoAlpha:1 }, 21.5);tl.from("#s4
.scene-num", { opacity:0, duration:.3 }, 21.6);tl.from("#s4 .scene-label",{ x:-30,
opacity:0, duration:.4, ease:"power3.out" }, 21.7);tl.from("#s4 .scene-title",{ y:50,

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opacity:0, duration:.6, ease:"expo.out" }, 21.9);tl.from("#s4 .scene-desc", { y:30,
opacity:0, duration:.5, ease:"power3.out" }, 22.3);tl.from("#s4 .pack-card",  { x:60,
opacity:0, stagger:.15, duration:.5, ease:"power3.out" }, 21.9);tl.from("#s4 .stat-val",
{ y:30, opacity:0, stagger:.15, duration:.5, ease:"back.out(1.5)" }, 23);tl.from("#s4
.stat-lbl",   { opacity:0, stagger:.15, duration:.3 }, 23.2);tl.to("#s4", { autoAlpha:0,
duration:.35, ease:"power2.in" }, 26.7);// === SCENE 5: TOOLS (27s-32.5s) ===tl.set("#s5",
{ autoAlpha:1 }, 27);tl.from("#s5 .scene-num",   { opacity:0, duration:.3 },
27.1);tl.from("#s5 .scene-label", { x:-30, opacity:0, duration:.4, ease:"power3.out" },
27.2);tl.from("#s5 .scene-title", { y:50, opacity:0, duration:.6, ease:"expo.out" },
27.4);tl.from("#s5 .scene-desc",  { x:40, opacity:0, duration:.5, ease:"power3.out" },
27.7);tl.from("#s5 .tool-card",   { y:50, opacity:0, stagger:.1, duration:.5,
ease:"power3.out" }, 28);tl.from("#s5 .tc-bar-fill", { scaleX:0, stagger:.1, duration:.5,
ease:"power3.out", transformOrigin:"left" }, 28.7);tl.to("#s5", { autoAlpha:0,
duration:.35, ease:"power2.in" }, 32.2);// === SCENE 6: CTA (32.5s-38s) ===tl.set("#s6", {
autoAlpha:1 }, 32.5);tl.from("#s6 .s6-logo",   { scale:.2, opacity:0, duration:.8,
ease:"back.out(1.7)" }, 32.8);tl.from("#s6 .s6-head",   { y:60, opacity:0, duration:.7,
ease:"expo.out" }, 33.5);tl.from("#s6 .s6-sub",    { y:30, opacity:0, duration:.5,
ease:"power3.out" }, 34);tl.from("#s6 .cta-primary", { x:-50, opacity:0, duration:.5,
ease:"back.out(1.5)" }, 34.5);tl.from("#s6 .cta-secondary",{ x:50, opacity:0, duration:.5,
ease:"back.out(1.5)" }, 34.65);tl.from("#s6 .px",           { scale:0, opacity:0,
stagger:.03, duration:.3, ease:"back.out(1.5)" }, 35.2);tl.from("#s6 .s6-foot",      {
opacity:0, duration:.4 }, 35.8);tl.to("#s6 .cta-primary", { boxShadow:"0 0 60px
rgba(106,170,58,.5)", duration:1, repeat:1, yoyo:true, ease:"sine.inOut" }, 36);//
Registerwindow.__timelines["versepc-promo"] = tl;&lt;/script&gt;&lt;/body&gt;&lt;/html&gt;
</div><div class="page-number">第 50 页</div><div class="separator"></div><div class="code-
block">============================================================文件: versepc-
promo\promo-
fastcut.html============================================================&lt;!DOCTYPE
html&gt;&lt;html lang="zh-CN"&gt;&lt;head&gt;&lt;meta charset="UTF-8"&gt;&lt;meta
name="viewport" content="width=device-width, initial-scale=1.0"&gt;&lt;title&gt;VersePC
FastCut&lt;/title&gt;&lt;link href="https://fonts.googleapis.com/css2?
family=Inter:wght@300;400;500;600;700;800&amp;display=swap"
rel="stylesheet"&gt;&lt;style&gt;*,*::before,*::after{margin:0;padding:0;box-sizing:border-
box}body{background:#fff;overflow:hidden;font-family:"Inter",-apple-
system,BlinkMacSystemFont,sans-serif}[data-composition-id="versepc-fastcut"]{
width:1920px;height:1080px;position:relative;overflow:hidden;background:#fff;}.scene{
position:absolute;inset:0;  display:flex;align-items:center;justify-content:center;
visibility:hidden;opacity:0;}/* ===== Act 1: Logo ===== */.a1-center{  display:flex;flex-
direction:column;align-items:center;gap:28px;}.a1-logo-wrap{  width:120px;height:120px;
border:1.5px solid rgba(0,0,0,.08);  display:flex;align-items:center;justify-
content:center;  box-shadow:0 4px 24px rgba(0,0,0,.04);}.a1-logo-wrap
img{width:64px;height:64px;image-rendering:pixelated;}.a1-title{  font-size:96px;font-
weight:700;color:#1D1D1F;  letter-spacing:-.03em;line-height:1;}.a1-title
.hl{color:#6AAA3A;}.a1-sub{  font-size:24px;font-weight:400;color:#86868B;  letter-
spacing:.08em;}/* ===== Bottom tag line (shared) ===== */.tagline{
position:absolute;bottom:80px;left:50%;transform:translateX(-50%);  font-size:18px;font-
weight:400;color:#86868B;  letter-spacing:.06em;white-space:nowrap;}</div><div class="page-
number">第 51 页</div><div class="separator"></div><div class="code-block">/* ===== Act 2:
Version ===== */.a2-wrap{  width:100%;padding:0 140px;  display:flex;align-
items:center;justify-content:space-between;}.a2-left{display:flex;flex-
direction:column;align-items:center;gap:36px;}.a2-ring-
wrap{position:relative;width:300px;height:300px;}.a2-ring{  position:absolute;inset:0;
border:2px solid rgba(0,0,0,.06);border-radius:50%;}.a2-ring-i{
position:absolute;inset:48px;  border:1.5px dashed rgba(0,0,0,.04);border-radius:50%;}.a2-
badge{  position:absolute;font-size:14px;font-weight:500;color:#1D1D1F;
background:#fff;border:1px solid rgba(0,0,0,.06);  padding:4px 14px;white-space:nowrap;
box-shadow:0 1px 6px rgba(0,0,0,.04);}.a2-
```

```
b1{top:0;left:50%;transform:translateX(-50%);}.a2-b2{right:-10px;top:38%;}.a2-
b3{bottom:0;left:50%;transform:translateX(-50%);}.a2-b4{left:-10px;top:38%;}.a2-play{
position:absolute;top:50%;left:50%;transform:translate(-50%,-50%);
width:72px;height:72px;border-radius:50%;  background:#6AAA3A;  display:flex;align-
items:center;justify-content:center;}.a2-play::after{  content:"";border:solid;border-
width:14px 0 14px 26px;  border-color:transparent transparent transparent #fff;margin-
left:5px;}.a2-right{display:flex;flex-direction:column;gap:16px;}.a2-loader{
display:flex;align-items:center;gap:12px;  padding:10px 20px;border:1px solid
rgba(0,0,0,.06);  background:#fff;box-shadow:0 1px 8px rgba(0,0,0,.03);}.a2-loader-
dot{width:8px;height:8px;border-radius:50%;flex-shrink:0;}.ldot-
f{background:#FFB347;}.ldot-a{background:#6AAA3A;}.ldot-n{background:#7C4DFF;}.a2-loader
span{font-size:15px;font-weight:500;color:#1D1D1F;letter-spacing:.02em;}/* ===== Act 3:
Mods ===== */.a3-wrap{  width:100%;padding:0 140px;</div><div class="page-number">第 52 页
</div><div class="separator"></div><div class="code-block">  display:flex;align-
items:center;justify-content:flex-end;}.a3-cards{display:flex;flex-
direction:column;gap:14px;}.a3-card{  display:flex;align-items:center;gap:16px;
width:420px;padding:16px 20px;  background:#fff;border:1px solid rgba(0,0,0,.06);  box-
shadow:0 2px 12px rgba(0,0,0,.04);}.a3-icon{  width:44px;height:44px;flex-shrink:0;
background:rgba(106,170,58,.08);border:1px solid rgba(106,170,58,.15);  display:flex;align-
items:center;justify-content:center;  font-size:20px;}.a3-info{flex:1;display:flex;flex-
direction:column;gap:4px;}.a3-name{font-size:16px;font-weight:600;color:#1D1D1F;}.a3-
meta{font-size:13px;color:#86868B;}.a3-bar{height:3px;background:rgba(0,0,0,.04);border-
radius:2px;overflow:hidden;margin-top:2px;}.a3-bar-
fill{height:100%;background:#6AAA3A;border-radius:2px;width:0%;}/* ===== Act 4: Batch
Download ===== */.a4-wrap{  width:100%;padding:0 200px;  display:flex;flex-
direction:column;gap:20px;}.a4-row{  display:flex;align-items:center;gap:16px;}.a4-check{
width:22px;height:22px;flex-shrink:0;  border:1.5px solid rgba(0,0,0,.12);
background:#fff;display:flex;align-items:center;justify-content:center;  font-
size:14px;color:#fff;}.a4-check.done{background:#6AAA3A;border-color:#6AAA3A;}.a4-
check.done::after{content:"✓";color:#fff;font-weight:700;font-size:13px;}.a4-row-name{
font-size:15px;font-weight:500;color:#1D1D1F;width:100px;}.a4-bar-
wrap{flex:1;height:6px;background:rgba(0,0,0,.04);border-radius:3px;overflow:hidden;}.a4-
bar-fill{height:100%;border-radius:3px;width:0%;}.a4-bar-fill.g{background:#6AAA3A;}.a4-
bar-fill.o{background:#FFB347;}.a4-btn{  align-self:center;  padding:10px 36px;margin-
top:12px;  background:#1D1D1F;color:#fff;border:none;  font-family:"Inter",sans-serif;font-
size:15px;font-weight:500;  letter-spacing:.02em;cursor:default;</div><div class="page-
number">第 53 页</div><div class="separator"></div><div class="code-block">}.a4-done-row{
display:flex;align-items:center;gap:10px;}.a4-done-row .check-icon{
width:28px;height:28px;border-radius:50%;background:#6AAA3A;  display:flex;align-
items:center;justify-content:center;  color:#fff;font-weight:700;font-size:15px;}/* =====
Act 5: Mystery ===== */.a5-wrap{  width:100%;padding:0 200px;  display:flex;flex-
direction:column;align-items:center;gap:32px;}.a5-blur-area{  position:relative;
width:500px;height:340px;  overflow:hidden;}.a5-blur-content{  position:absolute;inset:0;
display:flex;flex-direction:column;gap:12px;padding:30px;}.a5-blur-bubble{  max-
width:80%;padding:12px 16px;border-radius:8px;  font-size:15px;line-
height:1.5;color:#1D1D1F;  background:rgba(0,0,0,.03);border:1px solid rgba(0,0,0,.04);
backdrop-filter:blur(18px);-webkit-backdrop-filter:blur(18px);}.a5-blur-
bubble.bot{background:rgba(106,170,58,.04);border-color:rgba(106,170,58,.1);align-
self:flex-start;}.a5-blur-bubble.usr{background:rgba(0,0,0,.02);align-self:flex-end;}.a5-
glow{  position:absolute;bottom:-80px;left:50%;transform:translateX(-50%);
width:300px;height:120px;  background:radial-gradient(ellipse, rgba(106,170,58,.15) 0%,
transparent 70%);}.a5-flash-code{  position:absolute;top:30px;right:20px;  font-family:"SF
Mono","Menlo","Consolas",monospace;font-size:12px;  color:rgba(0,0,0,.2);line-height:1.6;
opacity:0;}.a5-tagline{  font-size:28px;font-weight:500;color:#1D1D1F;  letter-
spacing:.12em;}/* ===== Act 6: Modpacks ===== */.a6-wrap{</div><div class="page-number">第
54 页</div><div class="separator"></div><div class="code-block">  width:100%;padding:0
140px;  display:flex;align-items:center;gap:60px;}.a6-cards{display:flex;flex-
```

direction:column;gap:14px;flex:1;}.a6-card{ display:flex;align-items:center;gap:16px; padding:16px 20px; background:#fff;border:1px solid rgba(0,0,0,.06); box-shadow:0 2px 12px rgba(0,0,0,.04);}.a6-thumb{ width:48px;height:48px;flex-shrink:0; background:rgba(255,179,71,.08);border:1px solid rgba(255,179,71,.15); display:flex;align-items:center;justify-content:center;font-size:22px;}.a6-info{flex:1;display:flex;flex-direction:column;gap:4px;}.a6-name{font-size:16px;font-weight:600;color:#1D1D1F;}.a6-desc{font-size:13px;color:#86868B;}.a6-tag{ display:inline-flex;padding:2px 10px;font-size:11px;font-weight:500; background:rgba(106,170,58,.06);border:1px solid rgba(106,170,58,.12); color:#6AAA3A;}.a6-stats{display:flex;flex-direction:column;gap:20px;flex-shrink:0;}.a6-stat{text-align:center;}.a6-stat-val{font-size:48px;font-weight:700;color:#1D1D1F;line-height:1;}.a6-stat-lbl{font-size:14px;color:#86868B;margin-top:4px;}/* ===== Act 7: Finale ===== */.a7-wrap{ display:flex;flex-direction:column;align-items:center;gap:20px;}.a7-line{font-size:40px;font-weight:500;color:#1D1D1F;letter-spacing:.15em;}.a7-line.thin{font-weight:300;color:#86868B;}.a7-divider{width:40px;height:1px;background:rgba(0,0,0,.08);margin:8px 0;}.a7-logo-small{ width:56px;height:56px; border:1px solid rgba(0,0,0,.06); display:flex;align-items:center;justify-content:center;}.a7-logo-small img{width:32px;height:32px;image-rendering:pixelated;}.a7-date{ font-size:72px;font-weight:700;color:#1D1D1F; letter-spacing:.04em;line-height:1.1; margin-top:8px;}.a7-date .hl{color:#6AAA3A;}</style></head><body> 第 55 页 <div data-composition-id="versepc-fastcut" data-width="1920" data-height="1080" data-start="0"> <!-- ===== ACT 1: Logo 浮现 (0-3s) ===== --> <div class="scene" id="s1"> <div class="a1-center"> <div class="a1-logo-wrap"><img src="assets/icon.svg" alt="VersePC"></div> <div class="a1-title">VersePC</div> <div class="a1-sub">新一代 Minecraft 启动器</div> </div> </div> <!-- ===== ACT 2: 版本启动 (3-6.5s) ===== --> <div class="scene" id="s2"> <div class="a2-wrap"> <div class="a2-left"> <div class="a2-ring-wrap"> <div class="a2-ring"></div><div class="a2-ring-i"></div> <div class="a2-badge a2-b1">1.21.5</div> <div class="a2-badge a2-b2">1.20.1</div> <div class="a2-badge a2-b3">1.18.2</div> <div class="a2-badge a2-b4">1.16.5</div> <div class="a2-play"></div> </div> </div> <div class="a2-right"> <div class="a2-loader"><div class="a2-loader-dot ldot-f"></div>Forge</div> <div class="a2-loader"><div class="a2-loader-dot ldot-a"></div>Fabric</div> <div class="a2-loader"><div class="a2-loader-dot ldot-n"></div>NeoForge</div> </div> </div> <div class="tagline">一键启动 · 无限可能</div> </div> <!-- ===== ACT 3: 模组管理 (6.5-10s) ===== --> <div class="scene" id="s3"> <div class="a3-wrap"> <div class="a3-cards"> <div class="a3-card"> <div class="a3-icon">🔮</div> <div class="a3-info"> <div class="a3-name">OptiFine</div> <div class="a3-meta">v1.21.5 · 性能优化</div> <div class="a3-bar"><div class="a3-bar-fill"></div></div> </div> </div> <div class="a3-card"> <div class="a3-icon">📦</div> <div class="a3-info"> <div class="a3-name">Just Enough Items</div> 第 56 页 <div class="a3-meta">v19.21.0 · 物品查询</div> <div class="a3-bar"><div class="a3-bar-fill"></div></div> </div> </div> <div class="a3-card"> <div class="a3-icon">🌍</div> <div class="a3-info"> <div class="a3-name">JourneyMap</div> <div class="a3-meta">v6.0.0 · 小地图</div> <div class="a3-

bar"><div class="a3-bar-fill"></div></div> </div> </div> </div> </div> <div class="tagline">模组管理 · 尽在掌握</div> </div> <!-- ===== ACT 4: 批量下载 (10-14s) ===== --> <div class="scene" id="s4"> <div class="a4-wrap"> <div class="a4-row"><div class="a4-check"></div>OptiFine<div class="a4-bar-wrap"><div class="a4-bar-fill g"></div></div></div> <div class="a4-row"><div class="a4-check"></div>JEI<div class="a4-bar-wrap"><div class="a4-bar-fill o"></div></div></div> <div class="a4-row"><div class="a4-check"></div>JourneyMap<div class="a4-bar-wrap"><div class="a4-bar-fill g"></div></div></div> <div class="a4-row"><div class="a4-check"></div>Inventory Profiler<div class="a4-bar-wrap"><div class="a4-bar-fill o"></div></div></div> <div class="a4-row"><div class="a4-check"></div>Sodium<div class="a4-bar-wrap"><div class="a4-bar-fill g"></div></div></div> <div class="a4-btn">批量下载（5） </div> <div class="a4-done-row" style="opacity:0"><div class="check-icon">✓</div>5 个模组已安装</div> </div> <div class="tagline">批量下载 · 一键安装</div> </div> <!-- ===== ACT 5: ??? 神秘功能 (14-18s) ===== --> <div class="scene" id="s5"> <div class="a5-wrap"> <div class="a5-blur-area"> <div class="a5-blur-content" style="filter:blur(14px);-webkit-filter:blur(14px);"> <div class="a5-blur-bubble usr" style="max-width:75%">为什么我的 Forge 启动崩溃了？</div> <div class="a5-blur-bubble bot" style="max-width:80%">检测到模组冲突，正在分析日志…</div> <div class="a5-blur-bubble usr" style="max-width:60%">帮我修复</div> <div class="a5-blur-bubble bot" style="max-width:65%">已自动修复 ✅</div> </div> <div class="a5-glow"></div> <div class="a5-flash-code">[17:32:41] Detected mod conflict: OptiFine × Sodium
 → Auto-resolving... Done.</div> </div> <div class="a5-tagline">还有更多…</div> </div> </div> <!-- ===== ACT 6: 整合包 (18-21.5s) ===== --> <div class="scene" id="s6"> 第 57 页 <div class="a6-wrap"> <div class="a6-cards"> <div class="a6-card"> <div class="a6-thumb">⚔️</div> <div class="a6-info"> <div class="a6-name">RLCraft</div> <div class="a6-desc">高难度生存整合包 </div> <div>CurseForge</div> </div> </div> <div class="a6-card"> <div class="a6-thumb">🏭</div> <div class="a6-info"> <div class="a6-name">All the Mods 10</div> <div class="a6-desc">400+ 模组科技魔法</div> <div>Modrinth</div> </div> </div> <div class="a6-card"> <div class="a6-thumb">🧩</div> <div class="a6-info"> <div class="a6-name">Better MC</div> <div class="a6-desc">原版增强 · 轻量优化</div> <div>CurseForge</div> </div> </div> </div> <div class="a6-stats"> <div class="a6-stat"><div class="a6-stat-val">50+</div><div class="a6-stat-lbl">整合包 </div></div> <div class="a6-stat"><div class="a6-stat-val">1-Click</div><div class="a6-stat-lbl">一键导入</div></div> </div> </div> </div> <div class="tagline">整合包 · 一键安装</div> </div> <!-- ===== ACT 7: 收官 (21.5-30s) ===== --> <div class="scene" id="s7"> <div class="a7-wrap"> <div class="a7-line">免费</div>

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&lt;div class="a7-line"&gt;开源&lt;/div&gt;      &lt;div class="a7-line thin"&gt;永久更新
&lt;/div&gt;      &lt;div class="a7-divider"&gt;&lt;/div&gt;      &lt;div class="a7-logo-
small"&gt;&lt;img src="assets/icon.svg" alt="VersePC"&gt;&lt;/div&gt;      &lt;div
class="a7-date"&gt;&lt;span class="hl"&gt;6 月 21 日&lt;/span&gt; 见&lt;/div&gt;
&lt;/div&gt;  &lt;/div&gt;&lt;/div&gt;&lt;script
src="https://cdn.jsdelivr.net/npm/gsap@3.14.2/dist/gsap.min.js"&gt;&lt;/script&gt;</div>
<div class="page-number">第 58 页</div><div class="separator"></div><div class="code-
block">&lt;script&gt;gsap.defaults({overwrite:false, ease:"sine.inOut"});const tl =
gsap.timeline({paused:true});// ================================================// ACT 1:
Logo (0s - 3s)// ================================================tl.set("#s1",
{autoAlpha:1},0);tl.from("#s1 .a1-logo-wrap",
{scale:.85,opacity:0,duration:.8,ease:"sine.inOut"},.2);tl.from("#s1 .a1-title",
{y:30,opacity:0,duration:.7,ease:"power3.out"},.8);tl.from("#s1 .a1-sub",
{y:15,opacity:0,duration:.5,ease:"power3.out"},1.4);tl.to("#s1",
{autoAlpha:0,duration:.35,ease:"power2.in"},2.6);//
================================================// ACT 2: Version (3s - 6.5s)//
================================================tl.set("#s2",{autoAlpha:1},3);tl.from("#s2
.a2-ring",{scale:.9,opacity:0,duration:.55,ease:"sine.inOut"},3.1);tl.from("#s2 .a2-ring-
i",{scale:.8,opacity:0,duration:.4,ease:"sine.inOut"},3.3);tl.from("#s2 .a2-badge",
{scale:.8,opacity:0,stagger:.08,duration:.35,ease:"power3.out"},3.3);tl.from("#s2 .a2-
play",{scale:.8,opacity:0,duration:.4,ease:"power3.out"},3.8);tl.from("#s2 .a2-loader",
{x:30,opacity:0,stagger:.1,duration:.45,ease:"power3.out"},3.3);tl.from("#s2 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},4.5);// pulse the play buttontl.to("#s2 .a2-
play",{boxShadow:"0 0 40px
rgba(106,170,58,.25)",duration:.8,yoyo:true,repeat:1,ease:"sine.inOut"},4.2);tl.to("#s2",
{autoAlpha:0,duration:.3,ease:"power2.in"},6.2);//
================================================// ACT 3: Mods (6.5s - 10s)//
================================================tl.set("#s3",
{autoAlpha:1},6.5);tl.from("#s3 .a3-card",
{x:60,opacity:0,stagger:.12,duration:.5,ease:"power3.out"},6.6);tl.to("#s3 .a3-bar-fill",
{width:"100%",stagger:.12,duration:.5,ease:"power3.out",transformOrigin:"left"},7.3);tl.fro
m("#s3 .tagline",{y:12,opacity:0,duration:.4,ease:"power3.out"},7.8);tl.to("#s3",
{autoAlpha:0,duration:.3,ease:"power2.in"},9.7);//
================================================// ACT 4: Batch Download (10s - 14s)//
================================================tl.set("#s4",{autoAlpha:1},10);// stagger
checkbox checkstl.to("#s4 .a4-check",{
className:"+=done",stagger:.08,duration:.15,ease:"none"},10.2);// button
appearstl.from("#s4 .a4-btn",
{scale:.92,opacity:0,duration:.4,ease:"sine.inOut"},10.8);tl.to("#s4 .a4-btn",
{opacity:.4,duration:.2},11.1); // click feedback// progress bars all at once with
staggered delaystl.to("#s4 .a4-bar-fill",{
width:"100%",stagger:.15,duration:.6,ease:"sine.inOut",transformOrigin:"left"</div><div
class="page-number">第 59 页</div><div class="separator"></div><div class="code-
block">},11.2);// done rowtl.to("#s4 .a4-done-row",
{autoAlpha:1,duration:.4,ease:"sine.inOut"},12.2);tl.from("#s4 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},13);tl.to("#s4",
{autoAlpha:0,duration:.3,ease:"power2.in"},13.7);//
================================================// ACT 5: Mystery (14s - 18s)//
================================================tl.set("#s5",{autoAlpha:1},14);tl.from("#s5
.a5-blur-area",{scale:.92,opacity:0,duration:.8,ease:"sine.inOut"},14.1);// green glow
pulsetl.to("#s5 .a5-glow",
{scale:1.15,opacity:.6,duration:1.2,yoyo:true,repeat:1,ease:"sine.inOut"},14.6);// flash
code snippettl.to("#s5 .a5-flash-code",{autoAlpha:1,duration:.3},15.2);tl.to("#s5 .a5-
flash-code",{autoAlpha:0,duration:.3},15.9);tl.to("#s5 .a5-flash-code",
{autoAlpha:1,duration:.2},16.4);tl.to("#s5 .a5-flash-code",
{autoAlpha:0,duration:.3},16.9);tl.from("#s5 .a5-tagline",
{y:15,opacity:0,duration:.6,ease:"sine.inOut"},15.5);tl.to("#s5",
```

{autoAlpha:0,duration:.35,ease:"power2.in"},17.6);//
==========// ACT 6: Modpacks (18s - 21.5s)//
==========tl.set("#s6",{autoAlpha:1},18);tl.from("#s6
.a6-card",{y:30,opacity:0,stagger:.15,duration:.5,ease:"power3.out"},18.1);tl.from("#s6
.a6-tag",{opacity:0,stagger:.15,duration:.3,ease:"sine.inOut"},18.6);tl.from("#s6 .a6-stat-
val",{scale:.8,opacity:0,stagger:.15,duration:.5,ease:"back.out(1.4)"},18.8);tl.from("#s6
.a6-stat-lbl",{opacity:0,stagger:.15,duration:.3},19.1);tl.from("#s6 .tagline",
{y:12,opacity:0,duration:.4,ease:"power3.out"},19.5);tl.to("#s6",
{autoAlpha:0,duration:.3,ease:"power2.in"},21.2);//
==========// ACT 7: Finale (21.5s - 30s)//
==========tl.set("#s7",
{autoAlpha:1},21.5);tl.from("#s7 .a7-line",
{y:25,opacity:0,stagger:.25,duration:.6,ease:"power3.out"},21.6);tl.from("#s7 .a7-divider",
{scaleX:0,opacity:0,duration:.4,ease:"power3.out",transformOrigin:"center"},23.6);tl.from("
#s7 .a7-logo-small",{scale:.8,opacity:0,duration:.5,ease:"sine.inOut"},24.2);tl.from("#s7
.a7-date",{y:30,opacity:0,duration:.7,ease:"power3.out"},25);// keep finale on screen until
end// total timeline ~30s// Registerwindow.__timelines = window.__timelines ||
{};window.__timelines["versepc-fastcut"] = tl;</script></body></html>
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